

Independent Surreal Copies and Arithmetic Intersections

Prescribed Hahn cores, algebraic independence,
two transcendental descent obstructions, and composita

AI-assisted research draft

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Built from manuscripts of batches 31 and 34 of the collection, pinned to `bcac55a` and `7b256e0`

Abstract

Let H be any set-sized divisible additive subgroup of the surreal numbers, and let $L = \mathcal{K}_{\mathbb{R}}(H)$ be the full Conway–Hahn field with exponents in H . We construct an ordinal-indexed proper class of strong, real-fixing, monomial-preserving self-embeddings of $\mathbf{No}$ whose images are jointly linearly disjoint over L and have pairwise intersection exactly L . All embeddings fix L pointwise and preserve and reflect the omnific integers. The same construction gives complex-fixing, conjugation-compatible copies of $\mathbf{No}(i)$ and its Gaussian omnific ring. Every set of parameters can be fixed by choosing a suitable H .

The proof combines a cut-realization lemma that avoids an arbitrary set of previously chosen values with a coset-slice criterion for linear disjointness of full Hahn fields. It yields exact Boolean meet diagrams and descent of all polynomial relations across their squares. Two consequences distinguish this construction from mere universality. First, the fraction field of the intersection of two full omnific copies is a proper subfield of their common field, with a transcendental gap for every set-sized H . For $H = \mathbb{Q}$ a witness is $1 + \sum_{n \geq 1} \omega^{-n!}$; real constants already belong to the smaller fraction field. Second, explicit independent copies with intersection $\mathbb{R}$ have a full Hahn join containing an explicitly described series transcendental over their ordinary compositum. Both obstructions persist on complexification.

Status and scope. The article provides complete mathematical arguments relative to standard Conway normal form and Hahn-field foundations. The proposed new contribution is the combined prescribed-intersection theorem and its arithmetic and compositum applications. The ingredients and the bounded-support transcendence mechanism are not claimed as new. This is an unrefereed, non-formalized draft, not a certified priority claim or a claimed solution of a named longstanding conjecture. [write] Independent proof review and formalization are pending. Labels carry the prefix `isc:`. Conventions, the place in the collection, the non-claims and the provenance are in [sections 1.5, 1.6 and 14](#) and [Appendix A](#); added text is marked [write]. [merge] Sections 15–27 add a second manuscript (prefix `isc:cp:`).

Repository comparison is pinned to commit

`bcac55ae6fd3f2b354e568ba1d2e94d496a2cc59`.

Sources, limitations, and a concrete research program are included.

Contents

1	Three questions and their answers	3
1.1	Prescribing the common field	3
1.2	Arithmetic intersection is not field reconstruction	4
1.3	An ordinary compositum is smaller than a full Hahn join	4
1.4	What is new here, and what is not	4
1.5	Notation and conventions in this report	5
1.6	Place in the collection	6
2	Foundations, support, and size	9
2.1	A class framework with set-sized constructions	9
2.2	Full Hahn fields in decreasing-exponent convention	9
2.3	Strong summation and finite algebraic independence	10
3	Independent ordered-group embeddings	10
4	Strong field lifts and omnific compatibility	12
5	Coset slices and linear disjointness	13
6	Prescribed intersections and Boolean meet diagrams	15
6.1	All intersections in an entire diagram	16
6.2	A limited model-theoretic conclusion	17
7	Explicit copies with intersection exactly the constants	17
8	A transcendental gap above an ordinary compositum	19
8.1	Iterated Hahn coordinates	19
9	Omnific intersections and their fraction fields	21
9.1	The bounded-support field is exactly the shared fraction field	22
10	A self-contained cofinal gap argument	23
10.1	Where the two transcendence arguments differ	25
11	A set-sized version of the construction	25
12	Further research questions	26
13	Proof dependencies, verification boundary, and conclusion	28
13.1	A dependency map	28
13.2	A practical formalization decomposition	28
13.3	Claims deliberately not made	29
13.4	Conclusion	29
14	Consolidated ledger of non-claims	30
15	Beyond composita of surreal copies: a later manuscript	32
15.1	Field generation is not Hahn summation	33
15.2	Main conclusions	33
15.3	Scope, sources, and the meaning of new	34
15.4	What the addition settles in this report	34
15.5	Conventions and renamed symbols	35
15.6	Place in the collection	37

16 Conventions and foundational lemmas of the addition	38
16.1 Two support conventions	38
16.2 Euler derivations and differential independence	39
16.3 The coefficient-field envelope	39
17 An exact coefficient-rank test for the compositum	40
18 Linear disjointness under two enlargements	41
19 Differential independence from one coefficient variable	43
20 Independent coefficient blocks and tree amplification	45
20.1 Coefficient derivations with finite avoidance	45
20.2 The block theorem	46
20.3 Local use for class coefficient fields	47
21 Exact transcendence degrees for Laurent-series composita	48
21.1 Boundary cases that must not be suppressed	48
22 Separated surreal copies and explicit omnific witnesses	49
22.1 The inner and outer levels, printed once	49
22.2 The exact test for the explicit copies	50
22.3 A concrete Euler derivation	50
22.4 The countably supported family	50
23 A fixed tail survives every set of parameters	52
24 Arithmetic joins and the surcomplex extension	54
24.1 Fraction fields of the three integer rings	54
24.2 Surcomplex and Gaussian omnific versions	55
25 What the addition resolves, and what it does not	56
25.1 Relation to Research question 3	56
25.2 Provenance and proof dependencies	56
25.3 Critical proof checks	57
25.4 Limitations and nonclaims, as stated by the source	57
25.5 What the addition does not claim	58
26 Research questions of the addition	60
27 Finite verification and a formalization decomposition of the addition	62
27.1 Finite verification and reproducibility	62
27.2 A proposed formalization decomposition	63
A Provenance of this report	64
A.1 One manuscript	64
A.2 Repository statements at the pin	64
A.3 References checked for this report	65
A.4 Proofs read when the report was written	65
A.5 Theorem numbers	65
A.6 Files	66
A.7 Source 02 (batch 34)	66
A.8 Files of source 02, and their rerun	67

1 Three questions and their answers

The existence of an embedding into the surreal numbers says little about how two embedded copies meet. An intersection question asks for much more: which elements must be shared, which algebraic relations can pass from one copy to another, and whether their distinguished integer rings reconstruct the same common field. This article treats these questions for *full proper-class copies* of the surreal field, rather than only for small subfields.

The ambient presentation matters. A surreal has a unique Conway normal form

$$x = \sum_{g \in S} r_g \omega^g,$$

where $S \subseteq \mathbf{No}$ is a reverse-well-ordered *set*. For an additive subgroup $G \subseteq \mathbf{No}$, put

$$\mathcal{K}_{\mathbb{R}}(G) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq G\}, \quad \mathcal{K}_{\mathbb{C}}(G) = \mathcal{K}_{\mathbb{R}}(G)(i).$$

These are full Hahn fields: there is no additional bound on the cardinality or order type of a set-sized support. If G is divisible, the real field is real closed and the complex field is algebraically closed. These foundations are classical; see [1, 2, 3].

1.1 Prescribing the common field

Theorem A (Independent full copies over a prescribed Hahn core). *Let $H \subseteq \mathbf{No}$ be a set-sized divisible additive subgroup and write $L_k = \mathcal{K}_k(H)$ for $k = \mathbb{R}, \mathbb{C}$. There are class maps*

$$\Phi_{\alpha} : \mathbf{No} \longrightarrow \mathbf{No} \quad (\alpha \in \mathbf{On})$$

with the following properties.

(i) *Each Φ_{α} is a proper ordered field embedding, fixes $L_{\mathbb{R}}$ pointwise, sends Conway monomials to Conway monomials, and preserves and reflects Hahn summability of set-indexed families.*

(ii) *With $K_{\alpha} = \Phi_{\alpha}(\mathbf{No})$,*

$$K_{\alpha} \cap K_{\beta} = L_{\mathbb{R}} \quad (\alpha \neq \beta).$$

Every finite subfamily is jointly linearly disjoint over $L_{\mathbb{R}}$.

(iii) *For $R_{\alpha} = \Phi_{\alpha}(\mathbf{Oz})$,*

$$R_{\alpha} = K_{\alpha} \cap \mathbf{Oz}, \quad R_{\alpha} \cap R_{\beta} = L_{\mathbb{R}} \cap \mathbf{Oz}.$$

Moreover $\Phi_{\alpha}^{-1}(\mathbf{Oz}) = \mathbf{Oz}$.

(iv) *The coefficientwise complexifications have the same assertions over $L_{\mathbb{C}}$, commute with conjugation, and preserve and reflect $\mathbf{Oz}[i]$.*

The proof is given in [sections 3 to 6](#). “Jointly linearly disjoint” is stronger than pairwise intersection being small: finitely many independent choices from different copies satisfy no new polynomial relations over the common core. Proper-class terminology is made precise in [section 2](#); no tensor product indexed by a proper class is required.

A useful immediate specialization is $H = \{0\}$. It produces full copies of $\mathbf{No}$ intersecting exactly in $\mathbb{R}$, while their omnific rings intersect exactly in $\mathbb{Z}$. A second specialization, $H = \mathbb{Q}$, fixes not just ω and its rational powers but every Hahn series with rational exponents. In either case each copy is as large as the whole surreal field.

1.2 Arithmetic intersection is not field reconstruction

Theorem B (Transcendental failure of fraction-field descent). *Choose two copies in Theorem A. Then*

$$\text{Frac}(R_\alpha) = K_\alpha, \quad \text{Frac}(R_\alpha \cap R_\beta) \subsetneq K_\alpha \cap K_\beta = L_{\mathbb{R}}.$$

The latter extension is transcendental for every set-sized H , including $H = \{0\}$. For $H \neq 0$, the smaller field already contains every real constant. For $H = \mathbb{Q}$, the element

$$F = 1 + \sum_{n=1}^{\infty} \omega^{-n!}$$

is transcendental over $\text{Frac}(R_\alpha \cap R_\beta)$. The corresponding assertions hold for the Gaussian omnific copies inside $\mathbf{No}(i)$.

This is a failure of a particular descent operation, not a failure of $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. Indeed that classical identity is indispensable to the argument. Each copy separately supplies denominators for every element of its field, but the *shared* denominators need not suffice for their shared field. The phenomenon remains after the missing-real-constants explanation has been removed. Sections 9 and 10 prove the result.

1.3 An ordinary compositum is smaller than a full Hahn join

For full Hahn fields in the same normal-form presentation, their *Hahn join* is the full Hahn field on the sum of their exponent groups. Their *ordinary compositum* is the field generated using finitely many field operations. These should not be conflated.

Theorem C (A transcendental gap above the compositum). *There are explicit strong embeddings Φ_0, Φ_1 as in Theorem A with $H = 0$ for which, writing*

$$K_j = \Phi_j(\mathbf{No}), \quad E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1), \quad M = K_0 K_1,$$

the field E contains an explicit countably supported element S transcendental over M . In particular even the relative real algebraic closure of the ordinary compositum is a proper subfield of the Hahn join. The same S is transcendental over the compositum of the complexified copies inside $\mathcal{K}_{\mathbb{C}}(G_0 + G_1)$.

The two-level normal-form construction is explicit in section 7. The formula for S and the coefficient-field obstruction are proved in section 8. The latter argument is independent of the lacunary-series argument used for Theorem B.

1.4 What is new here, and what is not

The classical inputs are the set-cut property of $\mathbf{No}$, its normal-form presentation, Hahn multiplication and inversion, and the real/algebraic closedness criteria. The group extension argument is a concrete saturation construction, not a new saturation principle. The monomial lifting argument is standard Hahn functoriality. The coset-slice proof is presented as an elementary reusable lemma rather than as a certified new theorem of Hahn field theory.

The relevant repository comparison used the documentation catalogue and the reports on omnific-preserving automorphisms and bounded-support Hahn arithmetic at the pinned snapshot [6, 7, 8]. The former concerns stabilizers and rigidity; the latter already proposes much larger algebraically

independent families over bounded-support fraction fields. We neither repackage those results as discoveries nor use their unrefereed main theorems as black boxes. A self-contained one-witness gap argument is supplied here. In the published literature, the failure of fraction-field equality at set-sized cofinality already appears in L’Innocente–Mantova, Proposition 2.4.5 [3].

The proposed contribution is instead the *simultaneous arrangement*: full copies, a prescribed set-sized full Hahn intersection, strong arithmetic compatibility, independence of every finite subfamily, exact meet diagrams, and the two transcendental descent obstructions inside those arrangements. Targeted literature searches do not certify that this combined formulation is absent from all published or unpublished work. No claim of having audited every report or every later repository revision is made.

[write] This subsection is the source’s, written at its pin. Its description of the omnific-preserving report as concerning stabilizers and rigidity was accurate then: at the pin that report had no part on embeddings. Its Part III, written afterwards, classifies the strong exact-monomial embeddings that preserve $\mathbf{Oz}$, and the lifts used here are the real-fixing case of that classification. The bounded-support report was unchanged between the pin and batch 31; batch 32 added its Galois part (its Sections 11–19). The attribution to L’Innocente–Mantova was checked (Appendix A.3). The relation to the collection as it now stands, with exact labels, is in section 1.6.

1.5 Notation and conventions in this report

[write] This subsection and the next were written when the manuscript joined the collection. No symbol of the source was renamed and no normalization was changed. The meanings below are fixed once for the whole report, and the collisions with other reports are disclosed here rather than removed. Paths such as `surreal/omnific-preserving-automorphisms` are relative to `docs/`, and labels of other reports are cited by name.

- *Two exponent conventions.* Section 1 to section 8, section 11 and the beginning of section 9 use Conway’s decreasing large- ω convention $x = \sum r_g \omega^g$. There $\mathbf{Oz}$ consists of the series supported in $\mathbf{No}^{>0} \cup \{0\}$ with integral constant coefficient: the purely infinite part sits at *positive* exponents. From section 9.1 to the end of section 10 the source switches to $t^h = \omega^{-h}$, the convention of the omnific-preserving report’s Part III (`opa:as:sub:notation`) and of the bounded-support report (`bst:sec:setup`); there the purely infinite part has *negative* t -exponents and v is the least t -exponent. The tempting false reading of $\mathcal{B}_k(H)$ is “support bounded above in the Conway exponents”. The true reading is “bounded above in the t -exponents”, that is, Conway exponents bounded *below*.
- *Fields.* $\mathcal{K}_k(G)$ is the *full* Hahn field on G , support-closed and not generated by monomials; for $k \neq \mathbb{R}, \mathbb{C}$ it is formal. $L_k = \mathcal{K}_k(H)$, with H a set-sized divisible subgroup of $\mathbf{No}$. This H need not be convex. It is not a convex subgroup H_g of the omnific-preserving report (`opa:def:convex`), and not the large-cardinal report’s embedding H_j .
- *Rings.* $\mathcal{A}_k(H) = D_k \oplus \Pi_k(H)$ with $D_{\mathbb{R}} = \mathbb{Z}$, $D_{\mathbb{C}} = \mathbb{Z}[i]$ restricts the constant coefficient to D_k . It is the ring $\mathcal{R}_{\Gamma}$ (or $\mathcal{R}_{\Gamma}^{\mathbb{C}}$) of the omnific-preserving report’s Part III and the ring $\mathcal{R}_{D_k}(k, \Gamma)$ of the Diophantine report’s Section 11. It is *not* that Part’s $\mathcal{A}_{\Gamma} = \mathbb{R} \oplus \Pi_{\Gamma}$. Nor is it the Diophantine report’s $\mathcal{A}_k(\Gamma)$, whose constant coefficient ranges over all of k . D_k is a ring of integers and never a derivation; the omnific-preserving report reserves d and D for derivations. $\mathcal{B}_k(H)$ is the bounded-support report’s $\mathcal{B}_H(k)$, with the two arguments in the opposite order, and $\text{Frac } \mathcal{B}_k(H)$ is its $\mathcal{F}_H(k)$.

- *Maps.* Φ_h of (4.1) is the monomial map $M_{1,h} = J_{h,\text{id}}$ of `opa:as:eq:classification`. It is not the Taylor automorphism Φ_d of the surcomplex-automorphisms report (the omnific-preserving report's $\mathcal{T}_{d,\delta}$), and not the large-cardinal report's sign-sequence embedding J . The letters h_α , G_α , Φ_α , K_α denote the family of [theorem 3.4](#) in [sections 3 to 6](#) and the explicit family in [sections 7 and 8](#). There h_α is the *inner* map (7.1), which is the inner lift ι_{σ_α} of `opa:as:eq:innerlift`. The maps h_α fix H as a set of exponents, and Φ_α fixes $\mathcal{K}_{\mathbb{R}}(H)$ ([remark 3.6](#)). “Monomial-preserving” is the omnific-preserving report's “exact-monomial”: a Conway monomial goes to a Conway monomial with coefficient one. It does not mean commuting with the omega-map. The order isomorphism ψ of [lemma 7.1](#) is not that report's μ_{12} (its source 12's ψ), nor μ_{11} or μ_{13} . The slice $\lambda_{g,H}$ of (5.1) is the bounded-support report's shifted coset projection π_g (`bst:eq:projection`); it is not the class functional $\lambda_{\mathbf{On}}$.
- *Local letters.* L, R are the sides of a cut in (2.1) and [lemmas 3.1 and 3.2](#); L is also a generic base field in the definition of linear disjointness and in [lemma 5.3](#). $M = K_0K_1$ is an ordinary compositum, not a monomial map $M_{\chi,\tau}$. E is a span in the proof of [theorem 3.4](#), the field E_J of (6.1), and the Hahn join of [Theorem C](#). F is the witness of [Theorem B](#) and (10.2), and a polynomial in [corollary 5.6](#). It is not the image field F of `opa:as:prop:overlap`, and not the fraction field $\mathcal{F}_G(K)$. A is a set of parameters in [corollary 6.1](#), a matrix in [lemma 5.3](#) and a subgroup in [lemma 8.1](#), where B is a subgroup too; the calligraphic $\mathcal{A}, \mathcal{B}$ are the rings. D is a domain or subspace in [section 3](#) and a coefficient field in [section 8](#). S is a support set and the witness (8.3). I is an index set and I_α an interval.
- *Cardinals.* In [section 10](#), $\kappa = \text{cf}(H)$. In [section 11](#), κ is an arbitrary uncountable regular cardinal. Neither is the critical point κ of the large-cardinal report.
- *Strong* means preserving Hahn summability of set-indexed families; it is not convergence of partial sums.

1.6 Place in the collection

[write] The source compared itself with the catalogue, the omnific-preserving report and the bounded-support report at its pin ([Appendix A.2](#)). Checked against the collection at the placement commit, the relations are these.

Omnific-preserving automorphisms (`surreal/omnific-preserving-automorphisms`, prefix `opa:`). Its Part III (sub-prefix `opa:as:`) was written after the pin, so the source did not see it. Its bottom-gap classification `opa:as:thm:classification` lists the strongly additive field embeddings J with $J(t^\gamma) = t^{\tau(\gamma)}$ and $J(\mathcal{R}_\Gamma) \subseteq \mathcal{R}_\Delta$ as the maps $J_{\tau,\rho}$. Here ρ ranges over the ordered embeddings of $\mathbb{R}$ into the Hahn field of the bottom gap of τ .

1. *Lifts.* With $\Gamma = \Delta = \mathbf{No}$, $t^\gamma = \omega^{-\gamma}$ and $\tau = h$, the lift Φ_h of [theorem 4.1](#) is $J_{h,\text{id}}$, the case $\rho = \text{id}$. The same exponent map serves in both conventions because h is additive. The preservation and reflection of $\mathbf{Oz}$ in [proposition 4.2](#) is Step 5 of that proof and the display `opa:as:eq:classprops`. [Proposition 4.4](#) is the uniqueness clause of Step 2, for real-fixing maps; here the order of h is derived from real fixing. The floor, standard-part and modulus clauses of [proposition 4.2](#) are not stated there.
2. *Real fixing is a choice.* The explicit ranges G_α of [section 7](#) have a nonzero bottom gap: $\omega^{3\alpha}$ lies below every positive element of G_α , and for $\alpha = 0$ the gap is the group of finite surreals. By `opa:as:thm:taylorcriterion`, the composite of a Taylor automorphism $\mathcal{T}_{d,\delta}$ with Φ_α , for

$0 < \delta$ in that gap, therefore also preserves and reflects **Oz** and moves a transcendental real. The source takes $\rho = \text{id}$ throughout, which is consistent. For the family of [theorem 3.4](#) the construction does not determine the gap. [Theorem 6.2](#) depends on the real-fixing hypothesis: in the larger exact-monomial class, images meet $\mathbb{R}$ in the fixed field of ρ , for example $\ker d$ (`opa:as:prop:overlap`). No classification of intersections in that class is claimed.

3. *Inner and outer levels.* The inner map (7.1) is the inner lift ι_{σ_α} of `opa:as:eq:innerlift` and `opa:as:lem:inner`, and $\Phi_\alpha = M_{1, \iota_{\sigma_\alpha}}$. The order isomorphisms of `opa:as:eq:mus` have unbounded images; the explicit embeddings `opa:as:thm:explicit` combine them with Taylor sections in order to move a real. The maps σ_α here have bounded, pairwise disjoint images, which is what gives independence. The failure of omega-map commutation, $\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}$, is the same observation as the one recorded after `opa:as:cor:homogeneity`.
4. *Parameters.* [Corollary 6.1](#) extends `opa:as:thm:parameters`, which gives two proper strong exact-monomial copies fixing $A \cup \mathbb{R}$ and preserving and reflecting **Oz**. Here there is an **On**-indexed family whose pairwise intersection is exactly the full Hahn field $\mathcal{K}_{\mathbb{R}}(H) \supseteq A \cup \mathbb{R}$, jointly linearly disjoint over it. The mechanisms differ. That theorem fixes the inner supports through an order embedding (`opa:as:lem:parameters`); here the outer exponent group H is fixed by h_α . The topological assertions of that theorem (closed, discrete or nondiscrete images) are not claimed here, and that report does not study intersections.
5. *Complex version.* The complex clauses of [Theorem A](#) and [theorem 7.2](#) are the case $\varepsilon = 1$, $\rho = \text{id}$ of `opa:as:thm:complexembed`, parallel to `opa:as:cor:complexcopies`.
6. *Elementarity.* The ordered-field statement of [section 6.2](#) is the first assertion of `opa:as:thm:elementary`. That theorem's nonelementarity criterion needs a moved real, and the maps here fix $\mathbb{R}$.
7. *Questions.* None is answered. Research question 2 is the pair version, at a fixed Hahn core, of `opa:as:q:conjugacy`. Research question 9 asks `opa:as:q:elementary` for the maps here, with a prescribed intersection added. Research questions 6 and 7 lie inside `opa:as:q:exp`. [Theorem 6.3](#) decides image inclusion and intersection inside one independent family with $\rho = \text{id}$. That touches `opa:as:q:composition`, which asks for image inclusion of classified maps in general, but does not answer it. Research question 1 is related to `opa:as:q:units`, which asks to remove exact monomial preservation, but it asks about intersections and is new.

Bounded-support transcendence (`surreal/transcendence-over-bounded-support`, prefix `bst:`). The source cites it and says that [section 10](#) duplicates its one-witness case. That report was unchanged between the pin and batch 31; batch 32 added its Galois part (its Sections 11–19), which is not used here.

1. [Proposition 9.3](#) is the localization of `bst:prop:localdensity`, with the constant coefficient restricted to D_k . Since batch 32 that report also prints the general form, for every field K and every unital subring $D \subseteq K$, as `bst:gr:prop:fractions`.
2. [Theorem 10.2](#) is `bst:cor:onegap` when $\text{cf}(H) = \aleph_0$; that corollary's hypothesis forces a cofinal support of countable cofinality. In characteristic zero and at every cofinality it is the one-series case of `bst:thm:mainintro` (`bst:thm:independent`), which gives 2^κ independent series on one support of order type κ and proves that κ is optimal (`bst:cor:optimal`). The single-series form at uncountable cofinality over an arbitrary coefficient field, finite fields included, is not stated there; that report says that finite coefficient fields are not covered by its independence theorem.

Its proof is that report’s tail estimate and root count. [Lemma 10.1](#) is the second inequality of a separated exponent system (`bst:lem:skeleton`).

3. The factorial witness of [corollary 10.3](#) is `bst:cor:onegap` with $a_n = n!$ and all coefficients one; the constant 1 changes nothing. `bst:thm:actualreal` uses the same exponents, with integer codes, for a continuum of independent actual infinitesimals over the complex bounded-support fraction field. In the omnific-preserving report, `opa:as:ex:smallfrac` uses $\sum t^{n!}$ at $\Gamma = \mathbb{Z}$ to show that $\sum t^{n!}$ is not in $\text{Frac } \mathcal{R}_{\mathbb{Z}} = \mathbb{R}(t^{-1})$: non-membership, not transcendence.
4. Research question 8 is that report’s programme (`bst:cor:exactcard`, `bst:cor:realgroup`), as the source says. Only its uniformity over realized diagrams is new.
5. The coset slice $\lambda_{g,H}$ is that report’s projection π_g , and the linearity in [lemma 5.2](#) is `bst:lem:projection`. There it serves descent over a cofinal subgroup (`bst:thm:descent`). [Theorem 5.4](#) is not stated there: two full Hahn fields are linearly disjoint over the full Hahn field of the intersection of their groups, with no cofinality hypothesis. The source presents it as an elementary reusable lemma, not as a certified new theorem.

Omnific Diophantine geometry (`surreal/omnific-diophantine-geometry`, prefix `odg:`). [Lemma 9.1](#) is the global monomial clearing `odg:thm:fractions`, with the same proof.

Large-cardinal embeddings (`foundations-and-computation/large-cardinal-embedding-s-and-normal-forms`, title “Critical-Point Defects in Surreal Arithmetic”). It was placed after the pin, so the source did not see it. Its write phase is in progress, so it is cited here by name and question title, not by label. An elementary embedding with critical point κ gives there two ordered-field embeddings J and H_j of **No**. The strong one, H_j , is the lift Φ_h of [theorem 4.1](#) with h the restriction of J to $(\mathbf{No}, +, <)$, since $H_j(\omega^a) = \omega^{J(a)}$. The map J is not strongly additive, and $J(\mathbf{No})$ is not a full Hahn field on a subgroup. The common field there (surreals with fewer than κ terms and exponents in $J(\mathbf{No})$) is a proper class, not a full Hahn field. That report’s question “Joint algebraic dependence of the images” asks whether $J(\mathbf{No})$ and $H_j(\mathbf{No})$ are linearly disjoint over that common field. It is related to [theorem 5.4](#) but *not answered* by it: the slice theorem needs full Hahn fields on subgroups on both sides.

Formalization. No `isc:` label has a Lean mapping in `FORMALIZATION.md`. No theorem of this report is Lean-verified, and the identifiers proposed in [section 13.2](#) do not exist in the collection’s Lean library.

Status of the research questions. None of the collection’s named questions is answered. Research questions 1, 3, 4, 5 and 10 are new. Research question 2 is a pair version of `opa:as:q:conjugacy`, 6 and 7 lie inside `opa:as:q:exp`, 8 is the bounded-support programme, and 9 is an instance of `opa:as:q:elementary`. Research question 10 overlaps `opa:as:q:formal` in its Hahn-field kernel. [merge] *Status (batch 34)*. Research question 3 is now answered in the separated case with trivial common core, the case of [Theorem C](#), and in the natural Laurent-series cardinal cases, by the second manuscript printed in Sections 15–27 ([section 15.4](#)); the non-separated case and prescribed nonzero common cores stay open. Research questions 11–22 come from that manuscript and are new. The relations of the addition to the three-duals, tail-spans, bounded-support and dilation reports are in [section 15.6](#).

2 Foundations, support, and size

2.1 A class framework with set-sized constructions

We work in von Neumann–Bernays–Gödel class theory with Global Choice, with sets satisfying ZFC, as in the foundational convention of [2]. Classes are not members of other classes. Thus an ordinal-indexed family of class functions means one class relation whose sections are the functions. A field with carrier **No** is a *class field*; its elements, finite tuples, polynomial coefficient lists, and the supports of individual elements are sets.

The one surreal-specific order principle needed for the construction is:

$$L, R \subseteq \mathbf{No} \text{ sets, } L < R \implies \text{there is } y \in \mathbf{No} \text{ with } L < y < R. \quad (2.1)$$

Empty sides are permitted. The simplest such y , denoted $\{L \mid R\}$, is uniquely determined. We never apply (2.1) to proper-class sides. This restriction is central, not cosmetic.

Global Choice supplies a set-like class well-order of the surreals: order first by birthday and well-order each set-sized birthday level. It similarly supplies a set-like enumeration of $\mathbf{On} \times \mathbf{No}$. For the latter, group pairs by a bound on both their ordinal coordinate and the birthday of the surreal coordinate, then well-order each new set layer. Each initial segment is a set, and the resulting class order has order type **On**.

The recursion below has a *set as its entire state at every ordinal stage*. For any fixed ordinal, ordinary set transfinite recursion with the fixed class parameters constructs that initial segment. Uniqueness makes these initial segments coherent, and elementary class comprehension collects their stages. This is not an appeal to recursion with a proper-class-valued state at each stage, nor to an unmentioned class truth predicate.

2.2 Full Hahn fields in decreasing-exponent convention

If G is an ordered additive subgroup of **No** and k is a set-sized field, let $\mathcal{K}_k(G)$ denote the formal series

$$\sum_{g \in S} a_g \omega^g, \quad a_g \in k,$$

with reverse-well-ordered set support $S \subseteq G$. For $k = \mathbb{R}$ these are actual surreals. For $k = \mathbb{C}$, taking real and imaginary coefficients identifies them with actual elements of $\mathbf{No}(i)$. The union of the two real supports is again reverse well ordered, so this complexification is exact. For other k the notation is purely formal.

A nonzero series has a largest exponent, and its leading coefficient is nonzero. Addition is coefficientwise and multiplication is convolution. The Hahn support lemma guarantees that the support of a product is allowable and that each product coefficient is a finite sum. Normalizing a nonzero series as $a\omega^g(1 + \varepsilon)$ with all exponents of ε negative gives its inverse by a summable geometric family. Consequently $\mathcal{K}_k(G)$ is a field. These statements are understood locally in set-sized subgroups when G is a proper class.

The standard closedness theorem says that a Hahn field of characteristic zero with divisible exponent group is real closed when its coefficients are real closed, and algebraically closed when its coefficients are algebraically closed [3]. The same conclusion holds for our class fields: the supports of finitely many coefficients of a polynomial generate a set-sized divisible subgroup, and the standard

set-sized theorem provides the required roots there. In particular,

$$\mathcal{K}_{\mathbb{R}}(G) \text{ is real closed,} \quad \mathcal{K}_{\mathbb{C}}(G) \text{ is algebraically closed} \quad \text{if } G \text{ is divisible.}$$

If H is a set, then $\mathcal{K}_k(H)$ is a set: it is a subclass of the set of all functions $H \rightarrow k$. In contrast, if G is class-isomorphic to $(\mathbf{No}, +, <)$, then $\mathcal{K}_k(G)$ is a proper class, because it contains the proper class of monomials ω^g .

2.3 Strong summation and finite algebraic independence

Definition 2.1 (Hahn summability). A set-indexed family $(x_j)_{j \in J}$ of Hahn series is summable if the union of its supports is reverse well ordered and every exponent belongs to the support of only finitely many x_j . Its sum is the resulting coefficientwise finite sum at every exponent. A map is *strongly additive* if it sends such families to summable families and commutes with their sums.

The sum of a normal form is strong in this sense. Ordinary analytic convergence of a sequence of real constants is a different notion; it is not used in any proof. In particular strong additivity cannot be replaced by an unqualified assertion of continuity.

Definition 2.2 (Linear disjointness for class fields). Subfields P, Q of one class field are linearly disjoint over $L \subseteq P \cap Q$ if every finite L -linearly independent tuple from P remains linearly independent over Q . A finite family is jointly linearly disjoint if its multiplication map on finite tensors over L is injective. Equivalently, this injectivity holds after restricting each factor to any set-sized subfield containing the finitely many elements under consideration and the base when the base is a set.

All tensor arguments below can instead be read as finite matrix arguments. For a class-sized base, linear independence still quantifies only over a finite tuple of coefficients at a time. “Algebraically independent” also has its usual finite-polynomial meaning. We make no assertion about a proper-class-dimensional tensor object in the category of sets.

3 Independent ordered-group embeddings

The core construction is carried out on the *additive exponent group*, not initially on the field. Its flexibility comes from the ability to realize a set-sized cut while avoiding all previously used values.

Lemma 3.1 (A cut avoiding a set). *Let $L, R, E \subseteq \mathbf{No}$ be sets and suppose $L < R$. There is a surreal $y \notin E$ with $L < y < R$.*

Proof. Let $T = \{e \in E : L < e < R\}$. This is a set and $L \cup T < R$. Choose

$$y = \{L \cup T \mid R\}.$$

Then $L < y < R$. Were $y \in E$, it would belong to T , contradicting $T < y$. This also covers empty L , R , or T . $\square$

Lemma 3.2 (Ordered one-point extension with avoidance). *Let D be a set-sized $\mathbb{Q}$ -vector subspace of $\mathbf{No}$, let $f : D \rightarrow \mathbf{No}$ be an ordered additive embedding, and let E be a set-sized $\mathbb{Q}$ -vector subspace containing $f(D)$. For $x \notin D$, there is an ordered additive embedding*

$$f^+ : D + \mathbb{Q}x \longrightarrow \mathbf{No}$$

extending f and such that $f^+(x) \notin E$.

Proof. Set

$$L = \{f(d) : d \in D, d < x\}, \quad R = \{f(d) : d \in D, x < d\}.$$

These are sets with $L < R$. By lemma 3.1, choose $y \notin E$ realizing this cut. Define

$$f^+(d + qx) = f(d) + qy \quad (d \in D, q \in \mathbb{Q}).$$

The representation $d + qx$ is unique because $x \notin D$, so the formula is well defined and additive. If $q > 0$, positivity of $d + qx$ is equivalent to $x > -d/q$; by the defining cut this is equivalent to $y > f(-d/q)$, hence to $f(d) + qy > 0$. For $q < 0$, the same argument reverses the comparison. The case $q = 0$ follows from the order preservation of f . Thus f^+ preserves strict positivity and is an ordered embedding. Its new value is $y \notin E$. $\square$

Definition 3.3 (Independence over an additive subgroup). Subgroups $(G_i)_{i \in I}$ containing H are *independent over H* if, whenever $J \subseteq I$ is finite and $g_i \in G_i$,

$$\sum_{i \in J} g_i \in H \implies g_i \in H \text{ for every } i \in J.$$

Equivalently the images G_i/H have internal direct sum in the quotient additive group. The quotient here is algebraic; H need not be convex.

Theorem 3.4 (Independent group copies over any set-sized base). *Let $H \subseteq \mathbf{No}$ be a set-sized divisible additive subgroup. There is an ordinal-indexed class family of ordered additive embeddings*

$$h_\alpha : \mathbf{No} \longrightarrow \mathbf{No}$$

such that $h_\alpha|_H = \text{id}_H$ and the ranges $G_\alpha = h_\alpha(\mathbf{No})$ are independent over H .

Proof. A divisible subgroup of the torsion-free additive group of $\mathbf{No}$ is a $\mathbb{Q}$ -vector subspace. Fix the set-like enumeration of $\mathbf{On} \times \mathbf{No}$ described in section 2. We process each requirement (α, x) in that order, meaning that x must eventually be in the domain of the α th partial embedding.

At a stage the state consists of partial ordered $\mathbb{Q}$ -linear embeddings $f_\alpha : D_\alpha \rightarrow \mathbf{No}$ for a set of active indices. Their domains and graphs are sets, they contain the identity on H , and their ranges are independent over H . An inactive index has the implicit partial map id_H ; these implicit copies of H need not be stored separately.

Suppose the next requirement is (α, x) . Activate α if needed. If $x \in D_\alpha$, do nothing. Otherwise let

$$E = \text{span}_{\mathbb{Q}} \left(H \cup \bigcup_{\beta \text{ active}} f_\beta(D_\beta) \right).$$

The state is a set, so E is a set-sized vector space. Apply lemma 3.2 to extend f_α to $D_\alpha + \mathbb{Q}x$, choosing its new value y outside E .

We verify that independence survives. A new finite relation modulo H can involve at most one changed component. Write that component as $f_\alpha(d) + qy$. If $q \neq 0$, the relation would express y as a rational linear combination of old range elements and elements of H , contradicting $y \notin E$. Therefore $q = 0$ and the old independence assertion applies. This proves the invariant at successor stages.

At a limit ordinal, take unions of the compatible graphs. There have been only a set of preceding stages, so the resulting state is still a set. Every finite relation already occurred at an earlier stage, so independence is retained. The set-valued recursion is legitimate in the stated class framework.

Finally define h_α to be the union of its partial graphs over all stages. Each requirement was processed, hence its domain is all of $\mathbf{No}$. Every ordered-additive identity is witnessed at some stage, so h_α is an ordered additive embedding fixing H . Any finite relation among final range elements is also witnessed at one stage. Consequently the final ranges are independent over H . $\square$

Corollary 3.5. *For distinct α, β , one has $G_\alpha \cap G_\beta = H$. More generally, for $\alpha \notin J$,*

$$G_\alpha \cap \left(H + \sum_{\beta \in J} G_\beta \right) = H,$$

where the sum means finite sums, even when J is infinite. Every G_α is a proper subgroup of $\mathbf{No}$.

Proof. The first two assertions are exactly the independence condition. To prove properness, choose $\beta \neq \alpha$ and $x \notin H$. Injectivity and the identity on H imply $h_\beta(x) \notin H$. Thus it cannot lie in G_α , although it lies in $\mathbf{No}$. $\square$

Remark 3.6 (The role of the base). The maps h_α fix H as a set of *exponent values*. The field maps constructed next fix $\mathcal{K}_{\mathbb{R}}(H)$. These are different assertions: for a general subgroup H , it need not be true that $H \subseteq \mathcal{K}_{\mathbb{R}}(H)$. No identification of those two roles is used.

4 Strong field lifts and omnific compatibility

Theorem 4.1 (Monomial lift). *Let $h : \mathbf{No} \rightarrow \mathbf{No}$ be an ordered additive embedding. Then*

$$\Phi_h \left(\sum_g r_g \omega^g \right) = \sum_g r_g \omega^{h(g)} \quad (4.1)$$

defines a real-fixing ordered field embedding of $\mathbf{No}$ with image exactly $\mathcal{K}_{\mathbb{R}}(h(\mathbf{No}))$. It preserves and reflects Hahn summability, and its complexification has image $\mathcal{K}_{\mathbb{C}}(h(\mathbf{No}))$.

Proof. An order embedding sends a reverse-well-ordered set to a reverse-well-ordered set, so the formula defines a surreal. It preserves the leading coefficient and therefore the sign of a nonzero real-coefficient series. Coefficientwise addition is preserved. For multiplication, the identity $h(g + g') = h(g) + h(g')$ identifies each convolution coefficient on the two sides; the support lemma ensures that only finitely many terms contribute to each such coefficient. Since $h(0) = 0$, real constants are fixed. The map is injective by uniqueness of normal form, hence is a field embedding.

For a set-indexed family, the union of image supports is the h -image of the union of original supports. Reverse well-ordering and pointwise finiteness are equivalent under this injection. Thus summability is both preserved and reflected, and the sums agree coefficientwise.

Finally, a series supported in $h(\mathbf{No})$ can be pulled back term by term along the inverse of h on its image. Its support remains a set and is reverse well ordered. This proves that the image is the *full* Hahn field on $h(\mathbf{No})$, not merely the field generated by its monomials. Taking real and imaginary coefficients gives the complex assertion. $\square$

Proposition 4.2 (What the lift remembers). *The map Φ_h preserves the three normal-form projections to positive, zero, and negative exponents. It preserves and reflects membership in $\mathbf{Oz}$, and*

$$\Phi_h(\mathbf{Oz}) = \Phi_h(\mathbf{No}) \cap \mathbf{Oz}. \quad (4.2)$$

It commutes with the omnific floor map and with standard part on finite surreals. Its complexification commutes with conjugation, preserves and reflects $\mathbf{Oz}[i]$, and preserves the real-valued modulus in the sense

$$|\Phi_h^{\mathbb{C}}(z)| = \Phi_h(|z|).$$

Proof. Because h is ordered and additive, $h(g) > 0$, $h(g) = 0$, and $h(g) < 0$ are equivalent to the respective conditions on g . The coefficient at zero is unchanged. An omnific integer is precisely a series supported in $\mathbf{No}^{>0} \cup \{0\}$ with integral constant coefficient, so membership is preserved and reflected. The image formula follows by pulling back a series in the intersection using [theorem 4.1](#).

If $a = \lfloor x \rfloor_{\mathbf{Oz}}$, then $a \in \mathbf{Oz}$ and $a \leq x < a + 1$. Applying Φ_h gives an omnific integer in the corresponding unit interval for $\Phi_h(x)$, hence the unique omnific floor. If $x = r + \varepsilon$ is finite with $r \in \mathbb{R}$ and ε infinitesimal, the image has the same decomposition with infinitesimal part $\Phi_h(\varepsilon)$. This proves the standard-part assertion.

Complex conjugation is coefficientwise. Gaussian omnific membership is omnific membership of both real and imaginary parts. Finally $|z|$ is the unique nonnegative square root of $z\bar{z}$. The image of this equation, order preservation on the real part, and uniqueness of that square root prove the modulus identity. $\square$

Remark 4.3 (A floor boundary that must not be dropped). The positive-exponent truncation plus the ordinary floor of the constant coefficient is not always the omnific floor. For example, if $0 < \varepsilon$ is infinitesimal and $n \in \mathbb{Z}$, then $\lfloor n - \varepsilon \rfloor_{\mathbf{Oz}} = n - 1$. The order-and-uniqueness proof above retains this endpoint case automatically.

Proposition 4.4 (Classification within the monomial-preserving class). *Every strongly additive, real-fixing field embedding $\Phi : \mathbf{No} \rightarrow \mathbf{No}$ that maps each Conway monomial to a Conway monomial is uniquely Φ_h for an ordered additive embedding $h : \mathbf{No} \rightarrow \mathbf{No}$.*

Proof. Define $h(g)$ uniquely by $\Phi(\omega^g) = \omega^{h(g)}$. Multiplication of monomials proves additivity, and injectivity proves injectivity of h . A field embedding between real closed ordered fields preserves positivity: a positive element is a nonzero square. Hence it preserves order. For $g > 0$, the monomial ω^g exceeds every positive real number. Its image has the same property because the reals are fixed. A Conway monomial has this property exactly when its exponent is positive, so $h(g) > 0$. Thus h is ordered. Applying strong additivity to each normal form gives (4.1). Uniqueness is immediate from the monomials. $\square$

[write] [Theorem 4.1](#) and [propositions 4.2](#) and [4.4](#) are the case $\rho = \text{id}$ of the bottom-gap classification `opa:as:thm:classification` of the omnific-preserving report, written after the pin. Without the real-fixing hypothesis the exact-monomial embeddings preserving $\mathbf{Oz}$ are the maps $J_{h,\rho}$, which may move reals ([section 1.6](#)).

Warning 4.5 (Preserving monomials is not preserving the omega-map). The equality $\Phi_h(\omega^g) = \omega^{h(g)}$ does not say $\Phi_h(\omega^g) = \omega^{\Phi_h(g)}$. In general $h \neq \Phi_h$ as maps on surreal numbers. Nor have we shown compatibility with the Gonshor exponential, the Berarducci–Mantova derivation, birthdays, or simplicity. These extra structures are outside the assertions of this article.

5 Coset slices and linear disjointness

The next argument upgrades an exponent-group intersection calculation to linear disjointness of *full* series fields. The subgroups need not be convex, and no quotient ordering is needed.

Definition 5.1 (Normalized coset slice). Let $H \subseteq G$ be additive subgroups and $g \in G$. For $f = \sum_e a_e \omega^e \in \mathcal{K}_k(G)$ set

$$\lambda_{g,H}(f) = \omega^{-g} \sum_{e \in (g+H) \cap \text{supp}(f)} a_e \omega^e \in \mathcal{K}_k(H). \quad (5.1)$$

The value depends on the chosen representative g by the expected monomial normalization. We do not choose a global transversal of G/H .

Lemma 5.2 (Slice linearity and separation). *Every $\lambda_{g,H}$ is $\mathcal{K}_k(H)$ -linear, and the family of slices separates points of $\mathcal{K}_k(G)$.*

Proof. A subset of a reverse-well-ordered support is reverse well ordered, and translation preserves that property. Thus the definition is valid even when H is not convex. If $a \in \mathcal{K}_k(H)$, a product term with exponent $e + h$ belongs to $g + H$ exactly when e belongs to $g + H$. Therefore slicing the product af gives a times the slice of f . The equality is verified coefficientwise in finite convolution sums. Additivity follows directly. If $f \neq 0$, choose any g in its support. Its slice contains the nonzero coefficient a_g at exponent zero, so the slice is nonzero. $\square$

Lemma 5.3 (A finite separating matrix). *If $f_1, \dots, f_m \in \mathcal{K}_k(G)$ are linearly independent over $L = \mathcal{K}_k(H)$, there are $g_1, \dots, g_m \in G$ such that*

$$A = (\lambda_{g_a,H}(f_b))_{1 \leq a,b \leq m} \quad \text{is invertible over } L.$$

Proof. On the m -dimensional space $V = \text{span}_L(f_1, \dots, f_m)$, the restrictions of the slices have common kernel zero by lemma 5.2. Start with V . If the current common kernel is nonzero, choose a nonzero vector in it and a slice not vanishing on that vector. Intersecting with this new kernel strictly lowers dimension. After at most m steps the common kernel is zero. The resulting linear map from V to at most m copies of L is injective, so exactly m independent coordinate functionals can be used, and their matrix in the stated basis is invertible. This is a finite argument, also for class-sized L . $\square$

Theorem 5.4 (Full Hahn fields are disjoint over their full intersection). *Let G_1, G_2 be additive subgroups of one ordered abelian group and put $H = G_1 \cap G_2$. Inside $\mathcal{K}_k(G_1 + G_2)$,*

$$\mathcal{K}_k(G_1) \cap \mathcal{K}_k(G_2) = \mathcal{K}_k(H),$$

and the two fields are linearly disjoint over $\mathcal{K}_k(H)$. The assertion applies to set or class subgroups, with set-sized individual supports, and to any coefficient field k .

Proof. The intersection identity follows from uniqueness of coefficients and support. Put $L = \mathcal{K}_k(H)$ and let $f_1, \dots, f_m \in \mathcal{K}_k(G_1)$ be L -linearly independent. Choose the invertible slice matrix of lemma 5.3 with representatives $g_a \in G_1$.

On the larger field $\mathcal{K}_k(G_1 + G_2)$ take slices λ_{g_a,G_2} with values in $\mathcal{K}_k(G_2)$. They are $\mathcal{K}_k(G_2)$ -linear. For $f \in \mathcal{K}_k(G_1)$, the identity

$$G_1 \cap (g_a + G_2) = g_a + (G_1 \cap G_2) = g_a + H$$

shows that $\lambda_{g_a,G_2}(f) = \lambda_{g_a,H}(f)$. Suppose $\sum_b b_b f_b = 0$ with $b_b \in \mathcal{K}_k(G_2)$. Applying all of these enlarged slices gives $A(b_1, \dots, b_m)^T = 0$. Since A is invertible already over L , all b_b vanish. This is precisely linear disjointness. $\square$

Corollary 5.5 (Independence of any finite number of full Hahn fields). *If $(G_i)_{i \in I}$ are independent over H , then the full fields $\mathcal{K}_k(G_i)$ are jointly linearly disjoint over $\mathcal{K}_k(H)$. For $i \notin J$, the field $\mathcal{K}_k(G_i)$ is linearly disjoint over $\mathcal{K}_k(H)$ even from the full field $\mathcal{K}_k(H + \sum_{j \in J} G_j)$.*

Proof. The second assertion follows from [theorem 5.4](#) and the corresponding group intersection. For a finite family, induct on its size. The compositum of the preceding fields is contained in the full Hahn field on the sum of their groups, and the next field is linearly disjoint from that larger field. Extension of scalars over a field preserves an injection of vector spaces. Applying this to the already injective finite-tensor multiplication map proves injectivity for the enlarged family. Equivalently, choose a basis of the finite span of the preceding products and apply linear disjointness to that basis. $\square$

Corollary 5.6 (Polynomial-relation descent). *Let $P = \mathcal{K}_k(G_1)$, $Q = \mathcal{K}_k(G_2)$, and $L = \mathcal{K}_k(G_1 \cap G_2)$. For a finite tuple a from P , let $I_L(a) \subseteq L[X_1, \dots, X_n]$ be its polynomial relation ideal. Then*

$$I_Q(a) = I_L(a) Q[X_1, \dots, X_n].$$

Proof. Take a polynomial relation $F(a) = 0$ with coefficients in Q . A finite L -basis $b_1, \dots, b_r$ for the span of its coefficients gives $F = \sum_j b_j F_j$ with $F_j \in L[X_1, \dots, X_n]$. Linear disjointness is symmetric (as is the finite-tensor criterion), so the b_j remain independent over P . It follows that every $F_j(a) = 0$. Thus F belongs to the extended ideal. The other inclusion is immediate. $\square$

6 Prescribed intersections and Boolean meet diagrams

Proof of Theorem A. Apply [theorem 3.4](#) to obtain (h_α) and their independent ranges G_α , and set $\Phi_\alpha = \Phi_{h_\alpha}$. By [theorem 4.1](#), the image is $K_\alpha = \mathcal{K}_{\mathbb{R}}(G_\alpha)$. Since h_α is the identity on H , it fixes every monomial with exponent in H , and strong additivity makes Φ_α the identity on all of $L_{\mathbb{R}}$. Each G_α is proper by [corollary 3.5](#); a monomial with exponent outside it witnesses properness of the corresponding field.

The intersection and independence assertions follow from [theorem 5.4](#) and [corollary 5.5](#). [Proposition 4.2](#) gives the omnific assertions. The same groups and the same slice argument with coefficients in $\mathbb{C}$ give the entire complex package. $\square$

Corollary 6.1 (Fixing every element of a prescribed set). *For every set $A \subseteq \mathbf{No}$ there is a set-sized real closed full Hahn field $L \supseteq \mathbb{R} \cup A$ which is the exact pairwise intersection of an ordinal-indexed family of independent strong omnific-preserving copies of $\mathbf{No}$ fixing L pointwise. For a set $A \subseteq \mathbf{No}(i)$, the analogous statement holds with an algebraically closed $L \supseteq \mathbb{C} \cup A$.*

Proof. Take H to be the $\mathbb{Q}$ -span of the union of the normal-form supports of all elements of A . In the complex case take the real and imaginary supports, equivalently the complex support. Replacement and union make this a set, and rational span remains a set. Then $A \subseteq \mathcal{K}_k(H)$. Apply [Theorem A](#). $\square$

This corollary does *not* assert that the intersection can be prescribed to be the ordinary field generated by A , its real closure, or an arbitrary non-Hahn subfield. It prescribes the full support closure just constructed. The distinction is essential to both the proof and the classification below. [write] For two maps, and without intersection statements, proper strong exact-monomial copies fixing $A \cup \mathbb{R}$ and preserving and reflecting $\mathbf{Oz}$ are `opa:as:thm:parameters` of the omnific-preserving

report; the corollary extends that result to an **On**-indexed family with an exact common field (section 1.6). [merge] The same support-closure step localizes a set of parameters in two later results. In the dilation report, `adr:sr:thm:tail` places a set of surcomplex parameters in the Hahn field of the real span of their supports and finds a tail of an explicit ordinal family whose positive real dilation orbits are algebraically independent over that whole field, and `adr:sr:thm:kappa` gives set-sized families of independent orbits over any set-sized base. In this report, lemma 23.1 applies the step one normal-form level deeper, to the outer coefficients of a parameter set, and theorem 23.3 keeps a tail of a fixed ordinal family differentially independent over the compositum with any set of parameters adjoined.

Theorem 6.2 (Exact classification of set-sized intersections in this class). *The set-sized fields that occur as intersections of two images of strong, real-fixing, monomial-preserving self-embeddings of **No** are exactly the fields $\mathcal{K}_{\mathbb{R}}(H)$ with $H \subseteq \mathbf{No}$ a set-sized divisible additive subgroup.*

Proof. For necessity, proposition 4.4 writes the two images as full Hahn fields on divisible groups G_1, G_2 . Their intersection is $\mathcal{K}_{\mathbb{R}}(G_1 \cap G_2)$. In a torsion-free divisible ambient group, each divisible subgroup is closed under the uniquely defined rational division; therefore their intersection H is divisible. If the intersection field is a set, its monomials form a set and their exponents, recovered by the inverse of the omega-map on monomials, form a set. Thus H is set-sized. Sufficiency is Theorem A. $\square$

[write] The real-fixing hypothesis is used through proposition 4.4. Exact-monomial embeddings that preserve **Oz** but move reals exist whenever the bottom gap is nonzero (`opa:as:thm:taylor criterion`); their images meet $\mathbb{R}$ in a proper real closed subfield (`opa:as:prop:overlap`). The theorem says nothing about intersections in that larger class.

6.1 All intersections in an entire diagram

Fix a set of indices I from the family of Theorem A. For $J \subseteq I$ put

$$G_J = H + \sum_{j \in J} G_j, \quad E_J = \mathcal{K}_k(G_J), \quad G_{\emptyset} = H. \quad (6.1)$$

Every sum in G_J is finite; no infinite additive closure is intended.

Theorem 6.3 (Boolean meet realization). *The assignment $J \mapsto E_J$ is injective and order preserving. For $J, T \subseteq I$,*

$$E_J \cap E_T = E_{J \cap T}.$$

The analogous identity holds for every nonempty set-indexed intersection of subsets of I . Moreover E_J and E_T are linearly disjoint over $E_{J \cap T}$. Their full Hahn join is $E_{J \cup T}$. All E_J are real closed for $k = \mathbb{R}$ and algebraically closed for $k = \mathbb{C}$.

Proof. Independence over H means that an element of G_J/H has a unique finite set of nonzero components in the direct sum of the groups G_i/H . It lies in G_J/H exactly when all of these components have indices in J . This proves all the group intersection identities and $G_J + G_T = G_{J \cup T}$. Uniqueness of normal form transfers the intersection identities to full Hahn fields, and theorem 5.4 proves the asserted linear disjointness.

If $j \in J \setminus T$, choose $g \in G_j \setminus H$; then $\omega^g \in E_J \setminus E_T$. This proves injectivity and reflection of containment. The full Hahn join is, by definition, the full Hahn field on the sum of the two exponent groups, hence is $E_{J \cup T}$. Every G_J is divisible, which gives the closedness assertions. $\square$

Remark 6.4 (The word “join”). The theorem realizes a Boolean meet diagram with *Hahn* joins. It does not identify ordinary field composita with those joins. [Theorem C](#) shows that the difference can already be transcendental for two factors.

Corollary 6.5 (Independent choices and absence of new relations). *For distinct indices $i_1, \dots, i_m$, choose $a_j \in K_{i_j} \setminus L_{\mathbb{R}}$. Then $(a_1, \dots, a_m)$ is algebraically independent over $L_{\mathbb{R}}$. The same assertion holds over $L_{\mathbb{C}}$ for the complexified copies. More generally all polynomial relations across any square of [theorem 6.3](#) descend to its intersection field.*

Proof. A real closed subfield has no proper algebraic extension inside an ordered field, so each a_j is transcendental over $L_{\mathbb{R}}$. In the complex case use algebraic closedness. The polynomial algebras $L[a_j]$ therefore have no relations individually, and their finite tensor product is a polynomial algebra on m variables. Joint linear disjointness makes its multiplication map injective. The last statement is [corollary 5.6](#). $\square$

6.2 A limited model-theoretic conclusion

For each fixed first-order formula in the language of ordered fields, the inclusion of any E_J into **No** is elementary, by quantifier elimination for real closed fields. The corresponding complex statement follows from quantifier elimination for algebraically closed fields of characteristic zero [4]. This is a formula-by-formula use of ordinary field theory, not the construction of a class satisfaction predicate.

Nothing here asserts elementary embedding in the expanded language with a predicate for **Oz**, a symbol for the omega-map, or a derivation. The independence proved is algebraic independence and linear disjointness; it is not being silently identified with a model-theoretic independence notion in any richer theory.

7 Explicit copies with intersection exactly the constants

The general construction can fix an arbitrary set-sized Hahn core, but uses class bookkeeping. When $H = 0$ there is a direct formula. It also furnishes the separated scales needed for [Theorem C](#).

Lemma 7.1 (Compression of the surreal order). *The function*

$$\psi(x) = \frac{1}{2} + \frac{x}{2(1 + |x|)}$$

*is an order isomorphism from **No** onto the open interval $(0, 1)$.*

Proof. On the nonnegative half-line, $x/(1 + x)$ is strictly increasing and belongs to $[0, 1)$. On the nonpositive half-line, $x/(1 - x)$ is strictly increasing and belongs to $(-1, 0]$. The formulas agree at zero, proving strict monotonicity and the stated range containment. For $u \in (0, 1)$ put $s = 2u - 1$ and $x = s/(1 - |s|)$. Direct substitution on the two signs of s gives $\psi(x) = u$. All denominators are positive in the ordered field. $\square$

For every ordinal α let

$$I_\alpha = (3\alpha, 3\alpha + 1), \quad \sigma_\alpha(a) = 3\alpha + \psi(a),$$

where the arithmetic in 3α is surreal field arithmetic. These intervals are pairwise disjoint. Indeed $\alpha < \beta$ for ordinals implies $\alpha + 1 \leq \beta$ in the surreal order, hence $3\alpha + 1 < 3\beta$.

Define an *inner* map on normal forms by

$$h_\alpha \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{\sigma_\alpha(a)}. \quad (7.1)$$

This map is an ordered, real-linear, strongly additive *group* embedding. Additivity only uses the injective relabeling of coefficients; no additivity of σ_α is needed. It is not a unital field map:

$$h_\alpha(1) = \omega^{3\alpha+1/2}. \quad (7.2)$$

Its image is precisely the additive class of all surreal normal forms supported in I_α . Pulling back the support along the order isomorphism σ_α proves fullness of that image.

Now apply the *outer* construction of [theorem 4.1](#):

$$\Phi_\alpha \left(\sum_g c_g \omega^g \right) = \sum_g c_g \omega^{h_\alpha(g)}. \quad (7.3)$$

The distinction between (7.1) and (7.3) is crucial: the outer exponents g are themselves surreal numbers with their own normal forms. [write] The inner map is the inner lift ι_{σ_α} of the omnific-preserving report (`opa:as:eq:innerlift`, `opa:as:lem:inner`), and (7.3) is its monomial map $M_{1,\iota_{\sigma_\alpha}}$. That report's order isomorphisms $\mu_{11}, \mu_{12}, \mu_{13}$ have unbounded images; the bounded, pairwise disjoint images of the σ_α are what makes the ranges independent here.

Theorem 7.2 (An explicit independent ordinal family). *The maps (7.3) are strong, real-fixing, monomial-preserving proper field embeddings. Their image fields are jointly linearly disjoint over $\mathbb{R}$, have pairwise intersection $\mathbb{R}$, and their omnific image rings have pairwise intersection $\mathbb{Z}$. Their complexifications have the corresponding intersections $\mathbb{C}$ and $\mathbb{Z}[i]$.*

Proof. The inner ranges G_α have disjoint normal-form support intervals. In a finite sum of elements from different ranges, no nonzero coefficient from one interval can be canceled by another interval. Therefore these ranges are independent over zero. Apply [theorem 4.1](#), [proposition 4.2](#), and [corollary 5.5](#). $\square$

For example,

$$\Phi_0(\omega) = \omega^{\omega^{1/2}}, \quad \Phi_1(\omega) = \omega^{\omega^{7/2}}.$$

Both maps fix every real number. Since $\Phi_0(1) = 1$, the first identity also explicitly shows

$$\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}.$$

Thus preservation of the set of Conway monomials is much weaker than commutation with the omega-map as a unary operation on all of **No**.

Remark 7.3 (A tempting formula that does not give a proper copy). Replacing every exponent g by cg with a fixed $c > 0$ gives an automorphism, not a proper self-embedding, because $c\mathbf{No} = \mathbf{No}$. The compression above is applied one normal-form level below the field lift; that is what makes the exponent-group image genuinely proper.

8 A transcendental gap above an ordinary compositum

Take G_0, G_1 from [section 7](#). These groups are not merely independent. Their scales are separated:

$$|a| < b/n \quad (a \in G_0, 0 < b \in G_1, n \in \mathbb{N}_{>0}). \quad (8.1)$$

Indeed the inner leading exponent of a nonzero a lies in $(0, 1)$, whereas that of b lies in $(3, 4)$. Leading-monomial comparison proves (8.1). Thus G_0 is convex in the ordered group $G_0 + G_1$, and the sign of $a + b$ for $b \neq 0$ is the sign of b .

8.1 Iterated Hahn coordinates

Lemma 8.1 (Re-expansion at separated scales). *Suppose A, B are additive subgroups with $A \cap B = 0$ and every element of A is infinitesimal relative to every positive element of B , in the sense of (8.1). Then regrouping by B -coordinate gives a field isomorphism*

$$\mathcal{K}_k(A + B) \simeq \mathcal{K}_{\mathcal{K}_k(A)}(B), \quad \sum_{a+b} c_{a,b} \omega^{a+b} \mapsto \sum_b \left(\sum_a c_{a,b} \omega^a \right) \omega^b. \quad (8.2)$$

On the right the notation means iterated Hahn series; the coefficient field may be a class, but each individual series uses only a set of coefficients. The isomorphism is order preserving for ordered k .

Proof. The decomposition $a + b$ is unique, and the order on $A + B$ is lexicographic with B the dominant coordinate. The image in B of a reverse-well-ordered support is reverse well ordered: the inverse image of a nonempty subset of that image has a greatest element, whose projection is greatest in the subset. Each fiber is reverse well ordered in A .

Conversely, a reverse-well-ordered set of B -coordinates, with a reverse-well-ordered support in each fiber, yields a reverse-well-ordered lexicographic union. Its size is a set by replacement and union. The two regroupings are inverse. Addition and multiplication agree coefficientwise: there are finitely many pairs of B -coordinates contributing to each outer product coefficient, and within each resulting coefficient Hahn multiplication in A applies. The dominant B -coordinate and then the leading coefficient in $\mathcal{K}_k(A)$ also give exactly the original order. $\square$

For $k = \mathbb{R}$ or $\mathbb{C}$, put

$$K_0 = \mathcal{K}_k(G_0), \quad K_1 = \mathcal{K}_k(G_1), \quad E = \mathcal{K}_k(G_0 + G_1), \quad M = K_0 K_1.$$

In the iterated coordinates, K_0 is the coefficient field and K_1 has coefficients in the original k .

Lemma 8.2 (Finite coefficient-field control). *Every finite subset of the ordinary compositum M is contained, in the iterated coordinates, in a field $\mathcal{K}_D(G_1)$ where $D \subseteq K_0$ is a field finitely generated over k .*

Proof. Each of the finitely many chosen elements is obtained by finitely many field operations from finitely many elements of K_0 and K_1 . Let $a_1, \dots, a_r \in K_0$ include all the first-kind elements used, and put $D = k(a_1, \dots, a_r)$. The elements of K_1 have coefficients in $k \subseteq D$, while each a_j is a constant in the outer G_1 -series coordinate. The field $\mathcal{K}_D(G_1)$ is closed under the required field operations, proving the assertion. $\square$

This lemma does not claim that every series with coefficients in such a D is in M . It gives a necessary coefficient-field constraint, which is all that is needed. [merge] The constraint is far from sufficient: a series whose coefficients all lie in a field $\mathbb{R}(x)$ generated by one element can be differentially transcendental over M (corollary 19.5). The exact description of M in these coordinates is the finite coefficient-rank denominator test of corollary 22.1.

Theorem 8.3 (A coefficient-transcendence obstruction). *Let $s_1, s_2, \dots \in K_0$ be algebraically independent over k , where $k = \mathbb{R}$ or $\mathbb{C}$, and choose $0 < g \in G_1$. The well-defined series*

$$S = \sum_{n=1}^{\infty} s_n \omega^{-ng} \in E$$

is transcendental over M .

Proof. The outer support $\{-ng : n \geq 1\}$ is strictly decreasing, so the series is valid by lemma 8.1. Suppose a nonzero polynomial over M annihilates S . By lemma 8.2, all its coefficients lie in $\mathcal{K}_D(G_1)$ for one finitely generated field $D \subseteq K_0$ over k .

For $k = \mathbb{R}$, let D' be the relative algebraic closure of D in the real closed field K_0 . It is a set-sized real closed field. To see closedness, a square root of a positive element of D' and a root of an odd-degree polynomial over D' can be taken in K_0 ; each is algebraic over D , so belongs to D' . It is a set because a set of polynomials has only a set of finite root sets. For $k = \mathbb{C}$, the same argument with arbitrary polynomial roots gives a set-sized algebraically closed relative closure $D' \subseteq K_0$.

Since G_1 is divisible, the iterated Hahn field $\mathcal{K}_{D'}(G_1)$ is real closed in the real case, and algebraically closed in the complex case. For a proper-class G_1 this follows, as before, by working with the set-sized divisible subgroup generated by any finite collection of coefficient supports. Thus an element of E algebraic over $\mathcal{K}_D(G_1)$ must belong to $\mathcal{K}_{D'}(G_1)$: use relative algebraic closedness in an ordered field in the real case, or algebraic closedness in the complex case.

Applied to S , uniqueness of its outer expansion now gives $s_n \in D'$ for every n . But

$$\text{trdeg}_k D' = \text{trdeg}_k D < \infty,$$

whereas the s_n are an infinite algebraically independent family over k . This contradiction proves the theorem. $\square$

Explicit witness and proof of Theorem C. Take

$$g_n = h_0(\omega^n) = \omega^{\psi(n)}, \quad g = h_1(1) = \omega^{7/2}, \quad s_n = \omega^{g_n} \quad (n \geq 1).$$

The g_n are $\mathbb{Q}$ -linearly independent: they are distinct inner Conway monomials, so uniqueness of normal form precludes a finite rational linear relation. Consequently the s_n are algebraically independent over $\mathbb{R}$, and also over $\mathbb{C}$ in the complexified field. Indeed the distinct monomials of a polynomial in the s_n acquire distinct exponents $\sum_n m_n g_n$, and no nonzero coefficient can cancel.

The promised actual normal form is

$$S = \sum_{n=1}^{\infty} \omega^{\omega^{\psi(n)} - n\omega^{7/2}}. \tag{8.3}$$

Its exponents strictly decrease: the difference between consecutive exponents is $g + g_n - g_{n+1} > 0$ by scale separation. Thus (8.3) is a legitimate countably supported surreal, not a formal expression

awaiting an analytic interpretation. It belongs to E and is transcendental over the appropriate ordinary compositum by [theorem 8.3](#). Transcendence also excludes membership in its relative algebraic closure. $\square$

[merge] The second manuscript extends this witness. With $b = \omega^{15/4} \in G_1$, the element $\omega^b S$ is the first member Y_0 of an ordinal family Y_α of purely infinite omnific integers of order type ω ([theorem 22.2](#)); all their Euler jets are algebraically independent over M , and a tail of the family stays independent after any set of parameters is adjoined ([theorem 23.3](#)). In the real case S itself is then differentially transcendental over M for that Euler derivation.

Remark 8.4 (Why the infinite list of coefficients matters). A rational expression in the two fields can use arbitrary elements of each field, but only finitely many of them. In the separated-scale expansion this bounds the transcendence degree of its coefficient field over the constants. Algebraic extension does not remove that finite bound. The series (8.3) defeats it by displaying infinitely many independent coefficients. This argument is different from simply observing that a series is an infinite sum; many infinite sums, such as a geometric series, already belong to an ordinary compositum.

9 Omnific intersections and their fraction fields

For $k = \mathbb{R}, \mathbb{C}$ write $D_{\mathbb{R}} = \mathbb{Z}$ and $D_{\mathbb{C}} = \mathbb{Z}[i]$, and define

$$\Pi_k(H) = \{x \in \mathcal{K}_k(H) : \text{supp}(x) \subseteq H^{>0}\}, \quad \mathcal{A}_k(H) = D_k \oplus \Pi_k(H).$$

The direct sum is additive; multiplication makes $\Pi_k(H)$ a nonunital ring and an ideal in $\mathcal{A}_k(H)$. In particular,

$$\mathcal{A}_{\mathbb{R}}(H) = \mathcal{K}_{\mathbb{R}}(H) \cap \mathbf{Oz}, \quad \mathcal{A}_{\mathbb{C}}(H) = \mathcal{K}_{\mathbb{C}}(H) \cap \mathbf{Oz}[i].$$

We use the ordinary class fraction field of a class domain to mean the subfield of the ambient field consisting of ratios of two of its elements, with nonzero denominator. No quotient set of a proper class is invoked.

Lemma 9.1 (Common denominators for a set of surreals). *Every set $X \subseteq \mathbf{No}$ has a positive monomial $d \in \mathbf{Oz}$ such that $dX \subseteq \mathbf{Oz}$. The same holds for a set of surcomplex numbers with $\mathbf{Oz}[i]$ in place of $\mathbf{Oz}$, and a positive real monomial denominator. In particular*

$$\text{Frac}(\mathbf{Oz}) = \mathbf{No}, \quad \text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}(i).$$

Proof. The union S of the supports of elements of X is a set. Choose a surreal $a > 0$ with $a > -s$ for every $s \in S$, using the cut property. Then $d = \omega^a$ and every product dx have exclusively positive exponents, except that dx may be zero. Thus they lie in the required integer ring. In the complex case use the union of the real and imaginary supports. Applying the assertion to a singleton gives the fraction-field identities. These identities are classical; compare [3, Proposition 2.4.5]. $\square$

[write] In the collection this is the global monomial clearing `odg:thm:fractions` of the Diophantine report, with the same proof.

Transporting the lemma through an embedding gives more than merely one denominator for each individual element.

Proposition 9.2 (Separate uniform denominators, no shared uniform one). *Let $K_\alpha, R_\alpha, L_\mathbb{R}$ be as in Theorem A. Then $\text{Frac}(R_\alpha) = K_\alpha$, and there is a nonzero $d_\alpha \in R_\alpha$ with*

$$d_\alpha L_\mathbb{R} \subseteq R_\alpha.$$

For two distinct indices, no nonzero element of $R_\alpha \cap R_\beta$ can satisfy both such containments. The same conclusions hold in the complex and Gaussian setting.

Proof. The image under Φ_α of a ratio of omnific integers is a ratio of elements of R_α , and every element of K_α arises this way by lemma 9.1. Since $L_\mathbb{R}$ is a set and is fixed pointwise, apply that lemma to $L_\mathbb{R}$ and transport its denominator by Φ_α . This yields the uniform denominator d_α .

A hypothetical common nonzero denominator d belongs to $R_\alpha \cap R_\beta \subseteq L_\mathbb{R}$. If both containments held, then $dL_\mathbb{R} \subseteq \mathcal{A}_\mathbb{R}(H)$. But $d \neq 0$ is in the field $L_\mathbb{R}$, so $dL_\mathbb{R} = L_\mathbb{R}$, which is not contained in its omnific part (it contains $1/2$). This is impossible. The same proof uses $1/2 \notin \mathbb{Z}[i]$ in the complex case. $\square$

[write] The proof shows slightly more than the statement: since $dL_\mathbb{R} = L_\mathbb{R} \not\subseteq \mathbf{Oz}$, a nonzero element of $R_\alpha \cap R_\beta$ satisfies *neither* containment, not merely not both. The separate denominators d_α lie outside $L_\mathbb{R}$.

This proposition already separates two notions of a denominator. To prove Theorem B, however, we need an element with *no shared denominator at all*, not merely the absence of a shared denominator uniform on the entire common field. That stronger assertion is proved next.

9.1 The bounded-support field is exactly the shared fraction field

For this subsection and section 10, use the increasing-support Hahn convention

$$t^h = \omega^{-h}.$$

As $H = -H$, this changes the presentation of $\mathcal{K}_k(H)$, not its underlying field. The support of a nonzero t -series is well ordered and has a least element, its valuation v . Set $v(0) = +\infty$. Let

$$\mathcal{B}_k(H) = \{f \in k((t^H)) : \text{supp}_t(f) \text{ is bounded above in } H\}.$$

It is a subring: finite unions of upper bounds bound sums, and the sum of two upper bounds bounds a product. Its elements can have infinite support inside a bounded interval. In this convention $\mathcal{A}_k(H)$ consists of series with negative exponents and a constant coefficient in D_k .

Proposition 9.3 (Identification of the fraction fields). *If $H \neq 0$, then*

$$\text{Frac } \mathcal{A}_k(H) = \text{Frac } \mathcal{B}_k(H). \tag{9.1}$$

In particular this fraction field contains k and every single monomial t^h with $h \in H$.

Proof. One has $\mathcal{A}_k(H) \subseteq \mathcal{B}_k(H)$. Conversely, take $f \in \mathcal{B}_k(H)$ and choose an upper bound $u \in H$ of its support. Choose $a \in H$ with $a > u$ and $a > 0$, possible in any nonzero ordered abelian group. Both t^{-a} and $t^{-a}f$ have strictly negative support, so belong to $\mathcal{A}_k(H)$. Their ratio is f , giving the reverse inclusion of fraction fields. Constants and monomials have bounded support. $\square$

The identity explains why a mere missing constant is not the obstruction for nonzero H . For example, $r = (r\omega^h)/\omega^h$ for any $r \in \mathbb{R}$ and any positive $h \in H$. The unresolved issue is the unboundedness of supports *within the fixed set group* H . [write] [Proposition 9.3](#) is the localization of `bst:prop:localdensity` of the bounded-support report, with the constant coefficient restricted to D_k . Since batch 32 that report also prints the general form, for every field K and every unital subring $D \subseteq K$, as `bst:gr:prop:fractions`.

10 A self-contained cofinal gap argument

The mechanism in this section belongs to bounded-support Hahn arithmetic, not to the new embedding construction. It is included in full so that the transcendental conclusion in [Theorem B](#) does not depend on an unreviewed repository result. The repository's bounded-support report gives stronger independent-family assertions [8], and the non-fraction obstruction is already in [3]. [write] In that report, [theorem 10.2](#) is `bst:cor:onegap` at countable cofinality and, in characteristic zero, the one-series case of `bst:thm:independent`; the single-series form at uncountable cofinality over an arbitrary coefficient field is not stated there, but its proof is that report's tail estimate and root count ([section 1.6](#)). No priority is claimed for it.

Lemma 10.1 (Cofinal sequences with polynomial gaps). *Let $H \neq 0$ be a set-sized ordered abelian group and put $\kappa = \text{cf}(H)$. There is a strictly increasing cofinal family $(a_\alpha)_{\alpha < \kappa}$ in $H^{>0}$ such that for every positive integer d and every $c \in H$,*

$$a_{\alpha+1} - da_\alpha > c \quad \text{for all sufficiently large } \alpha < \kappa. \quad (10.1)$$

Proof. The cofinality of a nonzero ordered group is an infinite regular cardinal: a finite subset is bounded, and a cofinal union of fewer than κ subsets of size less than κ would contradict minimality of κ by taking bounds for those subsets. Fix a cofinal list $(u_\alpha)_{\alpha < \kappa}$.

When $\kappa = \aleph_0$, choose recursively $a_0 > 0, u_0$ and

$$a_{n+1} > u_{n+1}, \quad a_{n+1} > (n+2)a_n.$$

The list is cofinal. For fixed d , once $n+2 \geq d+1$ the difference in (10.1) exceeds a_n , and this eventually exceeds c .

When $\kappa > \aleph_0$, at stage α the set

$$\{0, u_\alpha\} \cup \{na_\beta : \beta < \alpha, n \in \mathbb{N}_{>0}\}$$

has cardinality less than κ and hence is bounded above in H . Choose a_α strictly above it. The resulting family is increasing and cofinal. At a successor stage, $a_{\alpha+1} > (d+1)a_\alpha$ for every positive integer d . Thus again the difference in (10.1) exceeds a_α and is eventually larger than c . Since an infinite cardinal has no last ordinal below it, all the successor indices used are still below κ . $\square$

Theorem 10.2 (One transcendental series over all bounded-support fractions). *For any field k and nonzero set-sized ordered abelian group H , choose a sequence from [lemma 10.1](#). Then*

$$F = 1 + \sum_{\alpha < \kappa} t^{a_\alpha} \quad (10.2)$$

is transcendental over $\text{Frac } \mathcal{B}_k(H)$.

Proof. The support is a well-ordered set, so F is a Hahn series and $v(F) = 0$. Suppose a nonzero polynomial over $\text{Frac } \mathcal{B}_k(H)$ annihilates F . Clearing its finitely many denominators gives

$$Q(Y) = \sum_{j=0}^d b_j Y^j \in \mathcal{B}_k(H)[Y] \setminus \{0\}, \quad Q(F) = 0,$$

where $d \geq 1$. Each nonzero coefficient has support bounded both below and above: below by its least exponent and above by the definition of $\mathcal{B}_k(H)$. Choose common bounds $b, c \in H$ for the supports of all nonzero coefficients, with $b \leq c$.

For $\alpha < \kappa$ let

$$P_\alpha = 1 + \sum_{\beta \leq \alpha} t^{a_\beta}.$$

These are distinct elements of the same Hahn field, including at limit indices; they need not be polynomials in t . Their supports lie in $[0, a_\alpha]$, and

$$v(P_\alpha) = 0, \quad v(F - P_\alpha) = a_{\alpha+1}.$$

Finite Hahn multiplication gives

$$\text{supp}_t Q(P_\alpha) \subseteq [b, c + da_\alpha]. \quad (10.3)$$

The elementary factorization

$$F^j - P_\alpha^j = (F - P_\alpha) \sum_{\ell=0}^{j-1} F^{j-1-\ell} P_\alpha^\ell$$

and the nonnegative valuations of the factors in the finite sum give

$$v(Q(F) - Q(P_\alpha)) \geq b + a_{\alpha+1}. \quad (10.4)$$

By [lemma 10.1](#), eventually

$$b + a_{\alpha+1} > c + da_\alpha.$$

Since $Q(F) = 0$, (10.4) says that every nonzero exponent of $Q(P_\alpha)$ is above the upper bound in (10.3). Therefore $Q(P_\alpha) = 0$ for every sufficiently large α . This gives infinitely many distinct roots of the nonzero finite-degree polynomial Q , a contradiction. The argument uses no characteristic-zero assumption. $\square$

Corollary 10.3 (The factorial witness). *In $\mathbb{R}((t^\mathbb{Q}))$ or $\mathbb{C}((t^\mathbb{Q}))$, the series*

$$F = 1 + \sum_{n=1}^{\infty} t^{n!}$$

is transcendental over the fraction field of all bounded-above-support series, hence over $\text{Frac } \mathcal{A}_k(\mathbb{Q})$.

Proof. The positive integers $n!$ are cofinal in $\mathbb{Q}$ and satisfy $(n+1)! - dn! = (n+1-d)n! \rightarrow +\infty$ for each fixed d . Thus the proof of [theorem 10.2](#) applies to this explicit sequence. For clarity, its partial sums start with $1+t$, $1+t+t^2$, $1+t+t^2+t^6$; there is no duplicate exponent from a $0!$ term. $\square$

[write] The same exponents carry the continuum of independent actual infinitesimals of `bst:thm:actualreal`. The omnific-preserving report's `opa:as:ex:smallfrac` uses $\sum t^{n!}$ at $\Gamma = \mathbb{Z}$ for the weaker statement that it is not a fraction of $\mathcal{R}_\mathbb{Z}$.

Proof of Theorem B. By Theorem A, the common field is $L_k = \mathcal{K}_k(H)$ and the common integer ring is $\mathcal{A}_k(H)$. Each full integer copy has fraction field equal to its field copy by proposition 9.2. If $H \neq 0$, combine proposition 9.3 and theorem 10.2 to obtain an element of L_k transcendental over $\text{Frac } \mathcal{A}_k(H)$. If $H = 0$, the common ring is $\mathbb{Z}$ or $\mathbb{Z}[i]$ and its fraction field is $\mathbb{Q}$ or $\mathbb{Q}(i)$. The ambient constant field $\mathbb{R}$ or $\mathbb{C}$ contains a transcendental element over that countable field: its algebraic elements are a countable union of finite root sets, whereas the constant field is uncountable. Finally corollary 10.3 supplies the specified witness for $H = \mathbb{Q}$. This proves every assertion. $\square$

Corollary 10.4 (Failure witnessed by the denominator equation). *For $H = \mathbb{Q}$ and F from corollary 10.3, there are nonzero $d_\alpha \in R_\alpha$ and $d_\beta \in R_\beta$ with $d_\alpha F \in R_\alpha$ and $d_\beta F \in R_\beta$, but there are no $a, d \in R_\alpha \cap R_\beta$ with $d \neq 0$ and $dF = a$.*

Proof. The separate denominators follow even uniformly from proposition 9.2. A shared solution would express $F = a/d$ in $\text{Frac } \mathcal{A}_{\mathbb{R}}(\mathbb{Q})$, contradicting transcendence. The same proof works for the Gaussian rings. $\square$

10.1 Where the two transcendence arguments differ

The gap above an ordinary compositum in section 8 is detected by an infinite algebraically independent *coefficient* family in a separated scale decomposition. The gap above the shared fraction field in this section is detected by a cofinal sequence of *support exponents*, even with all nonzero coefficients equal to one. Neither proof is a cardinality-only argument, and neither concludes merely that an element lacks a particular chosen representation: each excludes every finite-degree algebraic relation over the stated base.

11 A set-sized version of the construction

The class construction has a direct conditional set-sized analogue. This both isolates the exact saturation used and gives a more conventional target for formalization.

Definition 11.1. For an uncountable regular cardinal κ , a linear order is an η_κ -order if every cut $L < R$ with $|L \cup R| < \kappa$ has a strict intermediate point. Empty sides are allowed.

Theorem 11.2 (Independent embeddings in a saturated ordered group). *Let κ be uncountable regular, and let V be a divisible ordered abelian group of cardinality κ whose order is an η_κ -order. If $H \subseteq V$ is a divisible subgroup of cardinality less than κ and $2 \leq |I| \leq \kappa$, then there are ordered additive proper embeddings $h_i : V \rightarrow V$ fixing H whose ranges are independent over H .*

Proof. Enumerate the requirements $I \times V$ in κ stages. Before each stage, fewer than κ elements and partial graph entries have been introduced. Their rational span still has size less than κ ; uncountability and regularity ensure this also at limit stages below κ .

The one-point proof works verbatim. To avoid the current span E , form the set T of its elements realizing the current cut and realize the enlarged cut $L \cup T < R$. All these sets have size less than κ , so the η_κ hypothesis provides the new value. Independence is preserved by exactly the same finite-relation argument. At the end all domains are V , and another independent range witnesses properness of each one. $\square$

This theorem is conditional on the existence of the stated group; it does not assume that every cardinal admits such a structure. The uncountability hypothesis is purposeful: in a countable group, adjoining one rational vector already gives a countable domain, so mere realization of finite order cuts would not justify the same proof.

Lifting these embeddings to full set-sized Hahn fields gives the same intersection and disjointness statements. Those Hahn fields need not have cardinality κ ; the theorem places no unwarranted cardinality bound on all their allowable supports.

12 Further research questions

The questions below are proposed by the present theorems. They are not presented as a list of previously published open problems, nor as claims that their answers must be difficult. The first four most directly extend the proved results.

Research question 1 (Intersections outside the full Hahn category). Which set-sized real closed subfields $L \subseteq \mathbf{No}$ containing $\mathbb{R}$ can occur as the exact intersection of two strong, omnific-preserving self-embedding images if the monomial-preserving hypothesis is removed?

Why it is the next step. [Theorem 6.2](#) answers the monomial-preserving case exactly. A non-Hahn example, or an obstruction forcing Hahn closure for a larger class of embeddings, would identify the precise information carried by the chosen monomial cross-section. Strongness alone should not be assumed to settle it. [write] *Status in the collection: open and new.* It is related to `opa:as:q:units`, which asks to classify embeddings without exact monomial preservation, but that question does not concern intersections.

Research question 2 (Relative classification of pairs of copies). Suppose two independent pairs have the same core $\mathcal{K}_{\mathbb{R}}(H)$. Which invariants determine whether an omnific-preserving automorphism of $\mathbf{No}$ fixing that core carries one pair to the other?

Concrete first case. Compare pairs whose exponent groups form different ordered amalgams over H , even when their abstract additive quotients agree. The ambient ordering of mixed sums is an evident invariant for monomial lifts, but it is not proved here to be complete. [write] *Status in the collection: open.* It is the pair version, at a fixed Hahn core, of `opa:as:q:conjugacy`, whose partial answer `opa:as:thm:continuum` concerns single copies that move reals.

Research question 3 (An exact description of ordinary composita). For independent groups G_1, G_2 over H , characterize the subfield $\mathcal{K}_k(G_1)\mathcal{K}_k(G_2)$ inside $\mathcal{K}_k(G_1 + G_2)$. Determine the transcendence degree of the full Hahn join over the ordinary compositum in natural set-sized and class-sized examples.

Available starting point. In the separated case, [lemma 8.2](#) gives a necessary finite coefficient-field condition, and [theorem 8.3](#) supplies a transcendental obstruction even after algebraic closure. A necessary-and-sufficient condition would be much stronger. For a proper-class extension, formulate dimension statements as independent families of specified set cardinalities rather than casually assigning a set cardinal to the whole class. [merge] *Status (batch 34): answered in the separated case, open in general.* For the separated explicit copies of [Theorem C](#) (trivial core, G_0 infinitesimal relative to G_1), the compositum is characterized by a finite coefficient-rank denominator test ([theorem 17.2](#) and [corollary 22.1](#)), and the necessary condition of [lemma 8.2](#) is shown not to be sufficient

(corollary 19.5). In the natural set-sized Laurent examples the algebraic and Euler-differential transcendence degrees of $F((t))$ over $Fk((t))$ are both $|F|^{\aleph_0}$ when $|k| \leq \mathfrak{c}$ and $\text{trdeg}(F/k) > 0$ (theorem 21.1). In the class-sized explicit join, as this question asks, the answer is given by independent families of every set cardinal of countably supported omnific integers (theorem 22.2), a tail independent over any set of adjoined parameters (theorem 23.3), and the absence of set-sized generation (corollary 23.4). Open: groups over a common core H that are not separated, prescribed nonzero common cores, infinite algebraic coefficient extensions, and $|k| > \mathfrak{c}$ with small positive $\text{trdeg}(F/k)$ (Research questions 11, 12, 13 and 14).

Research question 4 (Descent for the arithmetic intersection). How do integral closure, complete integral closure, and finite-type algebraic geometry over $\mathcal{A}_k(H)$ compare with those over $\text{Frac } \mathcal{A}_k(H)$ and $\mathcal{K}_k(H)$? Which descent failures besides the single denominator equation are forced by the transcendental gap?

Important boundary. The elementary fraction-field failure proved here does not by itself calculate a normalization, a Picard group, or a nontrivial projective module. Those require additional arguments and should be compared with the repository's dedicated arithmetic reports.

Research question 5 (Large fixed bases). For which proper-class divisible subgroups $H \subsetneq \mathbf{No}$ can independent ordered-group copies fixing H be constructed? What cut-size or cofinality condition replaces the set-sized hypothesis of theorem 3.4?

Obstacle to the present proof. Its first partial domain and its forbidden range span must be sets. A proper-class H invalidates that step. This is a limitation of the argument, not a proof that every such extension is impossible.

Research question 6 (Compatibility with additional surreal operations). Can versions of the prescribed-intersection theorem be proved for embeddings commuting with the Gonshor exponential or with a fixed surreal derivation? What cores would necessarily be closed under those operations?

First necessary checks. If an operation is preserved by both embeddings, their images and intersection must be closed under that operation. The explicit maps of section 7 already fail to commute with the omega-map. Therefore they cannot simply be relabeled as solutions in richer languages. Published work on exponential surreal embeddings provides a relevant comparison [5]. [write] *Status in the collection: open.* It lies inside `opa:as:q:exp` of the omnific-preserving report, restricted to prescribed intersections.

Research question 7 (Simplicity and initiality). Which prescribed intersections remain possible when each image is required to be initial for the simplicity relation, or when a specified bound on birthdays is imposed?

Reason for separating this question. The saturation proof uses the simplest realizer of an enlarged cut to choose one point, but it does not preserve the global simplicity relation. Local use of a simplest number is not an initial-embedding theorem. [write] *Status in the collection: open.* Simplicity relations are also named in `opa:as:q:exp`.

Research question 8 (Uniform maximal transcendence at a fixed core). For a fixed nonzero set-sized H , obtain an independently checked sharp calculation of $\text{trdeg}(\mathcal{K}_k(H)/\text{Frac } \mathcal{A}_k(H))$ and relate it to $\text{cf}(H)$ and support cardinality. How much of this calculation can be made uniform across all realized intersection diagrams?

Prior work to retain. This is not an untouched direction: the repository’s bounded-support report already proposes a $2^{\text{cf}(H)}$ -sized independent family and optimal support thresholds [8]. A useful next contribution is verification, extension, or a genuinely different uniformity statement, not rediscovery of its one-witness special case. [write] *Status in the collection:* the programme of the bounded-support report (`bst:thm:independent`, `bst:cor:exactcard`, `bst:cor:realgroup`); its uniformity over realized diagrams is open.

Research question 9 (Elementary embeddings of the field–integer pair). Can the embeddings be chosen elementary in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$ while retaining a prescribed set-sized intersection and the stated algebraic independence? What extra restrictions does the integer-ring predicate impose?

What has actually been proved. The field embeddings preserve and reflect membership in $\mathbf{Oz}$ and commute with floor, but these facts do not establish preservation of all formulas with alternating quantifiers. Ordinary real-closed-field elementarity is a weaker assertion. [write] *Status in the collection:* open. For the maps here it is `opa:as:q:elementary`, with a prescribed intersection added; the nonelementarity criterion of `opa:as:thm:elementary` needs a moved real and does not apply.

Research question 10 (A verified set-level kernel). Can the coset-slice lemma, finite separating-matrix argument, and saturated ordered-group extension theorem be formalized as reusable library results, then connected to a universe-aware model of the surreal construction?

Concrete target. The next section gives declarations and obligations at the level of mathematical specifications. A verified finite support computation is not a substitute for these support and class-size proofs. [write] *Status in the collection:* open. Its Hahn-field kernel overlaps `opa:as:q:formal`; none of the proposed identifiers exists in the collection’s Lean library.

13 Proof dependencies, verification boundary, and conclusion

13.1 A dependency map

The proof of the main copy theorem follows the chain

$$\begin{aligned} & \text{set-cut property} \implies \text{one-point avoidance} \\ & \implies \text{independent group ranges} \implies \text{strong monomial lifts.} \end{aligned}$$

The upgrade from support intersections to algebraic independence is a separate chain:

$$\begin{aligned} & \text{coset slices} \implies \text{finite separating matrix} \\ & \implies \text{linear disjointness} \implies \text{polynomial-relation descent.} \end{aligned}$$

The arithmetic application adds the common-denominator lemma and the cofinal gap argument. The compositum application instead adds the explicit separated-scale construction, iterated Hahn coordinates, and finite coefficient-field control. Neither application is used to justify the other, and neither is needed to justify the group construction.

13.2 A practical formalization decomposition

A set-level implementation could first express full Hahn series in the usual increasing-support convention and prove the following specifications. The names below are proposed identifiers, not claims that declarations with these names already exist.

- (1) `cosetSlice_linear`: restriction of a Hahn series to one coset of a subgroup, followed by monomial normalization, is linear over the Hahn field on that subgroup. No convexity assumption is permitted.
- (2) `cosetSlice_separates` and `finite_slice_matrix`: the family separates points and yields an invertible finite matrix on a linearly independent tuple.
- (3) `hahnSubfields_linearlyDisjoint`: use the same slice matrix after enlarging the scalar subgroup from the intersection to the second subgroup. This is the main reusable field-theoretic lemma.
- (4) `orderedEmbedding_extend_avoid`: realize a cut of the current rational span while avoiding a prescribed small subspace, with explicit cardinality hypotheses for [theorem 11.2](#).
- (5) `boundedSupport_gap_transcendental`: formalize the two support bounds (10.3) and (10.4), followed by the finite root bound. Start with the factorial sequence over $\mathbb{Q}$ before the transfinite cofinality version.
- (6) `iteratedHahn_coefficientField`: formalize separated ordered sums, the regrouping isomorphism, and finite coefficient-field control of a compositum. This is the kernel of the second transcendence proof.

The final proper-class theorem additionally needs a class or universe framework, a set-like enumeration, and proof that every recursion stage and every support pullback is a set. A set-level Hahn formalization by itself does not discharge that final obligation. The repository contains relevant Hahn and normal-form infrastructure [6], but this article adds no Lean code and makes no machine-verification claim.

13.3 Claims deliberately not made

The following boundaries prevent several plausible but invalid upgrades. The common field is a *full Hahn field* on a *set-sized* subgroup, not an arbitrary prescribed real closed field. Group sums are finite sums, not hidden summable closures. Boolean joins are Hahn joins, not ordinary composita. Strongness concerns Hahn-summable set families, not convergence of all real series. The embeddings preserve omnific membership, not an unproved complete first-order theory of the pair. They are not asserted to preserve the omega-map, exponentiation, derivation, birthdays, or simplicity. The proofs establish the stated theorems but do not establish historical priority or settle a named general factorization conjecture for **Oz**. [write] [Section 14](#) collects these boundaries with every other limitation stated in the source, and adds those that come from the collection.

13.4 Conclusion

Full surreal copies can be arranged with far more control than a bare universality statement provides. An arbitrary set-sized full Hahn core can be fixed exactly, an ordinal-indexed family of copies can be made independent over it, and their arithmetic parts remain genuine copies of the entire omnific ring. The same construction simultaneously controls the surcomplex field and Gaussian omnific integers.

The two descent failures are complementary. Passing from intersecting omnific rings to their fraction field can lose transcendental elements that belong to both full field copies, even though each copy provides one uniform denominator for the whole common set field. Passing from two full

field copies to their ordinary compositum can likewise miss transcendental normal forms allowed by their combined exponent group. Thus support closure, ordinary field generation, and arithmetic localization are three distinct operations. The prescribed-intersection construction makes their difference visible within the actual surreal and surcomplex fields, rather than only in abstract auxiliary examples.

14 Consolidated ledger of non-claims

[write] The source states its limitations where they arise, and each is kept in place. This ledger collects them with their locations. “README” refers to the delivered README, which is not shipped; its content is in the collection README. Items S1–S24 are the source’s. Items W1–W8 were added when the report joined the collection. [merge] The limitations of the second manuscript (Sections 15–27) are listed separately, as items C1–C23 and M1–M5 of [section 25.5](#).

- S1. The common field is a *full* Hahn field on a *set-sized* divisible subgroup. It is not an arbitrary prescribed real closed or set-sized subfield, not the ordinary field generated by a parameter set, not its real closure, and not a non-Hahn subfield (after [corollary 6.1](#); [section 13.3](#); README).
- S2. Group sums G_J are finite sums, not hidden summable closures (after (6.1); [section 13.3](#)).
- S3. The joins of [theorem 6.3](#) are Hahn joins, not ordinary composita; [Theorem C](#) shows that the difference can be transcendental ([remark 6.4](#); [section 13.3](#)).
- S4. Strongness concerns Hahn-summable set-indexed families, not convergence of real series or continuity; ordinary analytic convergence is not used in any proof ([section 2](#); [section 13.3](#)).
- S5. Elementarity is claimed only formula by formula in the languages of ordered fields and of fields, by quantifier elimination for real closed and algebraically closed fields. Nothing is asserted in the language with a predicate for **Oz**, the omega-map or a derivation, and algebraic independence is not identified with a model-theoretic independence notion of a richer theory. Preservation and reflection of **Oz** and commutation with the floor do not give elementarity of the pair ([section 6.2](#); research question 9; [section 13.3](#)).
- S6. The embeddings are not asserted to preserve the omega-map, the Gonshor exponential, the Berarducci–Mantova derivation, birthdays or simplicity. Monomial preservation is not omega-map commutation: $h \neq \Phi_h$ in general, and $\Phi_0(\omega^1) \neq \omega^{\Phi_0(1)}$ for the explicit maps, so they cannot be relabelled as solutions in richer languages ([warning 4.5](#); [section 7](#); research question 6; [section 13.3](#)).
- S7. No proper-class base H is treated. The obstacle is a limitation of the argument, whose first partial domain and forbidden span must be sets, not a proof that such extensions are impossible (research question 5).
- S8. [Theorem 11.2](#) is conditional on the existence of the stated saturated group; it does not assume that every cardinal admits one. Uncountability is needed for the proof. The lifted Hahn fields need not have cardinality κ ([section 11](#)).
- S9. The fraction-field failure does not by itself compute a normalization, a Picard group or a nontrivial projective module (research question 4).
- S10. The bounded-support transcendence mechanism of [section 10](#) is not new. It belongs to bounded-support Hahn arithmetic; the bounded-support report gives stronger independent

families, and the non-fraction obstruction is Proposition 2.4.5 of [3]. It is reproved so that [Theorem B](#) does not depend on an unreviewed repository result, not claimed as a discovery ([section 1.4](#); [section 10](#); research question 8; README).

- S11. The ingredients are not claimed as new: the set-cut property, normal forms, Hahn multiplication and inversion, and the closedness criteria. The group extension is a concrete saturation construction, not a new saturation principle; the monomial lift is standard Hahn functoriality; the coset-slice lemma is an elementary reusable lemma, not a certified new theorem of Hahn field theory ([section 1.4](#); title page).
- S12. Priority is not certified. Targeted literature searches do not certify that the combined formulation is absent from published or unpublished work, and no claim of having audited every report or every later repository revision is made ([section 1.4](#); [section 13.3](#); README).
- S13. The article is unrefereed and not formalized. It adds no Lean code and makes no machine-verification claim. The identifiers of [section 13.2](#) are proposals, not existing declarations; a set-level Hahn formalization does not discharge the class-size obligations, and a verified finite support computation is no substitute for them (title page; [section 13.2](#); research question 10; README).
- S14. No named longstanding or published conjecture is claimed solved, and no general factorization conjecture for $\mathbf{Oz}$ is settled (title page; [section 13.3](#); README).
- S15. [Theorem B](#) is the failure of one descent operation, not of $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, which the argument uses (after [Theorem B](#)).
- S16. [Lemma 8.2](#) gives a necessary coefficient-field condition, not membership in M of every series with coefficients in D . The compositum argument is not the invalid inference that every infinite sum lies outside a compositum; a geometric series already lies in one (after [lemma 8.2](#); [remark 8.4](#); README).
- S17. The class framework uses no tensor product indexed by a proper class and no proper-class-dimensional tensor object, no class truth or satisfaction predicate, and no recursion with a proper-class state at a stage. The cut property is never applied to proper-class sides ([section 2](#); [section 6.2](#)).
- S18. [Theorem 6.2](#) is a classification inside the strong, real-fixing, monomial-preserving class only. Strongness alone should not be assumed to settle research question 1.
- S19. The maps h_α fix H as a set of exponent values and the Φ_α fix $\mathcal{K}_{\mathbb{R}}(H)$; the two roles are not identified ([remark 3.6](#)).
- S20. The positive-exponent truncation plus the ordinary floor of the constant coefficient is not always the omnific floor ([remark 4.3](#)). Scaling all exponents by a constant gives an automorphism, not a proper copy ([remark 7.3](#)).
- S21. The saturation proof uses a simplest realizer locally but does not preserve the simplicity relation; this is not an initial-embedding theorem (research question 7).
- S22. The research questions are proposed by these theorems; they are not presented as previously published open problems or as claims of difficulty. Research question 8 is not an untouched direction ([section 12](#)).
- S23. For coefficient fields other than $\mathbb{R}$ and $\mathbb{C}$ the Hahn-field notation is formal, not a subfield of $\mathbf{No}$ or $\mathbf{No}(i)$; fraction fields of class domains are ratios inside the ambient field, not quotients

of a proper class (section 2; section 9).

- S24. The encyclopedia page is orientation only; Ehrlich–Kaplan is a comparison, not a proof input; the repository reports were used for scope and nonduplication, not as dependencies. The delivered production checks (compilation, rendering and visual inspection of the PDF) are not formal verification of the mathematics (bibliography; README).
- W1. No `isc:` label has a Lean mapping in `FORMALIZATION.md`. Independent proof review and formalization are pending. The package shipped no verification code, data or audit, so the report has no suite to rerun (section 1.6; Appendix A.6). [merge] This holds for source 01; the second manuscript shipped a finite-check suite, rerun in Appendix A.8, which tests none of source 01’s theorems.
- W2. Theorem 4.1 and propositions 4.2 and 4.4 and the inner lift of section 7 are the real-fixing case of the omnific-preserving report’s classification and its inner lift; no priority is claimed for them relative to that report. Corollary 6.1 extends `opa:as:thm:parameters` but does not claim its topological assertions (section 1.6).
- W3. Lemma 9.1 is `odg:thm:fractions`, proposition 9.3 is the localization of `bst:prop:localdensity`, the coset slice is `bst:eq:projection`, and theorem 10.2 and corollary 10.3 duplicate `bst:cor:onegap` and the one-series case of `bst:thm:independent` (section 1.6).
- W4. Real fixing is a choice. For the explicit copies the bottom gap is nonzero, so exact-monomial embeddings with the same exponent maps that preserve **Oz** and move reals exist; nothing is claimed about them or about intersections in that larger class (section 1.6).
- W5. The large-cardinal report’s question “Joint algebraic dependence of the images” is not answered: its image $J(\mathbf{No})$ is not a full Hahn field on a subgroup (section 1.6).
- W6. None of the omnific-preserving report’s questions is answered; in particular `opa:as:q:composition`, `opa:as:q:conjugacy`, `opa:as:q:elementary` and `opa:as:q:exp` stay open. None of the bounded-support report’s questions is answered (section 1.6).
- W7. The source’s statements about the collection are those of its pin and are kept as provenance (section 1.4; Appendix A.2).
- W8. The proofs were read when the report was written (Appendix A.4). That reading is not an independent proof review.

15 Beyond composita of surreal copies: a later manuscript

[merge] Sections 15 to 27 print a second manuscript, written against this report. In the report’s local numbering (Appendix A.7) the original manuscript is source 01 and the addition is *source 02*: manuscript 05 of batch 34 of the collection, archive `Surreal_Composita_Research`, *Beyond Composita of Surreal Copies: Exact rank tests, differential transcendence, and omnific witnesses*, a 29-page PDF dated 23 September 2026 and pinned to repository commit `7b256e0792df7718995b816ad12a77bf56731f8f`. At that pin this report’s `article.tex` was the blob `7d5cfe644e7c902f2b594179c337a6c633350fc3`. Since then only three [write] sentences about the bounded-support report have changed (sections 1.4, 1.6 and 9.1), so the source’s description of sections 7, 8 and 12 is accurate. The source takes Research question 3 of section 12 as its target.

Section n of the source is Section $n + 14$ here for $n = 1, \dots, 12$, and every numbered statement keeps its position: the source’s Theorem $n.m$ is Theorem $(n + 14).m$ here, and likewise for lemmas,

corollaries, remarks, definitions and examples. Its Questions 1–12 are Research questions 11–22. Its Appendices A and B form [section 27](#); its notation index (Appendix C) is folded into the table of [section 15.5](#). Where the source cites this report by section number, the citation is replaced by a cross-reference to the same section; the numbering of Sections 1–12 has not changed since the pin. Where the source reproduces material already printed here (the explicit copies of [section 7](#) and the re-expansion of [section 8](#)), it is printed once and the source is credited ([section 22.1](#)). Text added in the merge is marked [merge]; symbols renamed from the source are listed in [section 15.5](#). Every result, proof, example, question and limitation of the source is printed.

15.1 Field generation is not Hahn summation

A Hahn field permits well-ordered, potentially infinite supports. An ordinary field compositum permits only finitely many field operations on finitely many elements at a time. These are different closure operations. Even when two Hahn subfields have a transparent intersection, their ordinary compositum need not contain all the series supported on the sum of their exponent groups.

This report constructs explicit strong surreal self-embeddings whose images K_0, K_1 intersect in the real constants ([theorem 7.2](#)), and [section 8](#) proves that the full Hahn join E contains an element transcendental over $M = K_0 K_1$ ([Theorem C](#)). Research question 3 asks for an exact description of ordinary composita and the transcendence degree of the join over the compositum. Source 02 takes that question as its target. The explicit copies and the one-series coefficient-transcendence obstruction of [theorem 8.3](#) are this report's; neither is claimed by source 02 as a new construction.

There are two genuinely different ways for a series to escape the compositum. Its coefficients can use infinitely much algebraically independent information. More subtly, its coefficients can all lie in one field $k(x)$ and nevertheless have complexity in x that defeats every finite algebraic or differential relation over the compositum. The second phenomenon shows that the necessary finite coefficient-field condition of [lemma 8.2](#) is far from sufficient.

15.2 Main conclusions

The first conclusion is an exact test. If the *coefficient rank* of a Hahn series means the dimension over k of the span of its coefficients, then

$$x \in Fk((t^\Gamma)) \iff \exists q \neq 0 (\text{crk}_k(q) < \infty \text{ and } \text{crk}_k(qx) < \infty). \quad (15.1)$$

This is an intrinsic test once the coefficient field and Hahn coordinates have been fixed. It does not assert decidability for arbitrary infinite input. The proof is elementary tensor algebra and is presented as such.

The second conclusion is a maximal-cardinality theorem. Write $\mathfrak{c} = 2^{\aleph_0}$. For $|k| \leq \mathfrak{c}$ and any coefficient extension F/k of positive transcendence degree,

$$\text{dtrdeg}(F((t))/Fk((t))) = \text{trdeg}(F((t))/Fk((t))) = |F|^{\aleph_0}. \quad (15.2)$$

Thus even $F = k(x)$, a single transcendental coefficient extension, produces the full continuum of differential freedom when k is $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$. Here the derivation acts only on the outer series variable.

The third conclusion concerns actual omnific integers. For the explicit separated copies of [section 7](#), with the order isomorphism $\psi(u) = \frac{1}{2} + \frac{u}{2(1+|u|)}$ of [lemma 7.1](#) and ordinal arithmetic used only in the label $\omega \cdot \alpha + n$, define

$$Y_\alpha = \sum_{n \geq 1} \omega^{\omega^{15/4} - n\omega^{7/2} + \omega^{\psi(\omega \cdot \alpha + n)}} \quad (\alpha \in \mathbf{On}). \quad (15.3)$$

Every exponent in this normal form is positive, and the exponents strictly decrease with n . Hence Y_α is a purely infinite omnific integer with countable support. All its Euler jets, together with the jets of all the other Y_α , are algebraically independent over the compositum in the precise finite-relation sense. Moreover, given *any* set $P \subseteq E$, there is an ordinal α_0 such that the tail $\{Y_\alpha : \alpha \geq \alpha_0\}$ remains differentially algebraically independent over $M\langle P \rangle$.

This last assertion is stronger than producing a fresh witness separately for each parameter set: a tail of one predefined family works. It also avoids assigning a set cardinal to a proper-class transcendence degree. [merge] For $\alpha = 0$ the label is $\beta_{0,n} = n$, and $Y_0 = \omega^{\omega^{15/4}}S$, where S is the witness (8.3) of Theorem C. Source 02 thus extends this report's witness to an ordinal family inside **Oz**, as it says.

15.3 Scope, sources, and the meaning of new

Hahn arithmetic, linear disjointness, algebraic extension of derivations in characteristic zero, and the surreal normal-form construction are established tools. Standard Hahn derivation frameworks appear in Kuhlmann–Matusinski [11]; recurrence descriptions of generalized series subfields are studied by Krapp–Kuhlmann–Serra [10]. The finite-rank characterization below is a coefficient-space counterpart, not a replacement for their recurrence theory. Omnific arithmetic has a substantial factorization literature, including L’Innocente–Mantova [3]; no factorization conjecture is addressed here.

The candidate contributions of source 02 are the rapid coefficient-degree differential-independence theorem, its combination with independent coefficient trees to give (15.2), and the explicit omnific tail-resilience theorem and arithmetic consequences. The literature comparison is targeted, not exhaustive. These claims are proved in the source but are not certified as previously unpublished. The differential statements use a specified diagonal Euler derivation, *not* the exponential-compatible surreal derivation of Berarducci–Mantova [12].

Source 02’s own source audit (shipped as `02-composita-SOURCE_AUDIT.md`) lists six candidate contributions: the rapid coefficient-degree theorem over $k((t))(x)$ with arbitrary Laurent coefficients in the relation; private coefficient-block independence over the whole finitely-generated-coefficient envelope and its tree amplification; their combination into the exact degrees $|F|^{\aleph_0}$; the explicit purely infinite omnific family with support of order type ω and the sharpness of its support cardinality; tail resilience of that predefined family under arbitrary set-sized parameter adjunction, including differential parameters; and persistent nonintegrality of the arithmetic join after set adjunction, with the Gaussian versions. It calls them proposed research contributions with written proofs; a targeted source search did not find them elsewhere, and that absence does not establish novelty.

15.4 What the addition settles in this report

[merge]

1. *Research question 3* (an exact description of ordinary composita) is answered in the *separated* case with trivial common core, which is the case of Theorem C, and in the stated cardinal cases. Its first part, the characterization of the compositum, is the finite coefficient-rank denominator test (theorem 17.2) and, for the explicit copies, corollary 22.1. Its second part, the transcendence degree of the join over the compositum, is answered in the natural set-sized Laurent examples by theorem 21.1: both the algebraic and the Euler-differential transcendence degree of $F((t))$ over $Fk((t))$ are $|F|^{\aleph_0}$ when $|k| \leq \mathfrak{c}$ and $\text{trdeg}(F/k) > 0$. In the class-sized explicit join it is answered

in the form the question asks for, by independent families of every set cardinal ([theorem 22.2](#)), a tail that survives every set of parameters ([theorem 23.3](#)), and the absence of set-sized generation ([corollary 23.4](#)); no set cardinal is assigned to the class extension. Open: independent groups that are not separated (neither infinitesimal relative to the other), copies with a prescribed nonzero common core H , infinite algebraic coefficient extensions, and $|k| > \mathfrak{c}$ with small positive $\text{trdeg}(F/k)$. These are Research questions [11](#), [14](#), [13](#) and [12](#).

2. [Lemma 8.2](#) is shown to be far from sufficient: one transcendental coefficient x already gives a series with all coefficients in $k(x)$ that is differentially transcendental over the compositum ([corollary 19.5](#)).
3. [Theorem 8.3](#) is strengthened for the explicit witness: $S = \omega^{-b}Y_0$, with $\omega^{-b} \in K_1$ a constant of the Euler derivation Δ of [section 22.3](#), so S is differentially transcendental over $\mathcal{C}_{\text{fg}}(K_0/\mathbb{R}; G_1) \supseteq M$ by [theorem 22.2](#) (see the [merge] note after the proof of [Theorem C](#)).
4. No other question of this report is answered. Research question 4 (descent for the arithmetic intersection $\mathcal{A}_k(H)$) is untouched: the nonintegrality result, [theorem 24.2](#), concerns the join of two copies, not their intersection, and computes no normalization.

15.5 Conventions and renamed symbols

[merge] Source 02 uses both exponent conventions of this report: the decreasing large- ω convention $\sum c_g \omega^g$ with reverse-well-ordered support, written $\mathcal{K}_k(G)$ as here, and the increasing small- t convention $F((t^\Gamma))$ with well-ordered support, identified by $t^\gamma = \omega^{-\gamma}$ as in [section 9.1](#). In [sections 16](#) to [21](#) the fields are abstract t -series fields over arbitrary coefficient fields, not subfields of **No**. From [section 22](#) on, the fields are the explicit surreal fields of [sections 7](#) and [8](#), with $k = \mathbb{R}$ (and $\mathbb{C}$ in [theorem 24.4](#)).

The following symbols of the source are renamed; everything else is as in the source.

Here	Source 02	Reason
$\Gamma, \Gamma_0, \Gamma'$: the outer ordered group, a subgroup, a set subgroup	H, H_0, H'	H is the prescribed common core of Theorem A , and the source itself uses H in that sense in its Question 4 (Research question 14)
$g > 0$ in Γ with $\lambda(g) = 1$; $g = \omega^{7/2}$ in sections 22 to 24	h	h, h_α are ordered additive embeddings here; this report already writes $g = h_1(1) = \omega^{7/2}$ (proof of Theorem C) and uses g in this role in theorem 8.3
ω^b , the shift multiplier	s	$s_n = \omega^{g_n}$ are the coefficients of (8.3)
$F[L]$, the ring generated by F and L	T	T is also the source's inner support set
$\Theta, G_\Theta, C_\Theta$: inner support set, its support group and localized coefficient field	T, H_T, C_T	as above; G_Θ is a subgroup of G_0
$R_0[R_1]$, the ring generated by $R_0 \cup R_1$; $R_0[R_1][S]$; $R_0[R_1][i]$	$B; B[S]; B_{\mathbb{C}}$	B is a subgroup in lemma 8.1 , and $\mathcal{B}_k(H)$ is the bounded-support ring
$j = 0, 1$, index of the two explicit copies	i	$i = \sqrt{-1}$ in section 24 ; Theorem C writes K_j
$\mathfrak{A}$, a family of infinite subsets of $\mathbb{N}_{>0}$	$\mathcal{A}$	$\mathcal{A}_k(H)$ is the integer ring of section 9
$p_{ij} = P_{Y_{ij}}(\mathbf{f})$, evaluated partials	A_{ij}	A_i are members of $\mathfrak{A}$ in the same proof

Here	Source 02	Reason
$W_i(X, x)$, the leading polynomials in (19.4)	$H_i(T, x)$	H and T as above
w_i , $\mathbf{w}$: tails in the proof of theorem 19.2	R_i , $\mathbf{R}$	R_j are the omnific copies; R is a cut side
ν , the degree in x in the same proof	D	D is a finitely generated coefficient field, as in lemma 8.2
$\gamma_0 + \gamma_1$ with $\gamma_j \in G_j$, and e : generic exponents from section 22 on	$g_0 + g_1, g$	g is $\omega^{7/2}$ above
crk_k , coefficient rank	crk_k , macro $\backslash \mathbf{rk}$	same symbol; only the macro name differs

No normalization is changed by these renamings. In particular $\omega^{7/2} = g$ and $\omega^{15/4} = b$ are the same surreals as in the source, and (15.3) is the source's formula verbatim.

Agreements. $\mathcal{K}_k(G)$, $E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1)$, $M = K_0 K_1$, $K_j = \mathcal{K}_{\mathbb{R}}(G_j)$, $G_j = h_j(\mathbf{No})$, $I_j = (3j, 3j + 1)$, ψ , h_j , Φ_j , the convention $t^\gamma = \omega^{-\gamma}$, the finitely generated field D of lemma 8.2, and a series $S_i = \sum a_{b_i(n)} t^{ng}$ of the shape of theorem 8.3 mean the same here as in sections 7 and 8. The source's $g_{\alpha,n}$ and $a_{\alpha,n}$ extend this report's $g_n = \omega^{\psi(n)}$ and $s_n = \omega^{g_n}$: $g_{0,n} = g_n$ and $a_{0,n} = s_n$.

Local letters and tempting false readings.

- In sections 16 to 21, $E = F((t^\Gamma))$, $L = k((t^\Gamma))$ and $M = FL$. This L is *not* the common field $L_k = \mathcal{K}_k(H)$ of Theorem A and not a cut side; this F is a coefficient field, not the witness F of Theorem B. In the surreal instance K_0 plays the role of F , $K_1 \simeq \mathbb{R}((t^{G_1}))$ that of L , and $\mathbb{R}$ that of k .
- $\lambda : \Gamma \rightarrow k$ is an additive *Euler weight*. It is not a coset slice $\lambda_{g,H}$ of (5.1), and not a class functional.
- ∂ and Δ are Euler derivations, d and d_b coefficient derivations. The omnific-preserving report reserves d and D for derivations; here D is always a field, and d is also a total degree, d_n a degree sequence, and $d \in G_j$ a denominator exponent in lemma 24.1; none of these is an Euler derivation. $D_{\mathbb{R}} = \mathbb{Z}$ and $D_{\mathbb{C}} = \mathbb{Z}[i]$ keep their meaning from section 9.
- $\mathcal{J}(y_i : i \in I)$ is a set of jets, not an index set J . I is an index set and I_j an interval, as in section 1.5. The index set B of a coefficient family $(a_b)_{b \in B}$ and the label maps b_i are local to section 20; $b = \omega^{15/4}$ is a positive element of G_1 , as the letter b is in (8.1).
- κ is an infinite cardinal in corollary 20.4 and a set cardinal in theorem 22.2; it is not $\text{cf}(H)$ of section 10. $\mu = |F|$ in theorem 21.1. $\mathfrak{c}$ is $2^{\aleph_0}$, with no assumption on its value as an aleph.
- P is a polynomial relation in sections 19 and 20 and a set of parameters in sections 23 and 24; $M\langle P \rangle$ is the differential field generated by M and all jets of P . $O = E \cap \mathbf{Oz}$. Y_α and Z_α are the omnific witnesses (the letter Y is also a jet variable Y_{ij}). $\tau = \omega^{-g}$ is a monomial, not an exponent map.

- *Tempting false reading:* “ $\text{crk}_k(x) < \infty$ is necessary for $x \in M$.” It is necessary only for $x \in F[L]$; the compositum M is the fraction field of $F[L]$ and contains series of infinite coefficient rank (example 17.4).
- *Tempting false reading:* “ Δ is a surreal derivation.” It is a diagonal Euler derivation of the explicit join, defined through the iterated coordinates, killing K_0 ; it is not the Berarducci–Mantova derivation, not Hardy type, and not compatible with the exponential.
- *Strong* means preserving Hahn summability, as in section 1.5, not convergence in an order topology.

Notation index of the source (its Appendix C), with the names used here. $F((t^\Gamma))$: small- t Hahn field, well-ordered support in Γ . $\mathcal{K}_k(G)$: large- ω Hahn field, reverse-well-ordered support in G . E, L, M in the abstract sections: $F((t^\Gamma))$, $k((t^\Gamma))$ and the ordinary field compositum FL . $F[L]$: the finite coefficient-rank ring, the image of $F \otimes_k L$. $\text{crk}_k(z)$: dimension over k of the span of all coefficients of z . $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$: union of $D((t^\Gamma))$ over finitely generated coefficient fields D/k inside F . ∂ : abstract outer Euler derivation, with weight map $\lambda : \Gamma \rightarrow k$. h_j, Φ_j : inner additive embedding and outer surreal field embedding. G_j, K_j, R_j : exponent-group image, full field image, and omnific image of copy j . E, M from section 22 on: the explicit Hahn join $\mathcal{K}_{\mathbb{R}}(G_0 + G_1)$ and its ordinary compositum $K_0 K_1$. g, b, ω^b : $\omega^{7/2}$, $\omega^{15/4}$, and the monomial that shifts all witness exponents into the positive cone. Δ : Euler derivation extracting the inner $\omega^{7/2}$ coefficient from the dominant exponent coordinate, with the sign in (22.5). $\beta_{\alpha, n}$: the ordinal label $\omega \cdot \alpha + n$, the only place where ordinal arithmetic is used. $g_{\alpha, n}, a_{\alpha, n}$: independent inner exponents and the corresponding outer monomial coefficients. Y_α, Z_α : real omnific witness and its Gaussian multiple $(1 + i)Y_\alpha$. $\Theta, G_\Theta, C_\Theta$: set of inner support labels, its full additive support group, and the localized coefficient field. $R_0[R_1], O$: the ring generated by the two omnific copies, and the full omnific ring inside their Hahn join. $M\langle P \rangle$: differential field generated by M and all jets of the set P .

15.6 Place in the collection

[merge] Source 02 cites only this report from the collection. Checked against the collection at the placement commit a7a435f, the following reports bear on it; none is a dependency of its proofs.

1. *This report.* Lemma 18.3 is a variant of the coset-slice result, theorem 5.4: there two full Hahn fields with the same coefficient field are disjoint over the field of the intersection group; here $F((t^{\Gamma_0}))$ and $k((t^\Gamma))$, with different coefficient fields and $\Gamma_0 \subseteq \Gamma$, are disjoint over $k((t^{\Gamma_0}))$, by the same slice argument. The source credits the coset slice as part of the elementary Hahn toolkit. The first clause of lemma 24.1, $\text{Frac}(R_j) = K_j$, is the first clause of proposition 9.2 for the explicit copies, by the monomial clearing of lemma 9.1, as the source says. Lemma 16.5 is lemma 8.2 in the abstract setting. The localization of a parameter set in lemma 23.1 is the support-closure step of corollary 6.1, applied one normal-form level deeper.
2. *Three duals of Hahn vector spaces* (surcomplex/three-duals-of-hahn-vector-spaces, prefix duals:). Parts (i)–(ii) of theorem 17.2 are the injectivity paragraph before duals:prop:algebraic and that proposition, for the k -vector space $V = F$: the scalar extension $K \otimes_k V$, $K = k((t^\Gamma))$, embeds in $V_\Gamma = F((t^\Gamma))$ with image the vectors of finite total coefficient rank. Source 02 does not cite that report and presents the tensor fact as standard. Part (iii), the fraction-field test, is new but immediate from (ii).

3. *Tail spans and differential transcendence* (`surreal/tail-spans-and-differential-transcendence`, prefix `tail:`). `tail:thm:surreal` uses the same binary-prefix almost disjoint family as [corollary 19.4](#), private indices and a Vandermonde argument to give $\mathfrak{c}$ differentially independent series. The base $(\mathbb{Q}((t^{\mathbb{Q}})))$, the derivation (Berarducci–Mantova, via Euler jets) and the mechanism (the tail-span theorem `tail:thm:tail` for multiquadratic coefficients) differ from those of [theorems 19.2](#) and [20.2](#), whose base is a compositum or a finite-generation envelope and whose tool is coefficient degree or a coefficient derivation. Source 02 does not cite it; the two are related, not duplicates.
4. *Bounded-support transcendence* (`surreal/transcendence-over-bounded-support`, prefix `bst:`). `bst:thm:independent` obtains $2^{\text{cf}(G)}$ independent series over the bounded-support fraction field by coefficient codes realized at cofinally many indices, a private-index device over a different base, with a support-gap estimate instead of a derivation. Source 02 does not cite it.
5. *Dilation rigidity* (`surcomplex/autonomous-dilation-relations`, prefix `adr:`). `adr:sr:thm:tail` localizes a set of surcomplex parameters in the Hahn field of the real span of their supports and finds a tail of an explicit ordinal family whose dilation orbits are independent over that whole field; `adr:sr:thm:kappa` gives set-sized families over a prescribed set base. [Theorem 23.3](#) has the same shape, with Euler jets instead of dilation orbits and localization at the inner normal-form level.
6. *Omnific-preserving automorphisms* (`opa:`). The explicit copies are the maps $M_{1,t_{\sigma_j}}$ of that report’s Part III ([section 1.6](#)). Research questions [17](#) and [18](#) lie inside `opa:as:q:exp`.

No `isc:cp:` label has a Lean mapping in `FORMALIZATION.md`.

16 Conventions and foundational lemmas of the addition

16.1 Two support conventions

For a field F and an ordered abelian group Γ , the Hahn field $F((t^\Gamma))$ consists of formal sums

$$x = \sum_{\gamma \in \Gamma} x_\gamma t^\gamma$$

whose nonzero coefficient support is a well-ordered set. All supports are sets. Addition is coefficient-wise. In a product, the coefficient at γ is the finite sum of $x_\alpha y_\beta$ over pairs with $\alpha + \beta = \gamma$.

Surreal normal forms instead use the large- ω convention $x = \sum_g c_g \omega^g$, with reverse-well-ordered support: every nonempty subset has a greatest element. As in [section 2](#), $\mathcal{K}_k(G)$ is the full field of these normal forms with coefficients in k and exponents in an additive subgroup $G \subseteq \mathbf{No}$. Whenever the conventions change, the identification is

$$t^\gamma = \omega^{-\gamma}, \tag{16.1}$$

as in [section 9.1](#). The reversal in (16.1) is essential in the final omnific construction.

For set-sized groups, ordinary Hahn field algebra suffices. For proper-class exponent groups, each finite calculation occurs in the subgroup generated by a set of supports. The inverse of a nonzero series can likewise be constructed in a set-sized Hahn field containing its support. Thus the formulas define class fields without allowing class-sized sums. The class assertions are formulated in NBG with global choice, with ZFC for sets, as in [section 2](#). The displayed class families and derivations are

given by formulas; no class-state recursion, class transcendence basis, or class satisfaction predicate is used.

16.2 Euler derivations and differential independence

Definition 16.1. Let $k \subseteq F$ have characteristic zero and let $\lambda : \Gamma \rightarrow (k, +)$ be additive. The associated outer Euler derivation is

$$\partial\left(\sum_{\gamma} a_{\gamma} t^{\gamma}\right) = \sum_{\gamma} \lambda(\gamma) a_{\gamma} t^{\gamma}. \quad (16.2)$$

It kills F . When $\Gamma = \mathbb{Z}$ and $\lambda(n) = n$, it is $t d/dt$.

Lemma 16.2. *Formula (16.2) is a derivation of $F((t^{\Gamma}))$. It preserves Hahn-summable families and their sums. If $d : F \rightarrow F$ is a k -derivation, its coefficientwise extension to $F((t^{\Gamma}))$ is also a derivation and commutes with ∂ .*

Proof. Both maps only delete terms or change coefficients, so the support condition is preserved. For a product coefficient, finiteness of the convolution and $\lambda(\alpha + \beta) = \lambda(\alpha) + \lambda(\beta)$ give the Leibniz identity. The same finite calculation works for d . A Hahn-summable family has a well-ordered union of supports and only finitely many nonzero contributions at each exponent; neither property is destroyed. Finally $d(\lambda(\gamma)) = 0$ because $\lambda(\gamma) \in k$, proving commutation coefficient by coefficient. $\square$

Definition 16.3. A family $(y_i)_{i \in I}$ in a differential field extension E/M is *differentially algebraically independent over M* if the family

$$\mathcal{J}(y_i : i \in I) = \{\partial^j y_i : i \in I, j \geq 0\}$$

is algebraically independent over M . Every relation and every collection of variables in a polynomial are finite. We write $M\langle P \rangle$ for the field generated by M and all $\partial^j p$, $p \in P$, $j \geq 0$.

For set fields, differential transcendence degree has its usual meaning. All lower bounds below are proved by exhibiting independent families of the stated cardinality. For class fields, only the finite-relation definition and set-indexed subfamilies are used; no class transcendence basis is postulated, and no cardinal equal to the size of **No**.

16.3 The coefficient-field envelope

Throughout the set-sized theory, put

$$E = F((t^{\Gamma})), \quad L = k((t^{\Gamma})), \quad M = FL. \quad (16.3)$$

The compositum is taken in E and is a field. In contrast, the ring $F[L]$ generated by F and L consists of finite sums of products.

Definition 16.4. Define the finite-generation coefficient envelope by

$$\mathcal{C}_{\text{fg}}(F/k; \Gamma) = \bigcup_{\substack{k \subseteq D \subseteq F \\ D/k \text{ finitely generated}}} D((t^{\Gamma})). \quad (16.4)$$

Here “finitely generated” means as a field, not as a ring.

Lemma 16.5. *The union in (16.4) is a field containing M and is stable under every Euler derivation (16.2). Every finite subset of M is contained in one $D((t^\Gamma))$ with D/k finitely generated.*

Proof. The compositum of two finitely generated coefficient fields is again finitely generated, so the union is directed. Each constituent is a Hahn field. A finite field expression in elements of F and L uses only finitely many elements $a_1, \dots, a_r$ of F , and therefore belongs to $k(a_1, \dots, a_r)((t^\Gamma))$. The same argument handles finitely many expressions simultaneously. Formula (16.2) introduces only scalars from k into coefficients. $\square$

The last lemma is the finite coefficient-field control of lemma 8.2, in the abstract setting. The later constructions show both strictness of its necessary condition and independence over the entire envelope.

17 An exact coefficient-rank test for the compositum

Definition 17.1. For $x = \sum x_\gamma t^\gamma \in E$, define

$$\text{crk}_k(x) = \dim_k \text{span}_k \{x_\gamma : \gamma \in \Gamma\}.$$

In particular, $\text{crk}_k(0) = 0$. Let $F[L] \subseteq E$ be the ring generated by F and L .

Theorem 17.2 (Finite rank and finite-rank denominators). *For arbitrary fields $k \subseteq F$ and any ordered abelian group Γ :*

- (i) *The natural multiplication map $F \otimes_k L \rightarrow E$ is injective, with image $F[L]$.*
- (ii) $F[L] = \{x \in E : \text{crk}_k(x) < \infty\}$.
- (iii) $M = \text{Frac}(F[L])$, and an element $x \in E$ belongs to M exactly when there is $q \in E \setminus \{0\}$ for which both $\text{crk}_k(q)$ and $\text{crk}_k(qx)$ are finite.

Proof. A finite tensor can be written as $\sum_{i=1}^r b_i \otimes \ell_i$ with the $b_i \in F$ linearly independent over k . If its image vanishes, then comparison at each exponent γ gives

$$\sum_{i=1}^r b_i (\ell_i)_\gamma = 0.$$

All $(\ell_i)_\gamma$ belong to k , so each is zero. Thus every $\ell_i = 0$, proving injectivity. The image is the finite sums of products of elements of F and L , namely $F[L]$.

For $x = \sum_{i=1}^r b_i \ell_i$, every coefficient of x lies in the finite-dimensional space spanned by the b_i . Conversely, suppose the coefficient span has basis $b_1, \dots, b_r$. Write uniquely

$$x_\gamma = \sum_{i=1}^r b_i c_{i,\gamma}, \quad c_{i,\gamma} \in k.$$

For every i , the support of $\ell_i = \sum_\gamma c_{i,\gamma} t^\gamma$ is contained in $\text{supp}(x)$, hence is well ordered. Therefore $\ell_i \in L$ and $x = \sum b_i \ell_i \in F[L]$.

Since $F[L]$ is a subring of a field, its fraction field inside E is precisely the smallest field containing F and L , which is M . Finally $x = p/q$ with $p, q \in F[L]$, $q \neq 0$, is equivalent to $q, qx \in F[L]$ for such a q . Part (ii) gives the asserted test. $\square$

[merge] Parts (i) and (ii) are, for the k -vector space $V = F$, the injectivity paragraph before `duals:prop:algebraic` of the three-duals report and that proposition ([section 15.6](#)); the source does not cite it and presents the tensor fact as standard.

Remark 17.3 (What the criterion does and does not do). The theorem describes the subfield exactly in terms of its coefficients, rather than merely as an unspecified compositum. Nevertheless, the existential search over q is an infinite-input problem. No algorithm deciding arbitrary Hahn-series membership follows. The tensor argument is a standard linear-disjointness mechanism; neither a new tensor-product theorem nor a new general theory of rational series is claimed. The comparison with recurrence criteria in [10] is conceptual, not a claim that the two formulations have identical inputs.

Example 17.4 (Infinite rank inside the compositum). Let $u \in F$ be transcendental over k and let $g \in \Gamma$ be positive. The geometric series

$$z = \frac{1}{1 - ut^g} = \sum_{n \geq 0} u^n t^{ng}$$

has infinite coefficient rank, because the powers of u are linearly independent over k . Yet $z \in M$: the multiplier $q = 1 - ut^g$ has rank two and $qz = 1$ has rank one. The support $\{ng : n \geq 0\}$ is well ordered in every ordered abelian group. No cofinality assumption on g is needed.

Corollary 17.5 (Finite coefficient extensions). *If $[F : k] < \infty$, then $E = F[L] = M$.*

Proof. Every coefficient span is a subspace of the finite-dimensional k -space F . Apply [theorem 17.2](#). □

For finite ranks one also has

$$\text{crk}_k(x + y) \leq \text{crk}_k(x) + \text{crk}_k(y), \quad \text{crk}_k(xy) \leq \text{crk}_k(x) \text{crk}_k(y).$$

Indeed a coefficient of the product lies in the span of pairwise products of basis elements for the two coefficient spaces. Infinite rank therefore obstructs membership in $F[L]$, but not in its fraction field.

18 Linear disjointness under two enlargements

The next lemmas prevent an independence proof over a small coefficient field or a cyclic exponent group from being lost when the ambient Hahn field is enlarged. The arguments are included instead of silently treating base extension as harmless. Linear disjointness has its usual definition via injectivity of the tensor multiplication map [13], as in [section 2](#).

Lemma 18.1 (Coefficient extension). *If $F_0 \subseteq F$, then F and $F_0((t^\Gamma))$ are linearly disjoint over F_0 inside $F((t^\Gamma))$.*

Proof. A finite relation with coefficients in $F_0((t^\Gamma))$ among elements of F linearly independent over F_0 vanishes coefficientwise only when every coefficient series is zero. This is the first argument of [theorem 17.2](#) with $k = F_0$. □

Lemma 18.2 (Intermediate base change). *Suppose fields A and B are linearly disjoint over K in an ambient field, and $K \subseteq A_0 \subseteq A$. Then A and A_0B are linearly disjoint over A_0 .*

Proof. Let $a_1, \dots, a_m \in A$ be linearly independent over A_0 and suppose $\sum_i a_i c_i = 0$ with $c_i \in A_0 B$. Clear a common nonzero denominator from the ring generated by A_0 and B . Each resulting coefficient is a finite sum of products. Choose a K -linearly independent list $b_1, \dots, b_r \in B$ spanning all the finitely many B -entries used. The relation becomes

$$\sum_{j=1}^r \left(\sum_{i=1}^m a_i a_{ij} \right) b_j = 0, \quad a_{ij} \in A_0.$$

Linear disjointness over K makes the b_j linearly independent over A . Hence every inner sum vanishes. Independence of the a_i over A_0 gives all $a_{ij} = 0$, so all the original c_i are zero. $\square$

Lemma 18.3 (Coset slices). *Let $\Gamma_0 \subseteq \Gamma$ be an additive subgroup. Inside $F((t^\Gamma))$, the fields*

$$E_0 = F((t^{\Gamma_0})), \quad L = k((t^\Gamma))$$

are linearly disjoint over $L_0 = k((t^{\Gamma_0}))$. No convexity assumption on Γ_0 is required.

Proof. Suppose $u_1, \dots, u_m \in E_0$ are linearly independent over L_0 and $\sum_i u_i v_i = 0$, $v_i \in L$. For a coset C of Γ_0 meeting one of the supports of the v_i , choose a representative $c \in C$. Let $v_{i,C}$ be the sum of the terms of v_i supported on C , multiplied by t^{-c} . Its support is a translate of a subset of a well-ordered set, and is contained in Γ_0 , so $v_{i,C} \in L_0$.

Multiplication by a series supported in Γ_0 does not change the coset of an exponent. Extracting the C -slice of the original relation and translating by $-c$ therefore gives

$$\sum_i u_i v_{i,C} = 0.$$

Independence over L_0 yields $v_{i,C} = 0$ for all i . Every support exponent belongs to a coset that was considered; hence every $v_i = 0$. Only a set of cosets is involved, even in the class-supported setting. $\square$

[merge] Up to the change of convention (16.1), $v_{i,C}$ is a normalized coset slice as in (5.1), and the lemma is a variant of [theorem 5.4](#) with different coefficient fields on the two sides ([section 15.6](#)).

Corollary 18.4 (Independence survives both enlargements). *Let $x \in F$ be transcendental over k , let $g > 0$ in Γ , and let $\lambda : \Gamma \rightarrow k$ satisfy $\lambda(g) = 1$. If a family in $k(x)((t))$ is differentially algebraically independent over $k((t))(x)$ for $t d/dt$, then replacing t^n by t^{ng} gives a family differentially algebraically independent over $Fk((t^\Gamma))$ in $F((t^\Gamma))$.*

Proof. First apply [lemma 18.1](#) to $k(x) \subseteq F$ and the cyclic exponent group. Then apply [lemma 18.2](#) with $A = k(x)((t))$ and $A_0 = k((t))(x)$: the field A is linearly disjoint from $Fk((t))$ over A_0 . Distinct monomials in any finite collection of jets, which were linearly independent over A_0 , remain so over $Fk((t))$.

Next identify this cyclic Hahn field with $F((t^{\mathbb{Z}g}))$. Apply [lemma 18.3](#), followed by [lemma 18.2](#) with the intermediate field $Fk((t^{\mathbb{Z}g}))$. This preserves independence after adjoining $k((t^\Gamma))$. Finally $\partial(t^{ng}) = nt^{ng}$, so the transported elements have exactly the transported jets. $\square$

19 Differential independence from one coefficient variable

The next theorem is the key to the small-transcendence-degree case. The exponents in the outer variable need not be sparse. Instead, the degrees in a single coefficient variable grow so fast that a late coefficient cannot be canceled by any fixed differential polynomial.

Definition 19.1. A family $\mathfrak{A}$ of infinite subsets of $\mathbb{N}_{>0}$ has the *finite private-index property* if, for every finite subfamily $A_1, \dots, A_r$ of distinct members and every i , the set

$$A_i \setminus \bigcup_{j \neq i} A_j$$

is infinite. In particular, every pairwise almost disjoint family of infinite subsets has this property.

Theorem 19.2 (Rapid coefficient-degree independence). *Let k have characteristic zero, let x be transcendental over k , and let $(d_n)_{n \geq 1}$ be a strictly increasing sequence of positive integers such that*

$$\frac{d_n}{d_{n-1}} \longrightarrow +\infty. \quad (19.1)$$

For a family $\mathfrak{A}$ with the finite private-index property, put

$$f_A = \sum_{n \in A} x^{d_n} t^n \in k(x)((t)). \quad (19.2)$$

Then $(f_A)_{A \in \mathfrak{A}}$ is differentially algebraically independent over $k((t))(x)$ for $\partial = t d/dt$ and $\partial x = 0$.

Proof. Suppose there is a relation in finitely many jets of distinct $f_{A_1}, \dots, f_{A_r}$. Among all nonzero polynomial relations over $k((t))(x)$ in finite lists of these jets, choose one of least total jet degree d . Write it as

$$P((\partial^j f_{A_i})_{1 \leq i \leq r, 0 \leq j \leq m}) = 0,$$

allowing unused variables. A degree-zero nonzero relation is impossible, so $d \geq 1$.

Clear the denominators in x of the finitely many coefficients of P , then multiply by a power of t large enough to remove their negative outer exponents. We may therefore assume

$$P \in k[[t]][x][Y_{ij} : 1 \leq i \leq r, 0 \leq j \leq m]. \quad (19.3)$$

Let ν be its degree in x . These operations do not change total jet degree or minimality. We use $\mathbf{f}$ to denote the tuple of all the indicated jets.

For each variable set

$$p_{ij} = P_{Y_{ij}}(\mathbf{f}) \in k[x][[t]].$$

If the formal polynomial $P_{Y_{ij}}$ is nonzero, its total jet degree is smaller than d , so minimality forces $p_{ij} \neq 0$. At least one formal partial is nonzero, because the characteristic is zero and $d \geq 1$. Consequently

$$\ell = \min_{p_{ij} \neq 0} \text{ord}_t(p_{ij})$$

is a nonnegative integer. Define polynomials in a new indeterminate X by

$$W_i(X, x) = \sum_{j=0}^m [t^\ell] p_{ij} X^j \in k[x][X]. \quad (19.4)$$

At least one W_i is nonzero: its coefficients occupy distinct powers of X , and at least one p_{ij} attains the valuation ℓ . Fix such an index i_* .

For an integer n , truncate every original series strictly before t^n :

$$v_i = \sum_{\substack{a \in A_i \\ a < n}} x^{d_a} t^a, \quad w_i = f_{A_i} - v_i.$$

Every jet of w_i has t -valuation at least n . Let $\mathbf{v}$ and $\mathbf{w}$ be the corresponding tuples of jets. Evaluating any polynomial from (19.3) at $\mathbf{f}$ or $\mathbf{v}$ changes it only by a multiple of t^n . Therefore, when $n > \ell$, the partial evaluations $P_{Y_{ij}}(\mathbf{v})$ have zero coefficients below ℓ and the same coefficient at ℓ as p_{ij} .

Choose n in the infinite private-index set

$$A_{i_*} \setminus \bigcup_{i \neq i_*} A_i.$$

We may require simultaneously

$$n > \max\{\ell, 1\}, \quad W_{i_*}(n, x) \neq 0, \quad d_n > \nu + d d_{n-1}. \quad (19.5)$$

The middle condition excludes only finitely many integers, since a nonzero polynomial in X over the field $k(x)$ has finitely many roots and characteristic zero gives distinct integer elements. The last condition holds eventually by (19.1).

The polynomial Taylor expansion about $\mathbf{v}$ is

$$P(\mathbf{v} + \mathbf{w}) = P(\mathbf{v}) + \sum_{i,j} P_{Y_{ij}}(\mathbf{v}) \partial^j w_i + \text{terms of tail degree at least two}. \quad (19.6)$$

Every omitted term is divisible by t^{2n} . Since $n + \ell < 2n$, these terms do not contribute at exponent $n + \ell$. In the linear sum, a tail coefficient after exponent n could contribute only by multiplying a partial coefficient before ℓ , and those coefficients vanish. At exponent n , only w_{i_*} is nonzero, and

$$[t^n] \partial^j w_{i_*} = n^j x^{d_n}.$$

Taking $[t^{n+\ell}]$ in (19.6), using $P(\mathbf{f}) = 0$, gives the exact polynomial identity

$$0 = [t^{n+\ell}] P(\mathbf{v}) + x^{d_n} W_{i_*}(n, x). \quad (19.7)$$

Every coefficient of every truncated jet has degree in x at most d_{n-1} . Thus the first summand in (19.7) has degree at most $\nu + d d_{n-1}$. The second is nonzero and has degree at least d_n . The final inequality in (19.5) makes cancellation impossible. This contradiction proves the theorem. $\square$

Remark 19.3 (Why arbitrary Laurent coefficients are allowed). The coefficients of P can be arbitrary elements of $k((t))(x)$, not just rational functions of t and x over k . Clearing denominators makes their degree in x uniformly finite, while their t -series can remain infinite. The proof uses only this finite x -degree and a common lower bound on outer exponents. It does not impose a recurrence on those coefficient series.

Corollary 19.4 (A continuum family in one coefficient field). *For every characteristic-zero field k and transcendental x , the field $k(x)((t))$ contains $\mathfrak{c}$ elements differentially algebraically independent over $k((t))(x)$, all with coefficients in $k(x)$.*

Proof. Take $d_n = n!$. Encode the nonempty finite binary strings injectively by positive integers. For each infinite binary sequence η , let A_η consist of the labels of its finite nonempty prefixes. Distinct branches have only finitely many common prefixes, so their sets are pairwise almost disjoint. There are $\mathfrak{c}$ branches. Apply [theorem 19.2](#). $\square$

[merge] The binary-prefix family is the one of `tail:eq:Aalpha` in the tail-spans report, where it serves `tail:thm:surreal` over a different base and with a different derivation ([section 15.6](#)).

Corollary 19.5 (Failure of the finite coefficient-field criterion). *Suppose F/k has positive transcendence degree and there are $g > 0$ in Γ and additive $\lambda : \Gamma \rightarrow k$ with $\lambda(g) = 1$. Then $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$ contains an element differentially transcendental over $M = Fk((t^\Gamma))$. In particular, containment of all coefficients in a single finitely generated field does not imply membership in M , or even algebraicity over M .*

Proof. Choose a transcendental $x \in F$ and use

$$f = \sum_{n \geq 1} x^{n!} t^{ng}.$$

The singleton family $\{\mathbb{N}_{>0}\}$ satisfies the private-index condition. Apply [theorem 19.2](#) and [corollary 18.4](#). The series belongs to $k(x)((t^\Gamma)) \subseteq \mathcal{C}_{\text{fg}}(F/k; \Gamma)$. $\square$

Example 19.6 (The first terms and the mechanism). The last witness starts as

$$xt^g + x^2t^{2g} + x^6t^{3g} + x^{24}t^{4g} + x^{120}t^{5g} + \dots$$

The outer support here is the entire positive cyclic progression, not a factorially sparse support. A polynomial differential relation of total jet degree d and coefficient degree ν sees only degree at most $\nu + d(n-1)!$ in the truncation before t^{ng} . A surviving linear response to the next coefficient has degree at least $n!$. This is the obstruction. Merely pointing to gaps in the outer support would not prove this example.

[merge] The source's table in [section 25](#) applies [corollary 19.5](#) to the explicit join of [section 8](#), with $F = K_0$, $k = \mathbb{R}$, $\Gamma = G_1$ and λ as in [section 22.3](#); the lemmas of [section 18](#) use only finite relations, so the class-sized F causes no difficulty. The example is then a series with every coefficient in the one-generator field $\mathbb{R}(x)$, $x \in K_0$ transcendental over $\mathbb{R}$. [Lemma 8.2](#) places every finite subset of M in some $\mathcal{K}_D(G_1)$ with D finitely generated; the example shows that the converse fails for a single generator.

20 Independent coefficient blocks and tree amplification

The preceding theorem uses one coefficient variable. The next results exploit many algebraically independent coefficients and obtain independence over a larger base than the compositum itself.

20.1 Coefficient derivations with finite avoidance

Lemma 20.1 (Avoiding a finitely generated coefficient field). *Let F/k have characteristic zero, let $(a_b)_{b \in B} \subseteq F$ be algebraically independent over k , and let $D \subseteq F$ be finitely generated over k . There is a finite set $B_0 \subseteq B$ such that, for every $b \in B \setminus B_0$, a k -derivation $d_b : F \rightarrow F$ exists with*

$$d_b(a_b) = 1, \quad d_b(a_c) = 0 \ (c \neq b), \quad d_b(D) = 0.$$

Proof. Extend the given independent set to a transcendence basis $\mathcal{T}$ of the set field F/k . Each member of a finite generating list for D/k is algebraic over $k(\mathcal{T}_0)$ for some finite $\mathcal{T}_0 \subseteq \mathcal{T}$; take one such finite set for the whole list. Let B_0 consist of the b with $a_b \in \mathcal{T}_0$.

On the rational function field $k(\mathcal{T})$ define the coordinate derivation that sends a_b to 1 and all other basis elements to 0. It extends uniquely over the algebraic extension $F/k(\mathcal{T})$, because this extension is separable. Explicitly, for an algebraic element z with minimal polynomial $p(X)$, differentiation of $p(z) = 0$ forces

$$d_b(z) = -\frac{(d_b p)(z)}{p'(z)}. \quad (20.1)$$

The denominator is nonzero. This formula gives the usual compatible extension through finite algebraic subextensions and hence through their union. If $b \notin B_0$, the derivation kills $k(\mathcal{T}_0)$, so (20.1) makes it kill each of the chosen generators of D . It therefore kills D . $\square$

20.2 The block theorem

Theorem 20.2 (Private coefficient blocks). *Let F/k have characteristic zero. Let $(a_b)_{b \in B} \subseteq F$ be algebraically independent over k , and choose $g > 0$ in Γ and additive $\lambda : \Gamma \rightarrow k$ with $\lambda(g) = 1$. For each $i \in I$, let*

$$b_i : \mathbb{N}_{>0} \longrightarrow B$$

be injective. Assume that for every finite set of distinct indices $i_1, \dots, i_r$ and every $i = i_q$, there are infinitely many n such that the label $b_i(n)$ occurs in no other sequence b_{i_p} , at any position. Put

$$S_i = \sum_{n \geq 1} a_{b_i(n)} t^{ng}. \quad (20.2)$$

Then $(S_i)_{i \in I}$ is differentially algebraically independent over $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$ for the Euler derivation associated to λ .

Proof. Every displayed series has well-ordered support contained in $\{ng : n \geq 1\}$. Suppose a nonzero polynomial relation exists among finitely many jets of finitely many distinct S_i , with coefficients in $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$. Choose a relation P of least total jet degree. Its coefficients all lie in one $D((t^\Gamma))$ with D/k finitely generated. The total degree is positive.

Fix one index i involved in the relation. By the private-label hypothesis, there are infinitely many positions n whose coefficient label occurs only in this series among those under consideration. After excluding the finite exceptional set of lemma 20.1, infinitely many remain. For such a position, extend $d_{b_i(n)}$ coefficientwise to the Hahn field. By lemma 16.2, it kills every coefficient of P , commutes with ∂ , and satisfies

$$d_{b_i(n)}(\partial^j S_i) = n^j t^{ng}, \quad d_{b_i(n)}(\partial^j S_{i'}) = 0 \quad (i' \neq i).$$

Differentiating $P = 0$ by this coefficient derivation and dividing by the nonzero monomial t^{ng} gives

$$\sum_{j=0}^m n^j P_{Y_{ij}}((\partial^a S_{i'})_{i',a}) = 0. \quad (20.3)$$

Here m bounds all jet orders used. The coefficients in (20.3) do not depend on n . Since it holds for infinitely many distinct integers in a characteristic-zero field, the polynomial in n is identically zero. Thus every evaluated partial $P_{Y_{ij}}$ is zero. Repeat for each index i in the relation.

A nonconstant polynomial in characteristic zero has some nonzero formal partial derivative. That partial has smaller total jet degree than P , yet its evaluation vanishes. This contradicts minimality. Therefore no relation exists. $\square$

Remark 20.3 (A finite Vandermonde version). For a relation involving jets only through order m , it is enough to use $m + 1$ distinct private positions outside the finite exceptional set. The matrix $(n_a^j)_{0 \leq a, j \leq m}$ has determinant $\prod_{a < b} (n_b - n_a) \neq 0$. The polynomial-root argument in the proof is simply a convenient way to avoid choosing a particular finite list.

Corollary 20.4 (Tree amplification). *If $\text{trdeg}(F/k) \geq \kappa$ for an infinite cardinal κ , and Γ, g, λ satisfy the preceding hypotheses, then $F((t^\Gamma))$ contains a family of cardinality $\kappa^{\aleph_0}$ differentially algebraically independent over $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$.*

Proof. The set of nonempty finite sequences in κ has cardinality κ . Choose an algebraically independent coefficient a_s for every such finite sequence s . For each branch $\eta \in \kappa^{\mathbb{N}_{>0}}$, use

$$S_\eta = \sum_{n \geq 1} a_{\eta|n} t^{n\eta}.$$

Two distinct branches share only finitely many prefix labels. Relative to any finite collection of other branches, every sufficiently long prefix is private. Apply [theorem 20.2](#). $\square$

20.3 Local use for class coefficient fields

Proposition 20.5 (Set-local form). *The private coefficient-block argument remains valid when F or Γ is a class, provided that k is a set field, every series has set support, and algebraic independence of the coefficient family means independence of every finite subfamily. The conclusion concerns every finite polynomial relation. No class transcendence basis is required.*

Proof. Fix a proposed relation. It involves finitely many S_i , so the union of their coefficient labels is a set. Its finitely many coefficient series belong to $D((t^\Gamma))$ for a single set field D finitely generated over k . Form the set field F' generated by D , k , and all the coefficients in the finitely many chosen S_i . The subgroup Γ' generated by g and all supports of the relation coefficients is a set. Restrict λ to Γ' .

Every expression in the relation lies in $F'((t^{\Gamma'}))$. The algebraically independent coefficient family used there is a set, so [lemma 20.1](#) applies to F'/k in ordinary set theory. All differentiations, finite convolution identities, and the contradiction now occur in this set-sized field. This proves the class statement one finite relation at a time. $\square$

Corollary 20.6 (Three distinct closure levels). *Assume $\text{trdeg}(F/k)$ is infinite and that Γ admits $g > 0$ and λ with $\lambda(g) = 1$. Then*

$$F[L] \subsetneq M \subsetneq \mathcal{C}_{\text{fg}}(F/k; \Gamma) \subsetneq E. \quad (20.4)$$

Moreover, the second strict inclusion has a differential-transcendental witness over M , and the third has one over $\mathcal{C}_{\text{fg}}(F/k; \Gamma)$.

Proof. The first strictness is [example 17.4](#), the second is [corollary 19.5](#), and the third follows from a single sequence of algebraically independent coefficients in [theorem 20.2](#). $\square$

21 Exact transcendence degrees for Laurent-series composita

Theorem 21.1 (Maximal Laurent transcendence). *Let $k \subseteq F$ be set fields of characteristic zero. Suppose*

$$|k| \leq \mathfrak{c} = 2^{\aleph_0}, \quad \text{trdeg}(F/k) > 0.$$

Inside $E = F((t))$, put $M = Fk((t))$ and use $\partial = t d/dt$, acting trivially on F . Then

$$\boxed{\text{dtrdeg}(E/M) = \text{trdeg}(E/M) = |F|^{\aleph_0}.} \quad (21.1)$$

No continuum hypothesis or other cardinal-arithmetic hypothesis is imposed.

Proof. Write $\mu = |F|$, an infinite cardinal. The set of Laurent series has cardinality $\mu^{\aleph_0}$: a series is specified by an integer lower bound and a sequence of coefficients, and arbitrary coefficient sequences already occur in $F[[t]]$. Thus

$$\text{dtrdeg}(E/M) \leq \text{trdeg}(E/M) \leq \mu^{\aleph_0}.$$

The first inequality also follows directly from the definition: the zeroth jets of a differentially independent family are algebraically independent.

If $\mu \leq \mathfrak{c}$, then

$$\mathfrak{c} \leq \mu^{\aleph_0} \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c},$$

so $\mu^{\aleph_0} = \mathfrak{c}$. Choose a transcendental $x \in F$. The continuum family from [corollary 19.4](#) remains differentially independent over M by [corollary 18.4](#), with $\Gamma = \mathbb{Z}$ and $g = 1$. This reaches the upper bound.

If $\mu > \mathfrak{c}$, let $\mathcal{T}$ be a transcendence basis for F/k . A field algebraic over $k(\mathcal{T})$ has cardinality

$$\max\{|k|, |\mathcal{T}|, \aleph_0\}.$$

This follows by counting polynomials and their finitely many roots. Since $|k| \leq \mathfrak{c} < \mu$, it follows that $|\mathcal{T}| = \mu$. The tree family from [corollary 20.4](#), with $\kappa = \mu$, has size $\mu^{\aleph_0}$ and is differentially independent even over $\mathcal{C}_{\text{fg}}(F/k; \mathbb{Z})$, hence over M . Again it reaches the upper bound. The two cases prove both equalities. $\square$

Corollary 21.2. *For $k = \mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$ and x transcendental over k ,*

$$\text{dtrdeg}(k(x)((t))/k((t))(x)) = \text{trdeg}(k(x)((t))/k((t))(x)) = \mathfrak{c}.$$

More generally, [theorem 21.1](#) applies to every positive-transcendence-degree extension of any of these three fields, irrespective of the cardinality of F .

21.1 Boundary cases that must not be suppressed

If F/k is finite, [corollary 17.5](#) gives $E = M$, so both transcendence degrees are zero. An infinite algebraic coefficient extension is not covered by either the finite-extension conclusion or [theorem 21.1](#); the formula is not silently extended to that case.

If $\Gamma = 0$, there is no outer series variable and $E = M = F$. If Γ is nonzero but no specified map $\lambda : \Gamma \rightarrow k$ takes a chosen positive element to 1, the particular differential statements here have not been formulated for it. The purely algebraic finite-rank test still applies. For divisible groups and

characteristic-zero coefficient fields, suitable maps can often be chosen by linear algebra over $\mathbb{Q}$, but the theorems keep the needed map explicit.

Characteristic zero is substantive. In characteristic $p > 0$, the standard Euler derivation on $F((t))$ satisfies

$$\partial^p = \partial,$$

because $n^p = n$ in the prime field. Every element then satisfies the differential polynomial $Y^{(p)} - Y^{(1)} = 0$. The differential-independence conclusion is therefore impossible for that derivation in positive characteristic. This does not settle the corresponding ordinary algebraic-transcendence problem.

The hypothesis $|k| \leq \mathfrak{c}$ enters only in the cardinal upper-bound matching when the coefficient transcendence degree is small. Both independence constructions themselves work over every characteristic-zero base field. For larger k , they still give their explicit lower bounds, but it is not claimed that the bounds always equal $|F|^{\aleph_0}$.

22 Separated surreal copies and explicit omnific witnesses

The set-local results now apply to actual surreal normal forms. The source reproduces the explicit two-copy construction of [section 7](#) to make its later proofs independent of any unreviewed general class-embedding theorem, in particular of [theorem 3.4](#). Only normal-form algebra and the elementary order comparison are needed.

22.1 The inner and outer levels, printed once

[merge] The source's Subsections 8.1 and 8.2 re-derive, with the same proofs, material printed in [sections 7](#) and [8](#); it is printed once, there, and the source is credited as an independent re-derivation. The correspondence is: the order isomorphism ψ and its surjectivity proof are [lemma 7.1](#); the intervals $I_j = (3j, 3j + 1)$, the inner maps h_j of [\(7.1\)](#), their non-unitality [\(7.2\)](#) and the description of $G_j = h_j(\mathbf{No})$ as all normal forms supported in I_j are in [section 7](#) (with $\alpha = j \in \{0, 1\}$); the outer lifts Φ_j of [\(7.3\)](#) with images $K_j = \mathcal{K}_{\mathbb{R}}(G_j)$ are [theorems 4.1](#) and [7.2](#); the scale separation is [\(8.1\)](#); and the re-expansion is [lemma 8.1](#), whose proof the source repeats. The source adds the following, each immediate from those statements.

Conway's omnific integers are

$$\mathbf{Oz} = \left\{ \sum_{e>0} r_e \omega^e + n : r_e \in \mathbb{R}, n \in \mathbb{Z}, \text{ supp reverse well ordered} \right\}. \quad (22.1)$$

A *purely infinite* omnific integer has constant term zero; the term does not require every coefficient to be positive. Since h_j preserves the sign and zero of an exponent, Φ_j preserves and reflects the condition in [\(22.1\)](#) ([proposition 4.2](#)), so

$$R_j := \Phi_j(\mathbf{Oz}) = K_j \cap \mathbf{Oz}, \quad R_0 \cap R_1 = \mathbb{Z}, \quad (22.2)$$

as in [theorem 7.2](#), and Φ_j preserves and reflects purely infinite elements. None of this says that Φ_j commutes with the unary omega-map: $\Phi_0(\omega) = \omega^{\omega^{1/2}} \neq \omega^{\Phi_0(1)} = \omega$ ([warning 4.5](#)). By [\(8.1\)](#), $G_0 + G_1$ is a direct sum as a group, ordered lexicographically with G_1 dominant. With

$$E = \mathcal{K}_{\mathbb{R}}(G_0 + G_1), \quad M = K_0 K_1,$$

as in [Theorem C](#), regrouping by the G_1 -coordinate ([lemma 8.1](#)) and using $t^\gamma = \omega^{-\gamma}$ identifies

$$E \simeq K_0((t^{G_1})), \quad K_1 \simeq \mathbb{R}((t^{G_1})). \quad (22.3)$$

Here the coefficient field K_0 is a class, but each individual outer series uses only a set of coefficients.

22.2 The exact test for the explicit copies

Corollary 22.1 (The exact test for these copies). *Expand $z \in E$ as a Hahn series in G_1 with coefficients in K_0 . Then $z \in M$ if and only if some nonzero $q \in E$ has both its coefficient space and the coefficient space of qz finite-dimensional over $\mathbb{R}$.*

Proof. Apply the coefficientwise proof of [theorem 17.2](#) in the coordinates (22.3). Every tensor, finite-dimensional span, and fraction used in that proof is set-sized even though the ambient fields are classes. $\square$

22.3 A concrete Euler derivation

Define

$$g = \omega^{7/2} \in G_1, \quad b = \omega^{15/4} \in G_1, \quad \tau = \omega^{-g}. \quad (22.4)$$

Here τ is the single cyclic monomial indexed by g in the full small- t notation $t^\gamma = \omega^{-\gamma}$, which is retained only for the full iterated Hahn field. The element g is $h_1(1)$, the same g as in the proof of [Theorem C](#).

Let $\lambda : G_1 \rightarrow \mathbb{R}$ extract the coefficient of the inner monomial $\omega^{7/2}$ from an element of G_1 . Coefficient extraction is additive, with

$$\lambda(g) = 1, \quad \lambda(b) = 0.$$

The associated derivation Δ on E kills K_0 and, in Conway notation, satisfies

$$\Delta(\omega^{\gamma_0 + \gamma_1}) = -\lambda(\gamma_1)\omega^{\gamma_0 + \gamma_1} \quad (\gamma_j \in G_j). \quad (22.5)$$

In particular $\Delta\tau = \tau$ and $\Delta\omega^b = 0$. It preserves both K_1 and M . By [lemma 16.2](#) it is a strong derivation for Hahn summation. It does not alter support exponents, and therefore maps $E \cap \mathbf{Oz}$ into its purely infinite ideal: any constant term is killed.

This is an explicitly specified Euler derivation adapted to the separated coordinates. It is not asserted to be a Hardy-type derivation, the Berarducci–Mantova derivation, or compatible with surreal exponentiation [[11](#), [12](#)].

22.4 The countably supported family

For every ordinal α and integer $n \geq 1$, let

$$\beta_{\alpha,n} = \omega \cdot \alpha + n, \quad g_{\alpha,n} = \omega^{\psi(\beta_{\alpha,n})}, \quad a_{\alpha,n} = \omega^{g_{\alpha,n}}. \quad (22.6)$$

The product and sum in the label $\beta_{\alpha,n}$ are ordinary *ordinal* operations. All arithmetic in $g_{\alpha,n}$, $a_{\alpha,n}$ and subsequent exponent formulas is surreal-field arithmetic. Distinct pairs (α, n) give distinct ordinal labels: the finite tail after $\omega \cdot \alpha$ lies strictly before $\omega \cdot (\alpha + 1)$.

Each $g_{\alpha,n}$ lies in G_0 . The family of all these $g_{\alpha,n}$ is linearly independent over $\mathbb{Q}$, because their inner normal forms are distinct monomials. Hence the family of all $a_{\alpha,n}$ is algebraically independent over $\mathbb{R}$: distinct monomials in any finite polynomial produce distinct sums of the $g_{\alpha,n}$ as outer exponents, and cannot cancel. [merge] For $\alpha = 0$ these are the $g_n = g_{0,n}$ and $s_n = a_{0,n}$ of the proof of [Theorem C](#), with the same independence argument.

Theorem 22.2 (Explicit omnific differential freedom). *Define*

$$Y_\alpha = \omega^b \sum_{n \geq 1} a_{\alpha,n} \omega^{-ng} \quad (22.7)$$

$$= \sum_{n \geq 1} \omega^{b-ng+g_{\alpha,n}}. \quad (22.8)$$

Then:

(i) Y_α is a purely infinite omnific integer in E whose support, read in decreasing order, has order type ω .

(ii) Its jets are

$$\Delta^j Y_\alpha = \sum_{n \geq 1} n^j \omega^{b-ng+g_{\alpha,n}} \quad (j \geq 0). \quad (22.9)$$

(iii) The family $(Y_\alpha)_{\alpha \in \mathbf{On}}$ is differentially algebraically independent over $\mathcal{C}_{\text{fg}}(K_0/\mathbb{R}; G_1)$, hence over M , in the finite-relation sense.

(iv) For every set cardinal κ , the subfamily indexed by $\alpha < \kappa$ has cardinality κ and has the same independence property.

Proof. The inner leading monomial of b has exponent $15/4$, that of g has exponent $7/2$, and all the $g_{\alpha,n}$ have inner leading exponent below 1. Consequently

$$b - ng + g_{\alpha,n} > 0$$

for every finite n . Moreover

$$(b - ng + g_{\alpha,n}) - (b - (n+1)g + g_{\alpha,n+1}) = g + g_{\alpha,n} - g_{\alpha,n+1} > 0$$

by (8.1). Thus (22.8) is a reverse-well-ordered normal form with distinct positive exponents indexed in decreasing order by the positive integers. Its coefficients are all 1, and it has no constant term. This proves (i).

Formula (22.5) gives eigenvalue n on the n th summand, since $\lambda(b) = 0$, $\lambda(g) = 1$, and Δ kills the G_0 -part. Strongness gives (22.9).

The coefficient sequences $n \mapsto a_{\alpha,n}$ are disjoint blocks in an algebraically independent family. Apply [theorem 20.2](#) and [proposition 20.5](#) to the unscaled sums in (22.7), using $k = \mathbb{R}$, $F = K_0$, and $\Gamma = G_1$. The multiplier ω^b is a nonzero member of M with $\Delta\omega^b = 0$. Multiplying every jet by ω^b is an invertible substitution over the base, so the scaled family remains independent. This proves (iii). Part (iv) follows by restriction; independence in particular implies distinctness. $\square$

[merge] Since $Y_0 = \omega^b S$ with S the witness (8.3) and $\omega^{-b} \in K_1$ is a Δ -constant, part (iii) contains the statement that S is differentially transcendental over $\mathcal{C}_{\text{fg}}(K_0/\mathbb{R}; G_1)$, a strengthening of [theorem 8.3](#) for this S in the real case. [Theorem 8.3](#) itself needs neither characteristic-zero derivations nor the explicit coefficients: it holds for any algebraically independent s_n and also over $\mathbb{C}$.

Corollary 22.3 (Countable support is the sharp cardinal threshold). *Every element of E with finite Conway normal-form support belongs to M . Nevertheless, $E \cap \mathbf{Oz}$ contains countably supported purely infinite elements differentially transcendental over M . Thus $\aleph_0$ is the least possible support cardinality for this phenomenon in the explicit join.*

Proof. A finite-support element is a finite sum of terms

$$c\omega^{\gamma_0+\gamma_1} = (c\omega^{\gamma_0})\omega^{\gamma_1} \in K_0K_1 \quad (\gamma_j \in G_j).$$

Apply [theorem 22.2](#) for the countable examples. The threshold is about support cardinality, not birthday or cofinality in the value group. $\square$

Remark 22.4 (Algebraic enlargements of the base). Every finite family of jets in [theorem 22.2](#) remains algebraically independent after any algebraic base extension within a common ambient extension field. Indeed, for a finite tuple $\mathbf{y}$, if N/M is algebraic, then $N(\mathbf{y})/M(\mathbf{y})$ is algebraic; transcendence degree in the two towers gives $\text{trdeg}(N(\mathbf{y})/N) = \text{trdeg}(M(\mathbf{y})/M)$. Only finitely many algebraic coefficients are involved in a proposed relation. In particular, the witnesses remain differentially independent over the relative algebraic closure of M in E , with the induced derivation: differentiating a minimal polynomial shows that this relative algebraic closure is stable under the derivation. This observation is not a computation of that closure.

23 A fixed tail survives every set of parameters

The class-sized strength of the preceding construction is not just that its indices run through the ordinals. Set-supported normal forms allow every set of parameters to be localized to a bounded collection of inner supports. All sufficiently late coefficient labels escape that collection.

Lemma 23.1 (Coefficient localization for a parameter set). *For every set $P \subseteq E$, there is a set $\Theta \subseteq (0, 1)$ such that, if G_Θ is the additive group of all inner surreal normal forms supported in Θ and*

$$C_\Theta = \mathcal{K}_{\mathbb{R}}(G_\Theta) \subseteq K_0,$$

then G_Θ and C_Θ are sets and

$$P \subseteq C_\Theta((t^{G_1})). \quad (23.1)$$

Proof. Expand each element of P in the iterated coordinates [\(22.3\)](#). Collect all its outer coefficients in K_0 . This is a set, since P and all its supports are sets. Expand each of these coefficients in its Conway normal form, whose exponents lie in G_0 , and collect the resulting set of exponents. Expand each such exponent once more in its inner normal form. Its support is a subset of $(0, 1)$. The union Θ of all these supports is again a set.

Every collected exponent belongs to G_Θ , so every original outer coefficient belongs to C_Θ . This gives [\(23.1\)](#). The class of inner normal forms supported in a fixed set Θ injects into the set of functions $\Theta \rightarrow \mathbb{R}$, and is closed under addition and negatives; it is therefore a set-sized additive group. Likewise the full Hahn field on the set group G_Θ injects into the set of functions $G_\Theta \rightarrow \mathbb{R}$ and is a set. All these statements use ordinary replacement and union, not a class-sized support operation. $\square$

[merge] This is the support-closure step of [corollary 6.1](#), applied to the outer coefficients and then to their exponents: there the parameters are placed in $\mathcal{K}_{\mathbb{R}}(H)$ with H the rational span of their supports; here their outer coefficients are placed in $\mathcal{K}_{\mathbb{R}}(G_\Theta)$.

Lemma 23.2 (A tail is independent over the localized coefficient field). *For Θ and C_Θ as above, there is an ordinal α_0 such that the family*

$$\{a_{\alpha,n} : \alpha \geq \alpha_0, n \geq 1\}$$

is algebraically independent over C_Θ .

Proof. Because ψ is injective, the ordinals β satisfying $\psi(\beta) \in \Theta$ form a set: each member of Θ has at most one inverse image, and replacement collects the ordinal inverse images. Choose α_0 greater than every such ordinal. If $\alpha \geq \alpha_0$, then $\beta_{\alpha,n} = \omega \cdot \alpha + n \geq \alpha$ and hence $\psi(\beta_{\alpha,n}) \notin \Theta$.

The corresponding $g_{\alpha,n}$ are distinct inner monomials supported outside Θ . Every nonzero rational linear combination of finitely many of them still has a nonzero coefficient outside Θ , and thus does not belong to G_Θ . They are therefore linearly independent over $\mathbb{Q}$ modulo G_Θ .

In a nonzero polynomial in finitely many $a_{\alpha,n}$ with coefficients in C_Θ , the support of each coefficient is contained in G_Θ . Multiplication by a monomial in the $a_{\alpha,n}$ translates that support by the corresponding integer linear combination of the $g_{\alpha,n}$. Distinct polynomial monomials produce distinct cosets modulo G_Θ . Their supports are consequently disjoint, so their nonzero terms cannot cancel. This proves algebraic independence over C_Θ . $\square$

Theorem 23.3 (Set-parameter tail resilience). *For every set $P \subseteq E$, there is an ordinal α_0 such that the predefined tail*

$$(Y_\alpha)_{\alpha \geq \alpha_0}$$

is differentially algebraically independent over $M\langle P \rangle$. In fact it is independent over the larger field

$$C_{\text{fg}}(K_0/C_\Theta; G_1) \tag{23.2}$$

for a set coefficient field C_Θ supplied by lemma 23.1.

Proof. Choose Θ , C_Θ , and α_0 from the preceding lemmas. Apply the private coefficient-block theorem with base field C_Θ , coefficient field K_0 , outer group G_1 , and the tail coefficients $a_{\alpha,n}$. Its hypotheses hold by lemma 23.2. The Euler weight map takes its values in $\mathbb{R} \subseteq C_\Theta$, so it is valid over this enlarged base. The set-local formulation proposition 20.5 applies because C_Θ is a set. This gives differential independence of the unscaled tail over (23.2), and then of the Y_α after multiplication by ω^b .

It remains to verify inclusion of the desired parameter field. The envelope (23.2) contains every element of K_0 as an outer constant, since an individual element generates a finitely generated extension of C_Θ . It contains K_1 because its coefficients lie in $\mathbb{R} \subseteq C_\Theta$. It contains P by (23.1). Finally it is stable under Δ , by the proof of lemma 16.5. Therefore it contains $M\langle P \rangle$, and independence descends to that smaller base. $\square$

Corollary 23.4 (No set-sized algebraic or differential generation). *For no set $P \subseteq E$ is E algebraic over $M(P)$. For no such P is E differential-algebraic over $M\langle P \rangle$. In particular, no set of ordinary or differential generators, even followed by the respective algebraic closure operation within E , generates E over M .*

Proof. A member of the tail supplied by theorem 23.3 is differentially transcendental over $M\langle P \rangle$ and therefore algebraically transcendental over its subfield $M(P)$. This contradicts either asserted algebraicity. $\square$

Remark 23.5 (Why the set qualification is sharp for the method). A proper class of parameters could use every inner support label $\psi(\beta)$. The localization set Θ would then no longer be a set, and the ordinal bound in [lemma 23.2](#) would not be available. The theorem is a genuine set-parameter result, not a statement about arbitrary proper-class parameter collections. It also does not assert independence over a differential-algebraic closure of $M\langle P \rangle$; that stronger base-change statement is unnecessary for [corollary 23.4](#).

24 Arithmetic joins and the surcomplex extension

24.1 Fraction fields of the three integer rings

Let $R_j = K_j \cap \mathbf{Oz}$ as in [\(22.2\)](#). Let

$$R_0[R_1] = \text{the subring generated by } R_0 \cup R_1, \quad O = E \cap \mathbf{Oz}. \quad (24.1)$$

Thus $R_0[R_1]$ is formed by finite ring operations, whereas O permits every set-supported omnific normal form whose exponents lie in $G_0 + G_1$. Both are class domains. Fractions and integrality are interpreted by their ordinary finite formulas, as in [section 9](#).

Lemma 24.1 (Monomial denominator clearing). *One has*

$$\text{Frac}(R_j) = K_j, \quad \text{Frac}(R_0[R_1]) = M, \quad \text{Frac}(O) = E. \quad (24.2)$$

Proof. For $x \in K_j$, its support is a set in G_j . Pull it back through the ordered-group isomorphism $h_j : \mathbf{No} \rightarrow G_j$. Every set of surreal numbers has an upper bound, so choose $d \in G_j$, $d > 0$, with $d + e > 0$ for every e in the support of x . Both ω^d and $\omega^d x$ have only positive exponents, and hence belong to R_j . Thus $x = (\omega^d x)/\omega^d$ is a fraction of elements of R_j . The reverse inclusion is immediate.

The fraction field of $R_0[R_1]$ contains both K_j and is contained in their compositum, giving $\text{Frac}(R_0[R_1]) = M$.

Finally take $x \in E$. Project its set of support exponents to a set in G_1 . Choose $d \in G_1$, $d > 0$, strictly greater than the negatives of all these projections. If $\gamma_0 + \gamma_1$ occurs in x , then $d + \gamma_1$ is positive in G_1 and dominates γ_0 by [\(8.1\)](#). Hence $d + \gamma_0 + \gamma_1 > 0$. The same denominator ω^d clears x into O , proving $\text{Frac}(O) = E$. $\square$

The first clearing mechanism is already part of this report's omnific fraction theory ([lemma 9.1](#), and $\text{Frac}(R_\alpha) = K_\alpha$ in [proposition 9.2](#)). It is included here because the exact fraction field of the *two-copy generated ring* is needed in the next result.

Theorem 24.2 (Arithmetic failure cannot be repaired by a set). *With $R_0[R_1] \subseteq O$ as in [\(24.1\)](#):*

- (i) *For every set $J \subseteq \mathbf{On}$, substitution $X_\alpha \mapsto Y_\alpha$ embeds the polynomial $R_0[R_1]$ -algebra in the variables $(X_\alpha)_{\alpha \in J}$ into O .*
- (ii) *For every set $S \subseteq O$, the extension $O/R_0[R_1][S]$ is not integral. In fact E is not algebraic over $\text{Frac}(R_0[R_1][S]) = M(S)$.*
- (iii) *The obstructing element in (ii) can be chosen from a tail of the same family [\(22.8\)](#); it has purely positive, countable Conway support.*

Proof. Since $R_0[R_1] \subseteq M$, algebraic independence over M implies the injectivity in (i); the image lies in O by [theorem 22.2](#). For (ii), apply [theorem 23.3](#) with $P = S$. Any tail member is transcendental over $M(S)$, while [lemma 24.1](#) gives $\text{Frac}(R_0[R_1][S]) = M(S)$. If it were integral over $R_0[R_1][S]$, a monic polynomial over that ring would make it algebraic over $M(S)$, a contradiction. The same element proves (iii). $\square$

Remark 24.3 (Not a normalization calculation). Nonintegrality of $O/R_0[R_1][S]$ does not compute the integral closure of $R_0[R_1]$ in M or in E . Nor does it compute a Picard group, a class group, or a projective module. The theorem isolates a much larger transcendental obstruction that no set of arithmetic parameters removes. Questions about normalization inside the fixed fraction field remain distinct.

24.2 Surcomplex and Gaussian omnific versions

Put

$$E_{\mathbb{C}} = E[i], \quad K_{j,\mathbb{C}} = K_j[i], \quad M_{\mathbb{C}} = M[i].$$

Because E is ordered, adjoining i gives a genuine quadratic field extension. Separating real and imaginary coefficients identifies these fields with the corresponding full complex-coefficient Hahn fields on the same groups. The union of two set supports is still a set, so this identification preserves the Hahn conventions.

Extend Δ by $\Delta i = 0$. Define

$$O_{\mathbb{C}} = E_{\mathbb{C}} \cap \mathbf{Oz}[i] = O[i].$$

The Gaussian integer images are $R_j[i]$, and the ring they generate is $R_0[R_1][i]$.

Theorem 24.4 (Surcomplex independence and Gaussian witnesses). *The following statements hold.*

- (i) *The real family (Y_{α}) remains differentially algebraically independent over $M_{\mathbb{C}}$ in $E_{\mathbb{C}}$.*
- (ii) *The genuinely nonreal elements*

$$Z_{\alpha} = (1 + i)Y_{\alpha} \tag{24.3}$$

belong to $\mathbf{Oz}[i]$, have countable support, and are differentially algebraically independent over $M_{\mathbb{C}}$.

- (iii) *For every set $P \subseteq E_{\mathbb{C}}$, a tail of each of these families is differentially algebraically independent over $M_{\mathbb{C}}\langle P \rangle$.*
- (iv) *$\text{Frac}(R_0[R_1][i]) = M_{\mathbb{C}}$ and $\text{Frac}(O_{\mathbb{C}}) = E_{\mathbb{C}}$. For every set $S \subseteq O_{\mathbb{C}}$, the ring $O_{\mathbb{C}}$ is not integral over $R_0[R_1][i][S]$.*

Proof. For a polynomial relation over $M[i]$ among real jets, write each coefficient uniquely as $a + ib$, $a, b \in M$. Its real and imaginary parts give polynomial relations over M . At least one is a nonzero polynomial if the original was nonzero. This contradicts [theorem 22.2](#), proving (i). Multiplication by the nonzero constant $1 + i$ is an invertible substitution on all jets, proving (ii).

For (iii), take the set $P_{\mathbb{R}} \subseteq E$ consisting of the real and imaginary parts of all members of P . Apply [theorem 23.3](#) to $P_{\mathbb{R}}$. Splitting coefficients as above gives independence over $M\langle P_{\mathbb{R}} \rangle[i]$, which contains $M_{\mathbb{C}}\langle P \rangle$. Again rescale to obtain the Z_{α} version.

For (iv), the displayed fraction fields follow from [lemma 24.1](#) after adjoining i , or by clearing real and imaginary parts with a common monomial. A tail element supplied by (iii) is transcendental over $\text{Frac}(R_0[R_1][i][S]) = M_{\mathbb{C}}(S)$, so cannot be integral over $R_0[R_1][i][S]$. $\square$

No statement here depends on extending analytic functions to all surcomplex arguments, choosing a complex logarithm, or assuming an exponential-compatible derivation. The passage from real to complex is finite algebraic extension and coefficient separation.

25 What the addition resolves, and what it does not

25.1 Relation to Research question 3

Research question 3 has two parts: an exact description of the ordinary compositum and a dimension calculation for the Hahn join over it. The results of the addition give the following answers in specified domains.

Problem component	Conclusion proved by source 02
Membership in separated coordinates	The finite coefficient-rank denominator criterion, theorem 17.2 and corollary 22.1 .
Finite coefficient-field condition	Not sufficient even for algebraicity; a witness uses only $k(x)$, corollary 19.5 .
Natural set-sized dimensions	Exact algebraic and differential dimensions for $F((t))/Fk((t))$ under the hypotheses of theorem 21.1 .
Explicit class-sized join	Independent families of every set cardinal, with countable purely infinite supports, theorem 22.2 .
Dependence on set parameters	A fixed tail remains independent after any set adjunction, theorem 23.3 .
Arithmetic and complex forms	Nonintegrality persists after set adjunction; the same phenomena occur in $\mathbf{Oz}[i]$, theorems 24.2 and 24.4 .

These answers do not solve the complete problem for arbitrary non-separated ordered amalgams of exponent groups, arbitrary coefficient-field cardinalities, and arbitrary prescribed common Hahn cores. The rank criterion applies whenever an iterated coefficient interpretation has been established. The cardinal theorem does not assign a set cardinal to the class extension. The question addressed is a question posed in this report, not a claimed resolution of a separately established, long-standing published conjecture.

25.2 Provenance and proof dependencies

The source compared itself with this report at commit `7b256e0792df7718995b816ad12a77bf56731f8f`, where the report's source had blob identifier `7d5cfe644e7c902f2b594179c337a6c633350fc3`. The key portions consulted were the explicit-copy and re-expansion material of [sections 7](#) and [8](#) and the question list of [section 12](#). The report's README and the collection's main catalogue were also consulted. The source's audit (shipped as `02-composita-SOURCE_AUDIT.md`) records the paths and distinguishes sources read for substance from sources consulted only for orientation.

The explicit maps [\(7.1\)](#) and [\(7.3\)](#), the scale separation, and the finite coefficient-field observation are credited to this report. The coset-slice argument belongs to the same elementary Hahn toolkit and is not presented as a priority claim. The proofs of the new class conclusions do not import this report's general transfinite independent-copy recursion ([theorem 3.4](#)), nor, in the source's words, the repository's proposed automatic-strongness results, curve-rigidity results, or unreviewed quotient theorems.

The main dependency chains are

rapid degrees $\implies$ continuum family $\implies$ small-cardinality case,
 coefficient derivations $\implies$ private blocks $\implies$ tree amplification,
 tree amplification $\implies$ large-cardinality case,
 separated copies and private blocks $\implies$ omnific witnesses,
 inner-support localization $\implies$ tail resilience,
 tail resilience $\implies$ arithmetic and complex consequences.

No numerical experiment replaces any arrow in these chains.

25.3 Critical proof checks

The rapid-degree argument depends on more than comparing leading terms informally. The polynomial relation is made integral in the outer variable before valuations are compared. Minimal total jet degree guarantees a nonzero evaluated partial derivative. The selected coefficient lies at $n + \ell < 2n$, so all quadratic tail terms vanish there. Partial coefficients below ℓ vanish, so later tail coefficients cannot interfere. Finally the coefficient degree ν is fixed even though the outer coefficient series are arbitrary Laurent series.

The private-block argument depends on a different finiteness principle. A finitely generated field D is algebraic over a field generated by finitely many elements of a transcendence basis. Coordinate derivations outside that finite set kill D after separable extension. This is why the argument controls whole coefficient series in $D((t^\Gamma))$, not merely their leading coefficients. Characteristic zero is used both for separability and for distinct integer eigenvalues.

The class argument uses two levels of support localization. An outer coefficient is a surreal series in exponents from G_0 ; those exponents themselves have inner supports in $(0, 1)$. Taking the union of the inner supports for a *set* of parameters produces a set Θ . The predefined ordinal coefficient labels outside Θ remain independent modulo G_Θ . Replacing this argument with an unproved claim that “a class is larger than a set” would not establish the required independence over the parameter field.

The source’s proof audit (shipped as `02-composita-PROOF_AUDIT.md`) adds three points. The auxiliary coordinate derivations of [theorem 20.2](#) need not preserve any *inner* surreal summation; they are extended coefficientwise only in the explicitly used *outer* Hahn representation. The rapid-degree argument does not require sparse outer support, but it does require a uniform finite degree in x after denominators are cleared. In [theorem 22.2](#) the $g_{\alpha,n}$ are inner monomials in $(0, 1)$ and the $a_{\alpha,n}$ are independent outer coefficient monomials; the Euler derivation has weight $-\lambda(\gamma_1)$ in Conway notation, so the n th witness term has eigenvalue n , and the shift multiplier has derivative zero.

25.4 Limitations and nonclaims, as stated by the source

The proofs are mathematical arguments, not Lean-checked results. The finite program supplied with the source tests identities and finite models only. Independent review should prioritize [theorems 19.2](#), [20.2](#) and [23.3](#). The source search did not certify novelty or exhaustive priority. The adjective “strong” refers to Hahn summability, not convergence in an order topology.

No claim is made about Gonshor-exponential compatibility, preservation of simplicity, initiality of the two images, birthday optimality, or the Berarducci–Mantova derivation. The full algebraic

closure of the compositum, its normalization, and its projective modules are not calculated. No general algorithm deciding compositum membership is proved. These distinctions are mathematically relevant, not merely editorial qualifications.

25.5 What the addition does not claim

[merge] This list collects every limitation stated by source 02, in its article, delivered README, source and proof audits, recorded run and build audit, with its location here, followed by those added in the merge.

- C1. The rank test is not a decision procedure: the existential search over q is an infinite-input problem, and no algorithm deciding arbitrary Hahn-series membership follows (remark 17.3; Research question 16).
- C2. The tensor fact is not new: it is a standard linear-disjointness mechanism. No new tensor-product theorem and no general theory of rational series is claimed (remark 17.3; source audit).
- C3. The comparison with Krapp–Kuhlmann–Serra [10] is conceptual, not a claim that the formulations have identical inputs; that paper was used for the literature comparison, not as a proof of the compositum theorems (remark 17.3; bibliography).
- C4. Infinite coefficient rank obstructs membership in $F[L]$, not in M ; a geometric series of infinite rank lies in M (example 17.4 and corollary 17.5).
- C5. Base extension is not treated as harmless; coefficient extension uses coefficientwise independence of constants, intermediate base change clears denominators, and exponent-group extension uses coset slices, not an unjustified claim that arbitrary base extension preserves algebraic independence. No convexity is required for the coset slice, but the surreal iterated presentation requires the explicit scale separation (section 18; proof audit).
- C6. The rapid-degree theorem does not rest on gaps in the outer support; merely pointing to such gaps would not prove example 19.6. It requires characteristic zero, a transcendental coefficient variable, unbounded ratios d_n/d_{n-1} , infinitely many private indices relative to each finite subfamily, and a uniform finite x -degree after clearing denominators (theorem 19.2 and remark 19.3; proof audit).
- C7. The formula $|F|^{\aleph_0}$ is not extended to infinite algebraic coefficient extensions, to $\Gamma = 0$, to groups without a specified λ with $\lambda(g) = 1$ (only the algebraic rank test applies there), or to $|k| > \mathfrak{c}$ with small positive transcendence degree, where only the explicit lower bounds are claimed; the rapid-degree family need not reach the cardinality of the Laurent field for large k (section 21.1; Research questions 12 and 13).
- C8. Characteristic zero is essential for the differential statements: in characteristic p the Euler derivation satisfies $\partial^p = \partial$. This does not settle the ordinary algebraic-transcendence problem in characteristic p (section 21.1).
- C9. The derivation Δ is an explicitly specified diagonal Euler derivation. It is not asserted to be Hardy type, the Berarducci–Mantova derivation, or compatible with exponentiation. It is proved to be a derivation directly and is not identified with a Hardy-type derivation of Kuhlmann–Matusinski [11], whose Hardy-type theory the source did not review in full (sections 15.3 and 22.3; source audit; Research question 17).

- C10. The explicit copies do not commute with the omega-map, and strong Hahn summability alone does not provide compatibility with it or with the exponential (section 22.1; Research question 18).
- C11. Remark 22.4 is not a computation of the relative algebraic closure of M in E .
- C12. The class results use only finite relations, set-indexed families and set-parameter resilience. No class transcendence basis is chosen, no set cardinal is assigned to the proper-class transcendence degree, and no cardinal equal to the size of $\mathbf{No}$ is postulated. No class-state recursion or class satisfaction predicate is used (section 16; proposition 20.5; delivered README; proof audit).
- C13. Tail resilience is for *set* parameters only; proper-class parameter collections are not covered. Independence is not asserted over the differential-algebraic closure of $M\langle P \rangle$ (remark 23.5).
- C14. Nonintegrality of $O/R_0[R_1][S]$ does not compute the integral closure of $R_0[R_1]$ in M or E , a normalization, a Picard group, a class group or a projective module (remark 24.3; Research question 19).
- C15. The complex versions use only finite algebraic extension and coefficient separation: no analytic continuation to surcomplex arguments, no complex logarithm, no exponential-compatible derivation (section 24).
- C16. Research question 3 is not solved for non-separated ordered amalgams, arbitrary coefficient-field cardinalities, or arbitrary prescribed common Hahn cores; the cardinal theorem assigns no set cardinal to the class extension; the question is this report's own, not a long-standing published conjecture (section 25).
- C17. The explicit maps, scale separation and finite coefficient-field observation are this report's; the coset slice is elementary. The proofs do not import this report's transfinite recursion, nor automatic-strongness, curve-rigidity or quotient theorems of the repository. No numerical experiment replaces a proof step (section 25).
- C18. Nothing is claimed on Gonshor-exponential compatibility, simplicity, initiality, birthday optimality, the full algebraic closure of the compositum, normalization or projective modules; no general membership algorithm is claimed (section 25; Research question 20).
- C19. Status: the proofs are not Lean-checked, not refereed and not independently reviewed; the source's proof audit is author-side, not a referee report; review should prioritize theorems 19.2, 20.2 and 23.3. Priority is not certified: the literature comparison was targeted, a search that finds nothing does not establish novelty, and not all related literature was read (sections 15.3 and 25; title page of the source; source and proof audits; build audit).
- C20. The source's comparison with this report was targeted: it read two line ranges of the pinned source for substance, not every proof of the report. The Stacks Project was used for terminology; the encyclopedia page on surreal numbers for orientation only; L'Innocente-Mantova for context, with no factorization result claimed (source audit; bibliography).
- C21. The research questions are proposed by the proved statements. Except for Research question 3 of this report, they are not represented as previously published open problems (section 26).
- C22. The finite checks are narrower than the theorems. They test polynomial identities, finite coefficient ranks, finite combinatorial examples and finite ordered-group models; they do not implement $\mathbf{No}$ or infinite Hahn summation, do not represent countably supported surreals, and prove no independence, cardinal or proper-class assertion. The lexicographic model is a

model of the comparisons, not an approximation to a surreal number. Neither typesetting nor finite computation is proof verification (section 27; recorded run).

- C23. The formalization decomposition names implementation targets, not declarations that exist (section 27.2).
- M1. Parts (i)–(ii) of theorem 17.2 are `duals:prop:algebraic` of the three-duals report and its preceding injectivity paragraph; part (iii) is new but immediate. The source does not cite that report. Lemma 18.3 is a variant of theorem 5.4 (section 15.6).
- M2. The private-index and tree mechanisms are related to `tail:thm:surreal` and `bst:thm:independent`, which use different bases, derivations and tools, and the tail resilience to `adr:sr:thm:tail`; the source cites none of them (section 15.6).
- M3. Research question 3 is answered only in the separated case with trivial core and in the cardinal cases of theorem 21.1; its remaining part is Research questions 11 and 14. No other question of this report is answered, and no question of another report (section 15.4).
- M4. No `isc:cp:` label has a Lean mapping in `FORMALIZATION.md`. The proofs were read when the addition was merged (Appendix A.7); that reading is not an independent review.
- M5. The recorded run was repeated in the merge on a copy, with identical counts; the shipped `code/02-composita-build.sh` does not run unchanged under the shipped file names (Appendix A.8).

26 Research questions of the addition

The following questions are proposed by the proved statements of source 02. Except for Research question 3, which the source takes from this report, they are not represented as previously published open problems. Each asks for a specific extension, obstruction, or effective refinement. [merge] The source’s Question n is Research question $10 + n$ here.

Research question 11 (Non-separated ordered amalgams). Let G_0, G_1 be independent ordered subgroups but assume neither is infinitesimal relative to the other. Determine the image of $\mathcal{K}_k(G_0)\mathcal{K}_k(G_1)$ inside $\mathcal{K}_k(G_0 + G_1)$ and its transcendence defect. Can a finite-rank denominator criterion be formulated without an iterated Hahn coefficient field?

The obstacle is not the finite tensor argument itself, but choosing slices with support properties strong enough to characterize the entire join. This is the main part of Research question 3 left untreated.

Research question 12 (Very large base fields). For $|k| > \mathfrak{c}$ and $0 < \text{trdeg}(F/k) < |k|$, must

$$\text{trdeg}(F((t))/Fk((t))) = |F|^{\aleph_0}$$

still hold? Does the same equality hold for differential transcendence degree under the outer Euler derivation?

The rapid-degree theorem gives a continuum family over any k , but that family need not reach the cardinality of the entire Laurent field. A stronger combinatorial coefficient construction or a counterexample would settle the gap in the cardinal argument.

Research question 13 (Infinite algebraic coefficient extensions). For an infinite algebraic extension F/k of characteristic zero, characterize $Fk((t))$ inside $F((t))$ and determine the algebraic and differential transcendence degrees of the latter over the former.

Finite extensions give equality of fields, while the transcendental-coordinate derivations no longer apply. Growth in the algebraic degrees needed to contain successive coefficients is a natural replacement invariant, but no theorem about it is claimed.

Research question 14 (Prescribed nonzero common Hahn cores). Can the explicit omnific tail-resilience theorem be extended to independent surreal copies with a prescribed set-sized common field $\mathcal{K}_{\mathbb{R}}(H)$? Which separation conditions on the ordered amalgam over H suffice?

This report already constructs copies with prescribed cores ([Theorem A](#)). What is needed is a compatible local coefficient decomposition and an independent-label family modulo the shared support group, not just an abstract pair of embeddings. [merge] Here H is the common core of [Theorem A](#), as in the source’s question.

Research question 15 (Presentation-independent coefficient rank). Which part of the finite-rank denominator criterion can be recovered from the pair of embedded fields alone, without naming a monomial cross-section or a chosen iterated Hahn presentation?

The compositum is intrinsic, but coefficient rank is a presentation-dependent certificate. An invariant description of the smallest possible certificate spaces could distinguish coordinate artifacts from structure of the pair.

Research question 16 (Effective membership on controlled inputs). For rational, recurrence-defined, automatic, or otherwise effectively presented supports and coefficients, when is the finite-rank denominator test decidable? Can one bound the denominator rank in a useful subclass?

The generalized recurrence framework of [10] is a relevant starting point. A useful algorithm must specify its representation of infinite series and should not treat a formal coefficient oracle as an effective finite input.

Research question 17 (Natural surreal derivations). Do comparable compositum-independence and set-parameter resilience statements hold for a Hardy-type surreal derivation, particularly the Berarducci–Mantova derivation, on subfields stable under it?

The proofs exploit a diagonal derivation that kills the entire lower-scale coefficient field. A natural surreal derivation generally differentiates those coefficients and may not preserve the explicit copy images. Establishing an appropriate stable pair is part of the problem, not an automatic corollary of [theorem 22.2](#). [merge] Research question 6 asks the analogous question for the prescribed-intersection theorem.

Research question 18 (Exponential-compatible copies). Can one obtain a similarly large gap for a pair of proper surreal self-embeddings commuting with the Gonshor exponential, or with the omega-map? Which common cores must then be closed under the additional operation?

The displayed copies fail to commute with the omega-map. Strong Hahn summability alone does not provide the desired compatibility. This question connects the compositum programme to the richer embedding questions already distinguished in this report (Research question 6).

Research question 19 (Normalization within the fraction field). For $R_0[R_1]$ from (24.1), compute or constrain its integral closure inside M . How does that ring compare with $M \cap \mathbf{Oz}$, and what can be said about finitely generated projective modules over it?

The nonintegrality of $O/R_0[R_1][S]$ comes from transcendence outside M and therefore does not answer this within-fraction-field problem. A normalization result would complement rather than restate theorem 24.2. [merge] Research question 4 asks the corresponding questions for the intersection ring $\mathcal{A}_k(H)$.

Research question 20 (Birthday complexity of minimal-support witnesses). Among countably supported omnific elements transcendental over the explicit compositum, what birthday bounds are possible? Can one lower the nested normal-form complexity while retaining set-parameter tail resilience?

The support cardinality $\aleph_0$ is already sharp by corollary 22.3. Birthday optimality is a different invariant and requires genuine estimates on surreal construction lengths.

Research question 21 (Several scales and commuting derivations). For three or more separated copies, construct several commuting Euler derivations and determine algebraic independence of mixed jets over the corresponding ordinary compositum. What is the correct tree-label condition for joint differential independence?

Coefficient derivations still offer finite avoidance, but one must separate multivariable eigenvalue interpolation from the ordered-group support problem. A multidimensional Vandermonde argument alone does not settle the Hahn support conditions.

Research question 22 (Formalization of the core mechanisms). Can the coefficient-rank theorem, the coefficient-extension and coset-slice lemmas, the rapid-degree Taylor argument, and the private-block theorem be formalized as reusable Hahn-field results before specializing to surreal numbers?

A promising order is to formalize the finite polynomial and coefficient-space identities first, then the set-sized Hahn statements, and only afterwards the class-local interface. This would test the most important new proofs without requiring a monolithic formalization of the entire report. [merge] It overlaps Research question 10 in the coset-slice kernel.

27 Finite verification and a formalization decomposition of the addition

27.1 Finite verification and reproducibility

The source's program, shipped as `code/02-composita-verify.py`, performs exact finite checks. Its recorded output, shipped as `data/02-composita-verification.json`, records the actual checks and their counts: 2,530 checks in 21 categories. These checks are deliberately narrower than the theorems. They do not represent countably supported surreal numbers and do not prove an independence or cardinality statement.

The checked mechanisms include finite geometric-series convolution and coefficient ranks, Euler and coefficient-derivation identities on polynomial examples, Vandermonde determinants at distinct integer eigenvalues, finite prefix-tree private-label behaviour, factorial growth inequalities used in (19.5), and a multivariable Taylor coefficient calculation with quadratic tails excluded at the relevant

order. A finite lexicographic exponent model tests the sign and strict descent used in (22.8). It is a model of the comparisons, not a numerical approximation to an actual surreal number.

A characteristic- p check illustrates the explicit boundary $\partial^p = \partial$ on monomials. Finite complex substitution checks confirm that multiplying variables by $1 + i$ is invertible; the proof for arbitrary polynomials remains the elementary field argument above.

The program uses the standard library and SymPy. It writes a file only when given `-output`. The source's build script also supports repeated `pdflatex` passes when `latexmk` is unavailable, and its build audit records the compiler result, page count, reference checks, and layout checks for the delivered PDF. Neither successful typesetting nor finite computation constitutes proof verification. [merge] The shipped file names, the rerun and its comparison with the recorded output are in Appendix A.8.

27.2 A proposed formalization decomposition

The following are implementation targets, not declarations claimed to exist in the collection.

Coefficient-space layer. Represent a finite-dimensional coefficient span and prove that its coordinate functions define Hahn series with supports contained in the original support. Prove tensor injectivity by coefficient comparison. Derive the rank-denominator criterion using the ordinary fraction field universal property.

Support and base-change layer. Define restriction to an exponent coset, prove its compatibility with multiplication by series on the subgroup, and establish the two linear-disjointness transfer lemmas. Make the chosen ambient field explicit to avoid conflating abstract tensor products with a specified compositum.

Polynomial perturbation layer. For a polynomial in finitely many jet variables over $k[[t]][x]$, prove a first-order Taylor congruence modulo t^{2n} after a perturbation divisible by t^n . Prove the coefficient-degree bound $\nu + d d_{n-1}$. These finite statements are the delicate reusable kernel of theorem 19.2.

Derivation and interpolation layer. Formalize coefficientwise Hahn derivations, their commutation with Euler derivations, extension of a derivation over separable algebraic fields, and the finite Vandermonde elimination of evaluated partials. This yields the set-sized private-block theorem.

Cardinal and surreal layer. Add the prefix-tree construction and cardinal counting. For the explicit surreals, prove the two-level normal-form relabeling, lexicographic comparison, and set-support localization. Each proposed relation must be transported into a set-sized field before invoking a transcendence basis. No unrestricted class basis should be introduced just to shorten the proof.

[merge] The coset-slice and iterated-Hahn items overlap items (1)–(3) and (6) of section 13.2.

A Provenance of this report

A.1 One manuscript

This report is built from *one* manuscript: the archive `Surreal_Independent_Copies`, manuscript 06 of batch 31 of the collection, with main file `independent_surreal_copies.tex`, a 25-page PDF, dated 23 September 2026 and pinned to repository commit `bcac55ae6fd3f2b354e568ba1d2e94d496a2cc59`. It was placed in commit `9d28e28` as a new report. It answers no named question of the collection, and its subject, independence, disjointness and descent for full copies of **No**, is the subject of no existing report (section 1.6). There was no second source. Nothing was merged and nothing was selected away: every result, proof, example, question and limitation of the manuscript is printed here. [merge] This subsection describes the report as written in batch 31. In batch 34 a second manuscript was added as source 02 (Appendix A.7); the report is now built from two manuscripts.

The package contained the article, its PDF and a README, and nothing else: no code, recorded data, audit or checksum list. The article was renamed `article.tex` and rewritten as described below; the README was replaced by the collection README; the delivered PDF is not shipped, and `article.pdf` is a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `isc:`. The source's 62 labels are kept, unchanged after the prefix. Twenty were added: `isc:sec:conventions`, `isc:sec:collection`, `isc:sec:nonclaims`, `isc:rem:base`, `isc:rem:floor`, `isc:warn:omega`, `isc:rem:join`, `isc:rem:scaling`, `isc:rem:infinite`, `isc:sub:model`, `isc:sub:bounded`, `isc:sub:formal`, `isc:sub:notmade`, `isc:app:report`, `isc:app:onesource`, `isc:app:pinned`, `isc:app:literature`, `isc:app:checked`, `isc:app:files` and `isc:app:numbers`. Labels on existing environments change no number.
- (2) The title page names the report as a single-source report of the collection, gives the batch and the pin, and extends the status paragraph. Sections 1.5 and 1.6, the ledger of section 14 and this appendix are new. Sentences marked [write] were added at the end of section 1.4; after proposition 4.4, corollary 6.1, theorem 6.2, (7.3), lemma 9.1, proposition 9.2 and proposition 9.3; at the start of section 10; after corollary 10.3; after research questions 1, 2, 6, 7, 8, 9 and 10; and in section 13.3.

No mathematical statement was changed and no symbol was renamed: the collisions are between reports, or between local letters of different sections, and section 1.5 fixes the meanings. Where the edit had to choose: the manuscript was placed as a new report, not as an addition to the omnific-preserving report, because it answers none of that report's questions and its spine (independence, exact intersections and descent) is not that report's (stabilizers, classification and coefficient motion). It was placed among the surreal reports because its fields are subfields of **No** and **No**(*i*). The title is kept.

A.2 Repository statements at the pin

The source compared itself with the catalogue, the omnific-preserving report and the bounded-support report at its pin, through links to that revision ([6, 7, 8]). Checked at the placement commit:

- At the pin the omnific-preserving report was the merged report of four manuscripts subtitled “Convex-scale stabilizers, definable constants, nondefinable monomials, and algebraic-parameter rigidity”. It had no part on embeddings and no `opa:as:` label. The source’s description of it as concerning stabilizers and rigidity was accurate. Its Part III, on automatic strongness and proper embeddings, was written in 20c4c9f, after the pin.
- The bounded-support report is unchanged between the pin and the placement commit (its last change, ce6715b, precedes the pin). The source’s description of it as proposing stronger independent families stands.
- The large-cardinal report was placed in 21375f8, after the pin, and is not cited by the source.
- No report of the collection supplies Theorem A, theorem 5.4, Theorem B for copies, or Theorem C; the overlaps are those of section 1.6.

A.3 References checked for this report

- *L’Innocente–Mantova* [3]. The arXiv page lists version 5 of 22 January 2024 and the journal DOI 10.1016/j.aim.2024.109513; the volume number was not checked. In the arXiv text, Fact 2.1.1 states that a Hahn field is algebraically closed, respectively real closed, if and only if its coefficient field is algebraically closed, respectively real closed, and its group is divisible. Section 2.3 introduces the surreal numbers as a Hahn field. Proposition 2.4.5 is stated for the Hahn field of series with supports of size below a cardinal κ : the fraction field of $\mathbb{Z} + K((G^{<0}))$ is the whole field if and only if $\text{cf}(G) \geq \kappa$ (or $G = 0$ and $\text{Frac } \mathbb{Z} = K$), and in particular $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. For a set-sized G and $\kappa > |G|$ this is the full Hahn field, so its proof, with a lacunary series along a cofinal sequence, gives the non-membership obstruction cited in section 1.4 and section 10. It proves non-membership in the fraction field, not transcendence. The citations are therefore accurate.
- *Ehrlich–Kaplan* [5]. The author-hosted PDF (37 pages) has the stated title and authors; only its first page was read.
- Gonshor [1], van den Dries–Ehrlich [2], Marker [4] and the encyclopedia page [9] were not checked.

A.4 Proofs read when the report was written

The proofs of lemmas 3.1 and 3.2, theorem 3.4, theorem 4.1, proposition 4.2, proposition 4.4, section 5, Theorem A, theorems 6.2 and 6.3, corollary 6.5, sections 7 and 8, sections 9 and 10 and theorem 11.2 were read line by line when the report was written. No error was found. The explicit formulas were rechecked: $\Phi_0(\omega) = \omega^{\omega^{1/2}}$, $\Phi_1(\omega) = \omega^{\omega^{7/2}}$, $g = h_1(1) = \omega^{7/2}$, and the consecutive exponents of (8.3) differ by $g + g_n - g_{n+1} > 0$. One understatement is recorded after proposition 9.2. This reading is not an independent proof review.

A.5 Theorem numbers

Numbers are those of this build and of the delivered PDF, which agree: labels were added and no environment was inserted before an existing one. Remarks and warnings share the theorem counter, so the classification of monomial-preserving embeddings is Proposition 4.4 (after Remark 4.3), and the saturated-group theorem is Theorem 11.2 (after a definition).

A.6 Files

File	Role
<code>article.tex</code>	this report (delivered as <code>independent_surreal_copies.tex</code>)
<code>article.pdf</code>	a build of this text, not the delivered PDF
<code>README.md</code>	the collection README, replacing the delivered one

The source shipped no verification program, recorded run or audit, so there is nothing to rerun. The delivered README reports a build with pdfTeX 1.40.26 and a visual inspection of rendered pages and contact sheets; those renderings are not shipped. [merge] This table lists the files of source 01. The seven shipped files of source 02, with the prefix `02-composita-`, and their rerun are in Appendix A.8.

A.7 Source 02 (batch 34)

[merge] Source 02 is manuscript 05 of batch 34 of the collection: the archive `Surreal_Composita_Research`, delivered in commit `a4611ad` and placed in commit `a7a435f` as an addition to this report with local number 02. Its main file `article.tex` (a standalone article with an internal bibliography) is titled *Beyond Composita of Surreal Copies: Exact rank tests, differential transcendence, and omnific witnesses*, dated 23 September 2026, with the author line “AI-assisted research draft prepared for Vladimir Reshetnikov”; its PDF has 29 pages. It is pinned to commit `7b256e0792df7718995b816ad12a77bf56731f8f`, at which this report’s `article.tex` was blob `7d5cfe644e7c902f2b594179c337a6c633350fc3`. The placement chose an addition rather than a new report because the manuscript answers this report’s Research question 3 in the separated case and extends its witness (8.3). Its manuscript, README and PDF are not shipped; its code, recorded data and two audits are shipped with the prefix `02-composita-` (Appendix A.8).

What it contributed. The finite coefficient-rank test for composita (theorem 17.2), the base-change lemmas of section 18, the rapid coefficient-degree theorem and its continuum family (theorem 19.2 and corollary 19.4), the failure of the finite coefficient-field criterion (corollary 19.5), the private-block theorem and tree amplification over the finite-generation envelope (theorem 20.2, corollaries 20.4 and 20.6, and proposition 20.5), the exact Laurent degrees $|F|^{\aleph_0}$ (theorem 21.1), the ordinal family Y_α of purely infinite omnific integers with independent Euler jets (theorem 22.2 and corollary 22.3), set-parameter tail resilience (theorem 23.3 and corollary 23.4), the arithmetic and Gaussian consequences (theorems 24.2 and 24.4), twelve research questions and a formalization decomposition.

Editorial changes.

- (1) Every label of the source carries the prefix `isc:cp:`, with the source’s name after it, except that its `sec:conventions` is `isc:cp:sec:foundations` and its appendix labels are replaced by `isc:cp:sec:verify` and `isc:cp:sub:formal`. The merge also labels the source’s unlabelled definition of coefficient rank, example, remarks and questions, and its new subsections and appendix subsections.
- (2) Symbols are renamed as in section 15.5; no normalization changed.
- (3) The source’s re-derivation of the explicit copies and of the re-expansion (its Subsections 8.1–8.2) is printed once, in sections 7 and 8, with the correspondence in section 22.1; the source’s added observations are kept there. Its duplicate of lemma 9.1 is kept inside lemma 24.1, where it serves a new statement.

- (4) The source’s citations of this report by section number are replaced by cross-references to the same sections. Its bibliography entries [10, 11, 12, 13] are added to this report’s; its entry for L’Innocente–Mantova is this report’s [3]. These four references were not checked in the merge.
- (5) Text marked [merge] was added in the merge: in section 15 (the first paragraph, the note on Y_0 , and sections 15.4 to 15.6), after theorems 17.2 and 22.2, lemmas 18.3 and 23.1, corollary 19.4, and example 19.6, in section 22.1, in the construction of the $a_{\alpha,n}$, in section 25.5, after Research questions 14, 17, 19 and 22, in section 27, and in this appendix. In source 01’s text, [merge] notes were added to the title page, to section 1.6 (the status of the research questions), after corollary 6.1 (with the reciprocal note for the dilation report), after lemma 8.2, after the proof of Theorem C, after Research question 3, at the start of section 14, in item W1, and in Appendices A.1 and A.6. No statement of source 01 was changed.
- (6) Cross-references to lemmas, propositions, corollaries, remarks and the warning were printed as “theorem” before the merge: the environments share the theorem counter, and cleveref named them by the counter. The merge adds cleveref’s type hint to the labels of these environments (26 of source 01 and 33 of source 02; for example the label of lemma 8.2 now carries the hint [lemma]) and a cleveref name for the warning. No label was renamed and no number changed; only the printed names of those references are corrected.

Where the merge chose. The addition is printed after the ledger of non-claims and before the appendix, so that no existing section, theorem, equation or question number changes; its sections keep the source’s numbering shifted by fourteen. The outer group is renamed Γ rather than keeping H , because H is this report’s common core and the source uses it in that sense too. The explicit copies are not reprinted.

Checks in the merge. The proofs of theorems 17.2, 19.2, 20.2, 21.1, 22.2, 23.3, 24.2 and 24.4, lemmas 18.2, 18.3, 20.1, 23.2 and 24.1, corollaries 18.4 and 20.4, and proposition 20.5 were read line by line when the addition was merged; no error was found. The following were rechecked: $g = \omega^{7/2}$ and $b = \omega^{15/4}$ lie in G_1 (inner exponents in (3, 4)), $\lambda(g) = 1$, $\lambda(b) = 0$, $\Delta\tau = \tau$ and $\Delta\omega^b = 0$ from (22.5); $\beta_{0,n} = n$, so $g_{0,n} = \omega^{\psi(n)}$ and $Y_0 = \omega^{\omega^{15/4}}S$ with S as in (8.3); the consecutive exponents of (22.8) differ by $g + g_{\alpha,n} - g_{\alpha,n+1} > 0$; and the source’s numbering (its Theorem 8.2 is the omnific theorem, its Theorem 10.4 the surcomplex one) agrees with its delivered PDF. This reading is not an independent proof review. The source’s statements about this report are those of its pin; the parts it used are unchanged since.

A.8 Files of source 02, and their rerun

[merge] The seven shipped files of source 02 are these.

Shipped path	Delivered name	Role
02-composita-SOURCE_AUDIT.md	SOURCE_AUDIT.md	pin, source scope, priority limits
02-composita-PROOF_AUDIT.md	PROOF_AUDIT.md	hypotheses, critical steps, non-claims
code/02-composita-verify.py	verify.py	exact finite checks (SymPy)
code/02-composita-build.sh	build.sh	runs the checks, then builds the source's PDF
data/02-composita-verification.json	verification.json	recorded run: 2,530 checks, 21 categories
data/02-composita-requirements.txt	requirements.txt	sympy==1.14.0
data/02-composita-BUILD_AUDIT.json	BUILD_AUDIT.json	build and PDF inspection record of the delivered PDF

All seven are byte-identical to the delivery. The manuscript, its README and its PDF are not shipped; the build audit therefore describes a PDF (29 pages, 499,044 bytes, pdfTeX 1.40.26) that is not in the collection.

The program writes a file only when given `-output`; otherwise it prints its JSON report. The delivered `build.sh` changes into its own directory and runs `python3 verify.py -output verification.json` before building `article.tex`. Under the shipped names it finds neither file in `code/`, and an adapted copy would write its record next to the script, not over the shipped data file. Rerun the checks on a copy, or with an explicit output path outside the report:

```
python code/02-composita-verify.py --output <scratch>/verification.json
```

In the merge the program was run on a copy with Python 3.14.4 and SymPy 1.14.0. It passed all 2,530 checks with the same 21 category counts as the recorded run; the only difference in the JSON record is the Python version (3.13.5 in the recorded run).

References

- [1] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. Standard background on normal forms and surreal arithmetic.
- [2] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. doi:10.4064/fm167-2-3. In particular pp. 174–176 for the class framework, set-cut property, normal forms, and Hahn subfields.
- [3] S. L’Innocente and V. Mantova, A factorisation theory for generalised power series and omnific integers, *Advances in Mathematics* **442** (2024), article 109513. arXiv:1710.07304, version 5, 22 January 2024. See Fact 2.1.1 for Hahn-field closedness, Section 2.3 for normal forms, and Proposition 2.4.5 for the cofinality/fraction-field obstruction. The arithmetic-intersection application here is distinct from that published proposition.
- [4] D. Marker, *Model Theory: An Introduction*, Graduate Texts in Mathematics, vol. 217, Springer, 2002. doi:10.1007/b98860.
- [5] P. Ehrlich and E. Kaplan, *Surreal ordered exponential fields*, author-hosted manuscript. https://elliottakaplan.github.io/Surreal_ordered_exponential_fields.pdf. Consulted as a

comparison for the additional exponential-embedding requirements, not as an input to any proof above.

- [6] VladimirReshetnikov/Surreal, documentation catalogue and project overview, repository snapshot `bcac55ae6fd3f2b354e568ba1d2e94d496a2cc59`, consulted 23 September 2026. [docs/REA DME.md](#); [README.md](#). The catalogue distinguishes AI-assisted manuscripts from the exact scope of checked formalization.
- [7] Surreal project, *Omnific-Preserving Automorphisms: Convex-scale stabilizers, definable constants, nondefinable monomials, and algebraic-parameter rigidity*, AI-assisted merged research draft, 23 September 2026, at the same pinned snapshot. [docs/surreal/omnific-preserving-automorphisms/article.tex](#). Used for scope and nonduplication comparison, not as an unproved dependency.
- [8] Surreal project, *Bounded-Support Hahn Arithmetic: Cofinal descent, optimal support thresholds, and maximal transcendence*, AI-assisted research draft, 22 September 2026, at the same pinned snapshot. [docs/surreal/transcendence-over-bounded-support/article.tex](#). Its main theorems propose stronger transcendence and descent results; [section 10](#) gives the needed one-witness argument independently.
- [9] Wikipedia contributors, *Surreal number*, user-supplied background page, consulted 23 September 2026. https://en.wikipedia.org/wiki/Surreal_number. This reference is orientation only; the proofs rely on the named mathematical foundations above rather than on this page.
- [10] L. S. Krapp, S. Kuhlmann and M. Serra, *Generalised power series determined by linear recurrence relations*, [arXiv:2206.04126v3](#), 2025. Cited by source 02 for the literature comparison with recurrence-defined series substructures, not as a proof of its compositum theorems.
- [11] S. Kuhlmann and M. Matusinski, *Hardy type derivations on generalized series fields*, [arXiv:0903.2197v4](#), 2012. [doi:10.1016/j.jalgebra.2011.11.024](#). Cited by source 02 for the background distinction between series derivations and Hardy-type conditions.
- [12] A. Berarducci and V. Mantova, *Surreal numbers, derivations and transseries*, *Journal of the European Mathematical Society* **20** (2018), 339–390. [arXiv:1503.00315v3](#). [doi:10.4171/JEMS/769](#).
- [13] The Stacks Project Authors, *The Stacks Project*, Section 9.27, “Linearly disjoint extensions”, Tag 09IC. <https://stacks.math.columbia.edu/tag/09IC>.